

Lattice Group Signatures, Revisited

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Abstract. Group signatures are one of the central privacy-preserving authentication mechanisms, offering an interesting trade-off between accountability and anonymity. Their versatility has led to many applications and even standardization at ISO/IEC. Unfortunately, they lack so far efficient quantum-safe constructions, despite several works implementing the seminal framework by Bellare, Micciancio and Warinschi (BMW) in the lattice setting.

In this work, we propose an alternative lattice-based construction that departs from the BMW blueprint by trying to minimize the number of elements to conceal in zero-knowledge proofs, the latter being quite complex in this setting. Concretely, it relies on delegated lattice bases, while avoiding the complex OR-proofs of some previous attempts in that direction. Combined with some tricks leveraging the peculiarities of a recent lattice sampler, it results in an efficient scheme that yet retains all the BMW security properties while only relying on standard lattice assumptions.

Keywords: Lattice-Based Cryptography · Group Signatures · Privacy · Static Groups

1 Introduction

Introduced by Chaum and Van Heyst [CvH91], group signatures quickly gained traction, with countless papers introducing concrete constructions, formal security models or new features. Far from being limited to the academic world, group signatures found their way into the industry [ISO13, TCG15, Int16] either directly or through its numerous variants, e.g., [BCC04, BL07].

The key to their success is the flexible balance between privacy and accountability they offer, making them amenable to most use-cases. Group signatures indeed enable a user to authenticate as an anonymous member of a group, while implementing safeguards, consisting in an opening process that allows a dedicated entity (the opener) to lift the anonymity of any group signature. It is particularly useful in scenarios where one needs to check that users own some rights, without identifying them (for privacy purposes). The group can then be

the one of all entities sharing these rights, in which a group signature acts as rights attestation. For example, the “group” can be the set of adult citizens, allowing to check whether a user is over 18, or the set of devices sharing a common certification level in the context of secure computing. Starting from this baseline, one can easily add some features, ranging from controlled linkability [BCC04] to revocation [BL07], threshold [CDL⁺20] or limited [SEH⁺12] opening, etc.

Arguably, the seminal work by Bellare, Micciancio and Warinschi [BMW03] (later abbreviated BMW) has been instrumental to the development of group signatures, by providing both a formal security model and a blueprint for concrete constructions. The core idea of the BMW approach is to deliver a certificate to each user that attests membership to the group. To ensure both anonymity and accountability, this certificate is concealed via encryption while validity of the resulting elements is demonstrated via non-interactive zero-knowledge proofs (NIZK).

This triptych “certificate, encryption, NIZK proof” is almost ubiquitous in existing constructions, both in the pre-quantum and the post-quantum worlds, but with varying degrees of success. Whereas pre-quantum constructions have proved remarkably efficient (see, e.g. [DP06]), the situation of post-quantum schemes is less satisfying, hampered by the difficulty of combining signature, encryption and NIZK proofs in this setting. Yet, several advances were achieved in the last few years [dPLS18, LNPS21, LNP22, JS25b] in the area of lattice cryptography, but mostly indirectly, that is, through improvements of the underlying building blocks. This has led the size of group signatures in this model to decrease to approximately 75 KB, which is still rather large. While further improvements of this approach can be expected, a significant cut of the size of a group signature is unlikely without a major breakthrough in the design of NIZK proofs that constitute its largest component. This calls for investigating alternative approaches to the BMW blueprint, that would prove more suitable to the lattice setting. We note that such alternatives already exist, e.g., [GKV10, LLNW16], but they are more proofs of concepts than practical solutions. Recently, Lyubashevsky and Nguyen [LN22] introduced another approach with remarkable size, but generating a signature requires performing a number of rather complex computational steps that grows linearly with the number of users, which quickly becomes hard to manage. Improving performance for medium to large scale groups then remains an open problem.

1.1 Our Contributions

In this paper, we propose a new lattice construction of static group signature addressing the above limitations. We significantly reduce the weight of the NIZK component by departing from the traditional BMW approach for proving membership. As we detail below, rather than grounding group membership in the possession of a certificate, we base it on the ability to generate fresh (short) preimages for specific syndromes. This removes the need for hiding said preimages in the NIZK proof. Meanwhile, we also leverage peculiarities of our scheme to optimize the encryption component. The latter, used for opening group signa-

tures, is quite seamless in our construction and only incurs a very small overhead to the size of group signatures, even when achieving the strongest anonymity level (CCA-anonymity). Let us now describe the main ideas behind our construction.

1.1.1 Proving Membership. We start by detailing how a user can prove membership to the group, which is the core essence of group signatures. In the BMW paradigm, membership is associated with the ownership of a certificate $\mathbf{v}$, which intrinsically places group members in a complex position: they need to prove that they own such a $\mathbf{v}$ without leaking any information on it, to preserve anonymity, which rules out most natural solutions, e.g., sending $\mathbf{v}$. Actually, users essentially have two options.

- (1) Either they conceal $\mathbf{v}$ and thus need to produce a NIZK proof that the concealed value is a valid certificate, as was done in most previous constructions;
- (2) or they re-randomize the certificate, that is, deriving from $\mathbf{v}$ a new certificate $\mathbf{v}'$ that is unlinkable to $\mathbf{v}$.

The second option is very attractive but, to our knowledge, randomizable signatures only exist in pairing-friendly settings (e.g. [CL04,PS16]) and have no lattice counterparts. It thus only remains option (1) and its heavy reliance on NIZK proofs, the latter then amounting for most of the group signature complexity (in size and runtime).

Delegating Preimage Sampling. Our approach avoids such costly NIZK proofs of the certificate $\mathbf{v}$ altogether by associating group membership with the knowledge of a basis of the q -ary kernel lattice $\mathcal{L}_q^\perp(\mathbf{A}_t)$ for some matrix $\mathbf{A}_t$ that remains to be defined. Thanks to this basis, and using well-known lattice sampling algorithms, e.g., [GPV08], the user can securely generate a short preimage $\mathbf{v}$ of any syndrome $\mathbf{u}$. With a few precautions that we describe later, $\mathbf{v}$ can be fully disclosed to the verifier, which considerably alleviates the NIZK relation. Moreover, $\mathbf{v}$ acts as a proof of the user’s capacities, thereby demonstrating that they are indeed a member of the group.

At this stage, we nevertheless need to make a few comments. The first one is that we have traded a certificate for a lattice basis in the group signing key, which is larger and more complex to generate. However, it leads to shorter group signatures, which is arguably an interesting trade-off as each group signing key is expected to produce many group signatures. The second one is that $\mathbf{A}_t$ cannot be any matrix as it must somehow depend on the public key of the group manager which should be the only entity able to generate the corresponding basis. Additionally, $\mathbf{A}_t$ must differ according to each group member while, at the same time, minimize the number of components specific to the group member (because these components then have to be concealed in the NIZK proof).

In our construction, we define $\mathbf{A}_t$ as $[\mathbf{I}_d | \mathbf{A} | q_L t \mathbf{G}_H - \mathbf{B} | \mathbf{a}_3] \in \mathcal{R}_q^{d \times m}$, where $\mathcal{R}$ is an integer ring (e.g., a power-of-two cyclotomic ring), $\mathbf{G}_H$ is a gadget matrix, $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{a}_3$ are part of the group public key, q_L divides q , and t is some element of an identity space $\mathcal{I}$ (concretely, $\mathcal{I}$ is a set of short elements of $\mathcal{R}_q$). This definition is chosen to perfectly match the recent sampler from [JS26], which

allows the group manager (that knows the trapdoor underlying $\mathbf{B}$) to generate short preimages of $\mathbf{0}$ for any t , from which a basis of $\mathcal{L}_q^\perp(\mathbf{A}_t)$ is constructed. Moreover, it allows for differentiating the matrix $\mathbf{A}_t$ according to each group member by merely changing t , which is the only information the group member must hide about $\mathbf{A}_t$. As t is only a short element of $\mathcal{R}_q$, it has almost no impact on the NIZK complexity, and its membership to $\mathcal{I}$ can be efficiently proven. We also note that even given a basis of $\mathcal{L}_q^\perp(\mathbf{A}_t)$, a user cannot sample short preimages for $\mathbf{A}_{t'}$ with $t' \neq t$, i.e., on behalf of another user.

How To Choose the Syndromes? A group member in our system is thus any entity that can invert any syndrome $\mathbf{u}$ with $\mathbf{A}_t$, for some tag $t \in \mathcal{I}$. To prove this capacity, the group member merely demonstrates it on some concrete $\mathbf{u}$ by providing a short preimage $\mathbf{v}$. However, security and privacy place some restrictions on $\mathbf{u}$. First of all, $\mathbf{u}$ cannot be known (let alone chosen) by the verifier. Otherwise, hiding t would not be sufficient: the verifier could simply test whether $\mathbf{A}_t \mathbf{v} = \mathbf{u}$ for all t , until it gets a match. One could hope to deter this exhaustive search by defining a very large group size $N = |\mathcal{I}|$. However, not only would it limit our construction to specific group size ranges, but it would unfortunately also conflict with security as our reduction from standard assumptions crucially requires a polynomial group size³. Similarly, allowing the user to freely choose $\mathbf{u}$ is also insecure, as they could simply choose any small $\mathbf{v}$ and sets $\mathbf{u} = \mathbf{A}_t \mathbf{v}$.

In our construction, we opt for a middle way and let the user generate $\mathbf{u}$ as a M-LWE instance $\mathbf{A}\mathbf{s} + \mathbf{e}$. The user then generates a short preimage $\mathbf{v}$ of $\mathbf{u}$ and sends $\mathbf{v}$ along with a NIZK proof that it satisfies $\mathbf{A}_t \mathbf{v} = \mathbf{A}\mathbf{s} + \mathbf{e}$ for some hidden t , $\mathbf{s}$ and $\mathbf{e}$. It thus prevents a user from choosing a random $\mathbf{u}$ as it needs to prove it knows the underlying short secret $(\mathbf{s}, \mathbf{e})$. Here, we note that we have essentially swapped the set of witnesses and public elements, compared to the BMW approach. Indeed, we have a public $\mathbf{v}$ and a secret $\mathbf{u}$, whereas BMW would have hidden $\mathbf{v}$ and revealed $\mathbf{u}$. However, we stress that these elements are by no means comparable. In particular, our witnesses $\mathbf{s}$ and $\mathbf{e}$ have a much smaller combined dimension and norm than $\mathbf{v}$, which is key in decreasing the complexity of the NIZK proof.

1.1.2 Adding the Opening Capability. Users are now able to privately demonstrate that they belong to the group but they still need to provide some concealed information about their identity, to enable the opening of group signatures. This could be done very simply and modularly by adding an encryption of t , for example, using a scheme that would be either IND-CPA secure or IND-CCA secure, depending on the anonymity level one targets. Besides adding a full ciphertext, one would have to prove that it is correctly formed, which would require proving knowledge of the underlying randomness, thereby increasing NIZK complexity.

³ We note that this restriction is common with prior lattice-based construction as the traceability proof strategy relies on guessing the targetted honest group member.

Leveraging the Composite Modulus. In our case, we can fortunately proceed more efficiently by noting that the syndrome $\mathbf{u} = \mathbf{A}\mathbf{s} + \mathbf{e}$ we built is actually very close to the largest component of a Dual-Regev ciphertext. At first sight, this does not look very helpful since we mentioned earlier that $\mathbf{u}$ must be unknown to the verifier. However, we can be a bit more specific here, by leveraging peculiarities of the sampler in [JS26]. Indeed, while the verifier cannot know the exact value of $\mathbf{u} \bmod q$, it can know $\mathbf{u} \bmod q_L$ as all identity dependencies vanish with this modulus due to the specific form of $\mathbf{A}_t$. Actually, the verifier already has access to this information from $\mathbf{v}$ by computing $\mathbf{A}_0\mathbf{v} \bmod q_L = [\mathbf{I}_d|\mathbf{A}|-\mathbf{B}|\mathbf{a}_3]\mathbf{v} \bmod q_L$, meaning there is nothing more to send. Moreover, the randomness underlying this ad-hoc partial ciphertext is already part of our NIZK witness, so there is no increase on this component either. It only remains to complete $\mathbf{u} \bmod q_L$ to get a full-fledged Dual-Regev ciphertext, that is adding one element of $\mathcal{R}_{q_L}$ to the group signature, as well as one short ring element to the witness of the NIZK.

Security of the Encryption. This, at a high level, successfully completes the opening procedure consisting of decrypting a Dual-Regev ciphertext modulo q_L . We however hint at a few subtleties which we explain in details in Section 4.1.

First, following the proof strategy of [JS26], we wish to reduce the traceability of our scheme to forging a preimage $\mathbf{x}$ of $\mathbf{0}$ for $\mathbf{A}_{t_h}$ where t_h is the identity of an honest user (let us call it U). Then, by guessing which t_h will be targetted by the adversary, we can tweak the key material (in particular by replacing $\mathbf{B}$ with $\mathbf{B} + q_L t_h \mathbf{G}_H$) so as to make the gadget vanish and use $\mathbf{x}$ to solve a specific instance of M-SIS. In doing so, we deprive the group manager of the ability to generate preimages of $\mathbf{0}$ for $\mathbf{A}_{t_h}$, let alone a basis to give to U . It means that to be able to carry this security reduction from M-SIS, we must simulate signatures produced by U without said basis, but in such a way that these simulated signatures coherently open to user U .

Because we created some dependencies between the ciphertext randomness and some of the public elements we provide, this simulation needs to be treated carefully in our security analysis. For the sake of clarity, our analysis leverages a hint variant of M-LWE introduced in [ENP24], which we generalize to cover our situation. We show that this variant accepts a reduction from M-LWE for the parameters we use in our construction. In other words, our scheme is secure under standard assumptions only, namely M-SIS and M-LWE.

This brings us to our second subtlety. The reduction, albeit efficient and not too constraining for the parameters, imposes Gaussian distributions on the ciphertext randomness. This naturally increases its size compared to a standalone Dual-Regev encryption like Kyber/ML-KEM [BDK⁺18]. In particular, to keep a correct decryption, this requires choosing a larger encryption modulus which conflicts with the goal of [JS26] of minimizing q_L to have shorter preimages and thus shorter bases in our case. We detail in Section 4.1 how we can lift the encryption component to a larger modulus $q_L q'$ while keeping q_L as in [JS26].

1.1.3 Enhancing Anonymity to Cover Opening Queries. At this stage, our scheme only satisfies the property of weak anonymity (or CPA-anonymity)

which corresponds to a setting where the adversary is not allowed to query opening of group signatures. Our approach and construction can be extended to the highest level of anonymity (CCA-anonymity), where the adversary can request the opening of all but one group signatures, at only a very mild cost on the group signature size. In previous lattice-based schemes [dPLS18,LNP22], CCA-anonymity is generally achieved by reusing the Naor-Yung strategy [NY90] of having two independent encryption of the same message, along with a NIZK for reconciliation. As these group signatures are built with a verifiable encryption (i.e., a public-key encryption with a NIZK proof of well-formedness, as we described also in our approach above), Lyubashevsky and Neven [LN17] showed one can obtain an IND-CCA2 verifiable encryption at a moderate cost, and thus a CCA-anonymous group signature scheme. In our case, we optimize this approach by following the Naor-Yung paradigm but with shared randomness [BMV16] in that we reuse the same first half of Dual-Regev for both ciphertexts (i.e., the same randomness $\mathbf{s}, \mathbf{e}$). This avoids increasing the witness of the NIZK proof, which in our case would essentially double its dimension.

Even in this setting, we are still able to carry out the Naor-Yung proof structure while dealing with opening queries. The anonymity proof presents requirements similar to the traceability proof discussed above on the dependencies between the different elements. We however show that under the very same hint M-LWE assumption (which, we insist, reduces from regular M-LWE in our parameter setting), we can tackle all these subtleties in a very efficient manner. Very concretely, reaching CCA-anonymity increases our group signature size by less than 1.5 KB.

1.1.4 Performance. In the end, we obtain a static group signature scheme based on well-understood lattice assumptions (M-SIS, M-LWE) leveraging this change of paradigm to reach more compact sizes. In particular, in its CPA-anonymous version, our group signatures amount to 52.49 KB for a group size of roughly 2^{32} . As the components we use do not scale⁴ with N , our performance extrapolates to other group sizes to cover different use cases. Our scheme is then almost twice as compact as that of [LNP22], and about 35% smaller than the latter with the most recent improvement of [JS25b]. When extending it to CCA-anonymity, we reach group signatures of 53.87 KB, only a 1.38 KB increase, which is significantly smaller than the overhead incurred by other constructions. A detailed comparison with prior works is given in Section 6.2, but we still discuss some of the techniques related to ours.

1.2 Related techniques

The idea of delegating a trapdoor to each group member is reminiscent of [GKV10], but the comparison stops there as their group public key concatenated all the N group members' public keys, which is hardly scalable. Moreover, the group

⁴ The group size only affects the tightness of our security proof, but in a very mild way.

signature itself contains N LWE-like tokens combined with an OR-proof that one of them is generated with a short error (sampled with the delegated trapdoor). Hence, not only does the size of their group signature scale with N , but the generation of the OR-proof can quickly become cumbersome for medium to large groups, at least computationally (see our discussion on [LN22] above).

The goal of minimizing the ciphertext overhead is shared with the construction of [LNPS21] that recycles some parts of the commitments they use in their NIZK proofs. However, this technique does not seem to interact nicely with the powerful NIZK framework in [LNP22] as the authors no longer use it in the follow-up group signature construction they describe there. While there might be some ways to adapt it to the [LNP22] framework, the exact outcome is unclear, and this would require a complete reassessment of the security of the resulting NIZK proof, which is far from trivial. Our solution to limit the ciphertext overhead is fundamentally different as it leverages the unique features of [JS26] and avoids meddling with the NIZK framework itself.

2 Preliminaries

Throughout this paper, we use the following standard notations. We denote by $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ the set of natural integers, integers, rationals and reals. For two integers $m \leq n$, we define $[m, n] = \{m, \dots, n\}$, and $[n] = [1, n]$. For a ring R , we use $R^\times$ to define the set of units of R , with the abuse of notation that $\mathbb{N}^\times = \mathbb{N} \setminus \{0\}$. For $q \in \mathbb{N}^\times$, we define $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$. When we explicitly need a choice of representative of a residue class $a \in \mathbb{Z}_q$, we may use $a \bmod^\pm q$ to denote the unique representative of a lying in $(-q/2, q/2] \cap \mathbb{Z}$. We denote by $[a^{-1}]_q$ the inverse of a in $\mathbb{Z}_q^\times$. Matrices and vectors are written in bold upper case and lower case letters respectively, and we use $[\mathbf{A}|\mathbf{B}]$ for the horizontal concatenation of matrices. The j -th column of a matrix $\mathbf{A}$ is denoted by $\mathbf{A}[j]$. By convention, vectors are column vectors and the vertical concatenation is then defined by $[\mathbf{a}^T | \mathbf{b}^T]^T$ where the superscript T is the transpose operator. We use $\|\mathbf{a}\|_p$ to denote the standard ℓ^p norm of $\mathbf{a} \in \mathbb{R}^n$, and for $\mathbf{A} \in \mathbb{R}^{m \times n}$, we define $\|\mathbf{A}\|_\infty = \max_{i \in [n]} \|\mathbf{A}\mathbf{e}_i\|_2$ (column norm) where $\mathbf{e}_i$ is the i -th canonical vector of $\mathbb{R}^m$. For a matrix $\mathbf{B}$ with linearly independent columns, we denote by $\text{GSO}(\mathbf{B})$ the Gram-Schmidt Orthogonalization of its columns.

2.1 Lattices

A lattice $\mathcal{L}$ is a discrete subgroup of $\mathbb{R}^d$ which can be defined by a basis $\mathbf{B} \in \mathbb{R}^{d \times r}$ as $\mathcal{L} = \mathcal{L}(\mathbf{B}) = \mathbf{B}\mathbb{Z}^r$. The parameter r is called the rank of $\mathcal{L}$, and defined irrespective of the choice of basis as $\dim \text{Span}_{\mathbb{R}}(\mathcal{L})$. When $r = d$, we say $\mathcal{L}$ is full-rank. Note that as opposed to vector spaces, if $\mathbf{B}' \in \mathbb{R}^{d \times r}$ is a matrix of linearly independent columns each belonging to $\mathcal{L} = \mathcal{L}(\mathbf{B})$, we have $\mathcal{L}(\mathbf{B}') \subseteq \mathcal{L}(\mathbf{B})$ but not necessarily $\mathcal{L}(\mathbf{B}') = \mathcal{L}(\mathbf{B})$. Equality only holds if both lattices have the same volume. Fortunately, the basis $\mathbf{B}'$ of the sublattice can efficiently be transformed into a basis of $\mathcal{L}$ of the same quality. This result, implicit in [Lov86], has been formalized in [MG02].

Lemma 2.1 ([MG02, Lem. 7.1]). *Let $d \in \mathbb{N}^\times$. There is a deterministic polynomial-time algorithm **ToBasis** that, given an arbitrary basis $\mathbf{B} \in \mathbb{R}^{d \times d}$ of a full-rank lattice $\mathcal{L} = \mathcal{L}(\mathbf{B})$ and a full-rank matrix $\mathbf{U} = [\mathbf{u}_1 | \dots | \mathbf{u}_d]$ with $\mathbf{u}_i \in \mathcal{L}$ and such that $\|\mathbf{u}_1\|_2 \leq \dots \leq \|\mathbf{u}_d\|_2$, outputs a basis $\mathbf{S}$ of $\mathcal{L}$ such that for all $i \in [d]$, $\|\text{GSO}(\mathbf{S})[i]\|_2 \leq \|\text{GSO}(\mathbf{U})[i]\|_2 \leq \|\mathbf{u}_i\|_2$.*

2.2 Algebraic Number Theory

Let us now give some notions of algebraic number theory needed for our paper. As we solely work over power-of-two cyclotomic rings, we specify the definitions to this case. A more complete background can be found in the extensive literature. We also use their isomorphic description as quotient of polynomial rings. For n a power-of-two, we define $\mathcal{K} = \mathbb{Q}[x]/\langle x^n + 1 \rangle$, and the ring of integers $\mathcal{R} = \mathbb{Z}[x]/\langle x^n + 1 \rangle$. We may also use the ring $\mathcal{K}_{\mathbb{R}} = \mathbb{R}[x]/\langle x^n + 1 \rangle$. For any $q \in \mathbb{N}^\times$, we define $\mathcal{R}_q = \mathcal{R}/q\mathcal{R} = \mathbb{Z}_q[x]/\langle x^n + 1 \rangle$.

We describe the following notations for $\mathcal{R}$ but they naturally extend to $\mathcal{K}, \mathcal{R}_q$ by changing $\mathbb{Z}$ with the corresponding ring or field. Elements of $\mathcal{R}$ naturally embed into $\mathbb{Z}^n$ via their coefficient vector representation. This embedding is denoted by τ , i.e., for $a = \sum_{i \in [0, n-1]} a_i x^i \in \mathcal{R}$, $\tau(a) = [a_0 | \dots | a_{n-1}]^T$. The conjugate of an element a is defined by $a^* = a(x^{-1})$ which means $\tau(a^*) = [a_0 | -a_{n-1} | \dots | -a_1]^T$. Through this coefficient embedding, we can extend the ℓ^p norms to $\mathcal{R}$ as $\|a\|_p = \|\tau(a)\|_p$. We also define $T_1 = \tau^{-1}(\{0, 1\}^n)$. Then, multiplication in $\mathcal{R}$ maps to a matrix-vector multiplication, which naturally defines a ring homomorphism M_τ by the relation $\tau(ab) = M_\tau(a)\tau(b)$ for all $(a, b) \in \mathcal{R}^2$. In our case, $M_\tau(a)$ is the nega-circulant matrix with first column $\tau(a)$, i.e.,

$$M_\tau(a) = \begin{bmatrix} a_0 & -a_{n-1} & \dots & -a_1 \\ a_1 & a_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & -a_{n-1} \\ a_{n-1} & a_{n-2} & \dots & a_0 \end{bmatrix}.$$

These definitions extend component-wise to vectors and matrices over $\mathcal{R}$, except that conjugation also transposes the matrix or vector. In particular, $\tau(\mathbf{a})$ is the vertical concatenation of all the $\tau([\mathbf{a}]_i)$, and $M_\tau(\mathbf{A})$ is the block-matrix formed by applying M_τ to each entry of $\mathbf{A}$. Note that $M_\tau(\mathbf{A}^*) = M_\tau(\mathbf{A})^T$. For $\mathbf{A} \in \mathcal{R}_q^{d \times m}$ and $\mathbf{u} \in \mathcal{R}_q^d$, we define the q -ary structured lattice coset $\mathcal{L}_q^{\mathbf{u}}(\mathbf{A}) = \{\mathbf{x} \in \mathcal{R}_q^m : \mathbf{A}\mathbf{x} = \mathbf{u} \bmod q\mathcal{R}\}$, and the corresponding structured lattice $\mathcal{L}_q^\perp(\mathbf{A}) = \mathcal{L}_q^{\mathbf{0}}(\mathbf{A})$. When embedded via τ , these correspond to $\mathcal{R}$ -module lattice (cosets).

2.3 Probabilities

For a distribution $\mathcal{P}$ over a set S , we use $x \leftarrow \mathcal{P}$ to say that $x \in S$ was sampled from distribution $\mathcal{P}$, and $x \sim \mathcal{P}$ to say that the random variable x is distributed according to $\mathcal{P}$. The later is generally used as subscript in probabilities to specify the distribution of interest, e.g., $\mathbb{P}_{x \sim \mathcal{P}}[E]$ is the probability that event E occurs

while x follows $\mathcal{P}$. For distributions $\mathcal{P}_1, \dots, \mathcal{P}_n$ of respective support $S_1, \dots, S_n$, we use $\otimes_{i \in [n]} \mathcal{P}_i$ to designate the joint distribution over $\times_{i \in [n]} S_i$ where the i -th component is independently drawn from $\mathcal{P}_i$. If all the $\mathcal{P}_i$ are identical and equal to $\mathcal{P}$, we may use $\mathcal{P}^n$ for conciseness. For a probabilistic algorithm A with inputs inp , we use $\text{out} \leftarrow A(\text{inp})$ to denote that out was computed using A on input inp and the randomness inside of A . Then, for two distributions $\mathcal{P}_0, \mathcal{P}_1$ over a countable set S (or exchangeably random variables following $\mathcal{P}_0, \mathcal{P}_1$), we recall the statistical distance between $\mathcal{P}_0$ and $\mathcal{P}_1$ by $\Delta(\mathcal{P}_0, \mathcal{P}_1) = \frac{1}{2} \sum_{x \in S} |\mathcal{P}_0(x) - \mathcal{P}_1(x)|$.

Specific Distributions. For a finite set S , we define the discrete uniform distribution over S by $U(S)$. By abuse of notation, we may use $\{0\}$ to denote the deterministic distribution outputting 0 with probability 1. For $N \in \mathbb{N}^\times$ and $w \in 2\mathbb{N}$ with $w \leq N$, we define $\mathfrak{W}_{N,w} = \{\mathbf{x} \in \{-1, 0, 1\}^N : \mathbf{x} \text{ has } w/2 \text{ ones and minus ones}\}$. We then define $\mathcal{W}_{d,w} = \tau^{-1}(U(\mathfrak{W}_{nd,w}))$, which corresponds to the uniform distribution over vectors of $\mathcal{R}^d$ whose coefficient embedding is a ternary vector with $w/2$ ones and $w/2$ negative ones. This distribution can be efficiently sampled from a variant of the Fischer-Yates shuffle algorithm [Knu98]. We also define ψ_1 to be the centered binomial distribution of parameter 1, i.e., that outputs 0 with probability 1/2 and ± 1 each with probability 1/4, and define $\mathcal{B}_1 = \tau^{-1}(\psi_1^n)$.

Relative Error Lemma. Our security proofs employ the relative error lemma of [Pre17] which efficiently bounds the Rényi divergence between marginally close distributions over the same support. Using properties of the Rényi divergence (see [vEH14] for details on the data processing inequality, multiplicativity and probability preservation properties of the Rényi divergence), we can adapt it into the following.

Lemma 2.2 ([Pre17, Lem. 3] adapted). *Let $\mathcal{P}_0, \mathcal{P}_1$ be two distributions over the same support S . Let $\delta \geq 0$ such that for all $x \in S$, $\mathcal{P}_0(x) \in [1 - \delta, 1 + \delta]\mathcal{P}_1(x)$. It then holds that for any $a > 1$, any $Q \in \mathbb{N}^\times$, any (possibly randomized) function f with domain S^Q and range S' , and any event $E \subseteq S'$, we have*

$$\mathcal{P}_0^{Q,f}(E) \leq \left(1 + \frac{a(a-1)\delta^2}{2(1-\delta)^{a+1}}\right)^{Q/a} \mathcal{P}_1^{Q,f}(E)^{1-1/a},$$

where $\mathcal{P}_i^{Q,f}$ is the distribution obtained by sampling $\mathbf{x} \leftarrow \mathcal{P}_i^Q$ and outputting $f(\mathbf{x})$, and where $\mathcal{P}_i^Q$ is the joint distribution of Q copies of $\mathcal{P}_i$.

Discrete Gaussian Measures. For a symmetric positive definite matrix $\mathbf{S} \in \mathbb{R}^{d \times d}$ and $\mathbf{c} \in \mathbb{R}^d$, we define the Gaussian function of center $\mathbf{c}$ and parameter $\mathbf{S}$ by $\rho_{\sqrt{\mathbf{S}}, \mathbf{c}}(\mathbf{x}) = \exp(-\pi(\mathbf{x} - \mathbf{c})^T \mathbf{S}^{-1}(\mathbf{x} - \mathbf{c}))$. For any countable set $A \subset \mathbb{R}^d$, we then define the discrete Gaussian over A , of center $\mathbf{c}$ and parameter $\mathbf{S}$, by $\mathcal{D}_{A, \sqrt{\mathbf{S}}, \mathbf{c}}(x) = \rho_{\sqrt{\mathbf{S}}, \mathbf{c}}(\mathbf{x}) \mathbf{1}_A(\mathbf{x}) / \rho_{\sqrt{\mathbf{S}}, \mathbf{c}}(A)$, where $\mathbf{1}_A$ is the indicator function of A . We extend the notation to Gaussians over algebraic modules via the coefficient embedding τ . Hence, for $A \subset \mathcal{R}^d$, $\mathbf{c} \in \mathcal{K}_{\mathbb{R}}^d$, and $\mathbf{S} \in \mathbb{R}^{nd \times nd}$ positive definite, we write $\mathcal{D}_{A, \sqrt{\mathbf{S}}, \mathbf{c}} = \tau^{-1}(\mathcal{D}_{\tau(A), \sqrt{\mathbf{S}}, \tau(\mathbf{c})})$. When $\mathbf{S} = s^2 \mathbf{I}_d$, we write $\mathcal{D}_{A, s, \mathbf{c}}$. Also, when $\mathbf{c} = \mathbf{0}$, we omit the subscript.

The smoothing parameter of a lattice, coined by Micciancio and Regev [MR07], is parameterized by a smoothing loss $\varepsilon > 0$ and defined as $\eta_\varepsilon(\mathcal{L}) = \min\{s > 0 : \rho_{1/s}(\mathcal{L}^*) = 1 + \varepsilon\}$, where $\mathcal{L}^*$ is the dual lattice of $\mathcal{L}$. For elliptical Gaussians with parameter $\mathbf{S}$, we say that $\sqrt{\mathbf{S}} \geq \eta_\varepsilon(\mathcal{L})$ if and only if $\rho_{\sqrt{\mathbf{S}^{-1}}}(\mathcal{L}^*) \leq 1 + \varepsilon$. This is equivalent to $\mathbf{S} \geq \eta_\varepsilon(\mathcal{L})^2 \mathbf{I}$ for the Loewner order. Then, by [GPV08], it holds that for any basis $\mathbf{B}$ of $\mathcal{L} \subset \mathbb{R}^n$, $\eta_\varepsilon(\mathcal{L}) \leq \|\text{GSO}(\mathbf{B})\|_\infty \eta_\varepsilon(\mathbb{Z}^n)$. The following result from [EWY23] also gives an exact expression with good approximations for remarkable lattices like $\mathbb{Z}^n$.

Lemma 2.3 ([EWY23, Lem. 5]). *For any lattice $\mathcal{L}$ and $\varepsilon > 0$, we have*

$$\eta_\varepsilon(\mathcal{L}) = \frac{1}{\lambda_1(\mathcal{L}^*)} \sqrt{\frac{1}{\pi} \ln \left(\frac{\kappa(\mathcal{L}^*)}{\varepsilon} (1 + o_\varepsilon(1)) \right)}$$

where $\lambda_1(\mathcal{L}^*)$ is the first minimum of the dual of $\mathcal{L}$, and $\kappa(\mathcal{L}^*)$ is the kissing number of the dual of $\mathcal{L}$. In particular, it holds that for any $n \in \mathbb{N}^\times$, $\eta_\varepsilon(\mathbb{Z}^n) \approx \sqrt{\ln(2n/\varepsilon)}/\pi$.

We later need the following standard lemma on the convolution of independent discrete Gaussian distributions. It can be seen as a generalization of [Reg09, Claim 3.9] by replacing the continuous Gaussian by a discrete one, and dealing with elliptical Gaussians instead of spherical. It is a specific case of [GMPW20, Thm. 4.6].

Lemma 2.4 ([GMPW20, Thm. 4.6] adapted). *Let d be in $\mathbb{N}^\times$, and $\varepsilon \in (0, 1)$. Let $\mathbf{S}_0, \mathbf{S}_1$ be symmetric positive definite matrices of $\mathbb{R}^{d \times d}$, $\mathbf{c}_0, \mathbf{c}_1, \mathbf{a}_0, \mathbf{a}_1$ be in $\mathbb{R}^d$, and $\mathcal{L} \subset \mathbb{R}^d$ be a full rank lattice. We define $\mathbf{a} = \mathbf{a}_0 + \mathbf{a}_1$, $\mathbf{c} = \mathbf{c}_0 + \mathbf{c}_1$, and $\mathbf{S} = \mathbf{S}_0 + \mathbf{S}_1$. We also define $\mathbf{T} = (\mathbf{S}_0^{-1} + \mathbf{S}_1^{-1})^{-1}$. We let $\mathcal{P} = \mathcal{D}_{\mathcal{L}+\mathbf{a}_0, \sqrt{\mathbf{S}_0}, \mathbf{c}_0} + \mathcal{D}_{\mathcal{L}+\mathbf{a}_1, \sqrt{\mathbf{S}_1}, \mathbf{c}_1}$. If $\sqrt{\mathbf{T}} \geq \eta_\varepsilon(\mathcal{L})$, it holds that for all $\mathbf{x} \in \mathcal{L} + \mathbf{a}$, $\mathcal{P}(\mathbf{x}) \in [\frac{1-\varepsilon}{1+\varepsilon}, \frac{1+\varepsilon}{1-\varepsilon}] \mathcal{D}_{\mathcal{L}+\mathbf{a}, \sqrt{\mathbf{S}}, \mathbf{c}}(\mathbf{x})$.*

We also use the standard tail bound on centered discrete Gaussians from [Ban93]. In our construction, we define bounds on discrete Gaussians as $c_d s \sqrt{d}$ where c_d corresponds to the tailcut in Lemma 2.5.

Lemma 2.5 ([Ban93, Lem. 1.5]). *Let $d \in \mathbb{N}^\times$ and $\mathcal{L} \subset \mathbb{R}^d$ be a full-rank lattice. For any $s > 0$ and $c > 1/\sqrt{2\pi}$, we have $\mathbb{P}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{L}, s}}[\|\mathbf{x}\|_2 > cs\sqrt{d}] < (c\sqrt{2\pi}ee^{-\pi c^2})^d$.*

2.4 Hardness Assumptions

The security of our group signature relies on the hardness of well-known module lattice problems, that is *Module Learning With Errors* and *Module Short Integer Solution* [LS15]. We recall their definition here.

Definition 2.1 (M-LWE). *Let $\mathcal{R}$ be a power-of-two cyclotomic ring of degree n . Let $d, m, q \in \mathbb{N}^\times$ and $\mathcal{D}$ be a distribution over $\mathcal{R}^{d+m}$. The (decisional) Module Learning With Errors problem, denoted $\text{M-LWE}_{n,d,m,q,\mathcal{D}}$ asks to distinguish the following distributions.*

$\mathcal{P}_0$	Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{b} \leftarrow U(\mathcal{R}_q^m)$, output $(\mathbf{A}, \mathbf{b})$.
$\mathcal{P}_1$	Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{x} = [\mathbf{s}^T \mathbf{e}^T]^T \leftarrow \mathcal{D}$, and compute $\mathbf{b} = [\mathbf{A} \mathbf{I}_m] \mathbf{x} \bmod q\mathcal{R} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q\mathcal{R}$. Output $(\mathbf{A}, \mathbf{b})$

We define $\text{Adv}_{\text{M-LWE}}[\mathcal{A}] = |\mathbb{P}_{(\mathbf{A}, \mathbf{b}) \sim \mathcal{P}_0}[\mathcal{A}(\mathbf{A}, \mathbf{b}) = 1] - \mathbb{P}_{(\mathbf{A}, \mathbf{b}) \sim \mathcal{P}_1}[\mathcal{A}(\mathbf{A}, \mathbf{b}) = 1]|$ as the advantage of a PPT adversary in solving M-LWE. We then call $\varepsilon_{\text{M-LWE}} = \sup_{\mathcal{A}} \text{PPT Adv}_{\text{M-LWE}}[\mathcal{A}]$ the hardness bound of M-LWE $_{n,d,m,q,\mathcal{D}}$, where the parameters are clear from the context. The search version corresponds to finding $\mathbf{X}$ given an instance drawn from $\mathcal{P}_1$.

Definition 2.2 (M-SIS). Let $\mathcal{R}$ be a power-of-two cyclotomic ring of degree n . Let $d, m, q \in \mathbb{N}^\times$ with $m > d$, and $B > 0$. The Module Short Integer Solution problem, denoted M-SIS $_{n,d,m,q,B}$ asks to find $\mathbf{x} \in \mathcal{L}_q^\perp([\mathbf{I}_d | \mathbf{A}])$ such that $0 < \|\mathbf{x}\|_2 \leq B$ given $\mathbf{A} \leftarrow U(\mathcal{R}_q^{d \times m-d})$. We define the advantage of an adversary $\mathcal{A}$ as $\text{Adv}_{\text{M-SIS}}[\mathcal{A}] = \mathbb{P}_{\mathbf{A} \sim U(\mathcal{R}_q^{d \times m-d}), \mathbf{x} \leftarrow \mathcal{A}(\mathbf{A})}[\mathbf{x} \in \mathcal{L}_q^\perp([\mathbf{I}_d | \mathbf{A}]) \wedge 0 < \|\mathbf{x}\|_2 \leq B]$. We then call $\varepsilon_{\text{M-SIS}} = \sup_{\mathcal{A}} \text{PPT Adv}_{\text{M-SIS}}[\mathcal{A}]$ the hardness bound of M-SIS $_{n,d,m,q,B}$, where the parameters are clear from the context.

2.5 Lattice Gaussian Samplers

Our group signature resorts to two Gaussian samplers over lattices for the generation of user keys and for user to generate group signatures. As described in Section 1, each group member is given a trapdoor $\mathbf{S}_i$ for a matrix of the form $\mathbf{A}_{t_i} = [\mathbf{I}_d | \mathbf{A} | q_L t_i \mathbf{G}_H - \mathbf{B} | \mathbf{a}_3]$. The matrix $\mathbf{A} \in \mathcal{R}_q^{d \times d}$ and vector $\mathbf{a}_3 \in \mathcal{R}_q^d$ are uniform, $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 \bmod q\mathcal{R}$ is seemingly uniform for the users, and $\mathbf{G}_H = [1 | b_H | \dots | b_H^{k_H-1}] \otimes \mathbf{I}_d$ is the gadget modulo $q_H = b_H^{k_H}$ (where $q = q_L q_H$). The overall dimension is then $m = 2d + dk_H + 1$. The trapdoor $\mathbf{S}_i$ is generated from a matrix $\mathbf{U}_i$ of m linearly independent preimages of $\mathbf{0}$ using **ToBasis**. Said preimages of $\mathbf{0}$ can be efficiently sampled by the group manager knowing $(\mathbf{R}_1, \mathbf{R}_2)$ using the Gaussian sampler **TSampler** of [JS26]. The latter produces secure preimages $\mathbf{x} = [\mathbf{x}_{1,1}^T | \mathbf{x}_{1,2}^T | \mathbf{x}_2^T | x_3]$ of arbitrary syndromes $\mathbf{u}$, which are statistically close to an elliptical Gaussian with covariance matrix $\mathbf{S} = \text{diag}(s_{1,1}^2 \mathbf{I}_{nd}, s_3^2 \mathbf{I}_{nd}, s_4^2 \mathbf{I}_{ndk_H}, s_4^2 \mathbf{I}_n)$ with

$$s_{1,1}^2 = \frac{s_L^4}{s_L^2 - \eta^2} \cdot \frac{b_L^{2k_L} - 1}{b_L^2 - 1} + 3 \frac{s_H^4}{s_H^2 - \eta^2} \|\mathbf{R}_1\|_2^2, \quad s_3^2 = 3 \frac{s_H^4}{s_H^2 - \eta^2} \|\mathbf{R}_2\|_2^2, \quad s_4^2 = 3 \frac{s_H^4}{s_H^2 - \eta^2}$$

with $q_L = b_L^{k_L}$, $s_x = b_x \eta_{\varepsilon/dk_x}(\mathbb{Z}^n)$ (for $x \in \{L, H\}$), and $\eta = \eta_{\varepsilon}(\mathbb{Z}^{nd(k_L+k_H)})$. In particular, these preimages do not leak information on $(\mathbf{R}_1, \mathbf{R}_2)$, which prevents user i (that is given a trapdoor $\mathbf{S}_i$ for $\mathbf{A}_{t_i}$) to derive a short basis $\mathbf{S}_j^*$ for $\mathbf{A}_{t_j}$ and frame user j .

As $\mathbf{S}_i$ is a short basis of $\mathcal{L}_q^\perp(\mathbf{A}_{t_i})$, user i can use the Gaussian sampler of Klein [Kle00, GPV08] to produce preimages $\mathbf{v}$ of arbitrary syndromes $\mathbf{u}$ for $\mathbf{A}_{t_i}$. Since Klein's sampler produces a Gaussian over a lattice, we slightly adapt it to be over a lattice coset. We define an algorithm **KleinSamplePre** that takes a short basis $\mathbf{S}$ of a q -ary lattice $\mathcal{L}_q^\perp(\mathbf{A}')$ associated to the matrix $\mathbf{A}'$, a syndrome $\mathbf{u}$,

and a Gaussian parameter s , and outputs a short Gaussian $\mathbf{v}$ such that $\mathbf{A}'\mathbf{v} = \mathbf{u} \bmod q\mathcal{R}$. To do so from the original Klein sampler, one simply has to compute an arbitrary solution $\mathbf{v}_0$ such that $\mathbf{A}'\mathbf{v}_0 = \mathbf{u} \bmod q\mathcal{R}$ (not necessarily short), sample $\mathbf{v}_1$ from $\mathcal{D}_{\mathcal{L}_q^\perp(\mathbf{A}'), s, -\mathbf{v}_0}$ using Klein (which itself uses $\mathbf{S}$, s and the center $-\mathbf{v}_0$), and output $\mathbf{v} = \mathbf{v}_0 + \mathbf{v}_1$. When the matrix $\mathbf{A}'$ is in Hermite Normal Form $[\mathbf{I}_d | \mathbf{A}'']$, one can directly choose $\mathbf{v}_0 = [\mathbf{u}^T | \mathbf{0}]^T$. We later discuss in Section 4.1 how $\mathbf{u}$ is chosen precisely, but for now we need the guarantee that the generated preimages $\mathbf{v}$ do not leak $\mathbf{S}_i$. We thus use the following lemma which essentially combines (adapted versions of) the analysis of Prest [Pre17, Lem. 6] with the regularity lemma of [GPV08, Cor. 2.8]. This result is well-known but we were not able to find the full loss written down in the literature. We therefore provide the proof for completeness. We present it over the ring $\mathcal{R}$ directly but the analysis actually embeds everything into $\mathbb{Z}$.

Lemma 2.6 ([Pre17, Lem. 6] & [GPV08, Cor. 2.8] adapted). *Let $\mathcal{R}$ be a ring of integers of degree n , d, q, m be in $\mathbb{N}^\times$, and $\varepsilon \in (0, 1)$. Let $\mathbf{A}' \in \mathcal{R}_q^{d \times m}$ be primitive (i.e., $\mathbf{A}'\mathcal{R}_q^m = \mathcal{R}_q^d$) and $\mathbf{S} \in \mathbb{R}^{nm \times nm}$ be a basis of $\mathcal{L}_q^\perp(\mathbf{A}')$ (when embedded with τ). We then let $s \geq \|\text{GSO}(\mathbf{S})\|_\infty \eta_\varepsilon(\mathbb{Z}^{nm})$. We define the following distributions.*

$\mathcal{P}_0$ Sample $\mathbf{u} \leftarrow U(\mathcal{R}_q^d)$ and $\mathbf{v} \leftarrow \text{KleinSamplePre}(\mathbf{S}, \mathbf{u}, s)$. Output $(\mathbf{v}, \mathbf{u})$.

$\mathcal{P}_1$ Sample $\mathbf{v} \leftarrow \mathcal{D}_{\mathcal{R}^m, s}$ and $\mathbf{u} \leftarrow \mathbf{A}'\mathbf{v} \bmod q\mathcal{R}$. Output $(\mathbf{v}, \mathbf{u})$.

It holds that for all $(\mathbf{v}, \mathbf{u}) \in \text{Supp}(\mathcal{P}_0) = \text{Supp}(\mathcal{P}_1) = \mathcal{R}^m \times \mathcal{R}_q^d$, we have $\mathcal{P}_0(\mathbf{v}, \mathbf{u}) \in [r^{-1}, r]\mathcal{P}_1(\mathbf{v}, \mathbf{u})$ with $r = \frac{1+\varepsilon}{1-\varepsilon} \left(\frac{1+\varepsilon/nm}{1-\varepsilon/nm}\right)^{nm}$. In particular, $\mathcal{P}_0 \in [1 - \delta_v, 1 + \delta_v]\mathcal{P}_1$ with $\delta_v = r - 1 \sim_{\varepsilon \rightarrow 0+} 4\varepsilon$.

Proof. We first analyze the distribution of $\mathbf{v}$ in $\mathcal{P}_0$ conditioned on a specific $\mathbf{u}$. Recall that $\text{KleinSamplePre}(\mathbf{S}, \mathbf{u}, s)$ first computes an arbitrary solution $\mathbf{v}_0$ satisfying the equation $\mathbf{A}'\mathbf{v}_0 = \mathbf{u} \bmod q\mathcal{R}$, and then calls Klein's sampler [Kle00, GPV08] to sample $\mathbf{v}_1 = \text{KleinSampler}(\mathbf{S}, s, -\mathbf{v}_0)$. As $s \geq \|\text{GSO}(\mathbf{S})\|_\infty \eta_\varepsilon(\mathbb{Z}^{nm})$, the proof of [Pre17, Lem. 6] shows that the distribution $\mathcal{D}_{v_1}$ of $\mathbf{v}_1$ satisfies

$$\mathcal{D}_{v_1} \in \underbrace{\left[\left(\frac{1 - \varepsilon/nm}{1 + \varepsilon/nm} \right)^{nm}, \left(\frac{1 + \varepsilon/nm}{1 - \varepsilon/nm} \right)^{nm} \right]}_{[r_0^{-1}, r_0]} \mathcal{D}_{\mathcal{L}_q^\perp(\mathbf{A}'), s, -\mathbf{v}_0}$$

Hence, the distribution of $\mathbf{v} = \mathbf{v}_0 + \mathbf{v}_1$ (conditioned on $\mathbf{u}$) denoted $\mathcal{P}_{0|\mathbf{u}}$ satisfies

$$\mathcal{P}_{0|\mathbf{u}} \in [r_0^{-1}, r_0] (\mathbf{v}_0 + \mathcal{D}_{\mathcal{L}_q^\perp(\mathbf{A}'), s, -\mathbf{v}_0}) = [r_0^{-1}, r_0] \mathcal{D}_{\mathcal{L}_q^\perp(\mathbf{A}'), s}$$

where the last equality stems from the fact that $\mathbf{x} + \mathcal{D}_{\mathcal{L}, s, -\mathbf{x}} = \mathcal{D}_{\mathcal{L}+\mathbf{x}, s}$ and $\mathcal{L}_q^\perp(\mathbf{A}') + \mathbf{v}_0 = \mathcal{L}_q^\perp(\mathbf{A}')$. Now looking at the joint distribution, we then have that for all $(\mathbf{v}', \mathbf{u}') \in \mathcal{R}^m \times \mathcal{R}_q^d$,

$$\mathbb{P}_{(\mathbf{v}, \mathbf{u}) \sim \mathcal{P}_0}[(\mathbf{v}, \mathbf{u}) = (\mathbf{v}', \mathbf{u}')] = \mathbb{P}_{\mathbf{u} \sim U(\mathcal{R}_q^d)}[\mathbf{u} = \mathbf{u}'] \mathbb{P}_{\mathbf{v} \sim \text{KleinSamplePre}(\mathbf{S}, \mathbf{u}', s)}[\mathbf{v} = \mathbf{v}']$$

$$\in q^{-nd}[r_0^{-1}, r_0] \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}'}(\mathbf{A}'), s}(\mathbf{v}').$$

Let us now look at distribution $\mathcal{P}_1$. We have

$$\begin{aligned} & \mathbb{P}_{(\mathbf{v}, \mathbf{u}) \sim \mathcal{P}_1}[(\mathbf{v}, \mathbf{u}) = (\mathbf{v}', \mathbf{u}')] \\ &= \mathbb{P}_{\mathbf{v} \sim \mathcal{D}_{\mathcal{R}^m, s}}[\mathbf{A}'\mathbf{v} = \mathbf{u}' \bmod q\mathcal{R}] \mathbb{P}_{\mathbf{v} \sim \mathcal{D}_{\mathcal{R}^m, s}}[\mathbf{v} = \mathbf{v}' \mid \mathbf{A}'\mathbf{v} = \mathbf{u}' \bmod q\mathcal{R}] \\ &= \mathbb{P}_{\mathbf{v} \sim \mathcal{D}_{\mathcal{R}^m, s}}[\mathbf{A}'\mathbf{v} = \mathbf{u}'] \mathbb{P}_{\mathbf{v} \sim \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}'}(\mathbf{A}'), s}}[\mathbf{v} = \mathbf{v}'] \\ &= \mathbb{P}_{\mathbf{v} \sim \mathcal{D}_{\mathcal{R}^m, s}}[\mathbf{A}'\mathbf{v} = \mathbf{u}'] \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}'}(\mathbf{A}'), s}(\mathbf{v}'), \end{aligned}$$

where the second equality holds by the second part of [GPV08, Lem. 5.2]. For the first factor, we first observe that $s \geq \eta_\varepsilon(\mathcal{L}_q^\perp(\mathbf{A}'))$ by [GPV08, Lem. 3.1]. Then, a finer look at the proof of the first item of [GPV08, Lem. 5.2], which relies on [GPV08, Cor. 2.8], we have that $(\mathbf{A}'\mathcal{D}_{\mathcal{R}^m, s} \bmod q\mathcal{R}) \in [r_1^{-1}, r_1]q^{-nd}$ with $r_1 = (1 + \varepsilon)/(1 - \varepsilon)$. It then means $\mathbb{P}_{\mathbf{v} \sim \mathcal{D}_{\mathcal{R}^m, s}}[\mathbf{A}'\mathbf{v} = \mathbf{u}'] \in [r_1^{-1}, r_1]q^{-nd}$ and therefore that $\mathcal{P}_1(\mathbf{v}', \mathbf{u}') \in [r_1^{-1}, r_1]q^{-nd} \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}'}(\mathbf{A}'), s}(\mathbf{v}')$.

Combining both then yields $\mathcal{P}_0 \in [(r_0 r_1)^{-1}, r_0 r_1] \mathcal{P}_1$ as claimed. Because $1 - (r_0 r_1 - 1) \leq (r_0 r_1)^{-1}$, it holds that $\mathcal{P}_0 \in [1 - \delta_v, 1 + \delta_v] \mathcal{P}_1$ for $\delta_v = r_0 r_1 - 1$. $\square$

2.6 Group Signature Definitions

Following the definition by Bellare, Micciancio, and Warinschi [BMW03], a static group signature scheme involves five polynomial-time algorithms $\mathcal{GS} = (\text{GS.Setup}, \text{GS.KeyGen}, \text{GS.Sign}, \text{GS.Verify}, \text{GS.Open})$ defined as follows.

- **GS.Setup**($1^\lambda, 1^N$): takes as inputs the security parameter λ and the group size N in unary encoding. It outputs the public parameters pp which are used by the other algorithms. In particular, pp contains a description of the set of identities $\mathcal{I}$ (of size N).
- **GS.KeyGen**(pp): takes as inputs the public parameters pp . It then outputs $(\text{gpk}, \text{gmsk}, \text{gsk})$ where gpk is the group public key, gmsk the group manager secret key, and gsk is the list of the N secret signing keys, with $\text{gsk}[i]$ being the secret key of group member i .
- **GS.Sign**($\text{gsk}[i], \text{M}, \text{gpk}, \text{pp}$): takes as inputs the secret signing key of group member i , a message M , the group public key gpk , and the public parameters pp . It outputs a group signature gsig .
- **GS.Verify**($\text{gpk}, \text{M}, \text{gsig}, \text{pp}$): takes as inputs the group public key gpk , a message M , a group signature gsig , and the public parameters pp . It outputs either 1 if the signature is valid, or 0 otherwise.
- **GS.Open**($\text{gmsk}, \text{M}, \text{gsig}, \text{gpk}, \text{pp}$): takes as inputs the group manager secret key gmsk , a message M , a group signature gsig , the group public key gpk , and the public parameters pp . It outputs either an identity $\text{id} \in \mathcal{I}$ or a failure symbol $\perp$.

A static group signature must achieve three properties: correctness, full-anonymity, and full-traceability, which are defined below. We start with the correctness property which requires that an honestly generated group signature passes verification and opens to the user that produced it.

Definition 2.3. A static group signature $\mathcal{GS} = (\text{GS.Setup}, \text{GS.KeyGen}, \text{GS.Sign}, \text{GS.Verify}, \text{GS.Open})$ is correct if for any security parameter λ , group size N , any $\text{pp} \leftarrow \text{GS.Setup}(1^\lambda, 1^N)$, any $(\text{gpk}, \text{gmsk}, \text{gsk}) \leftarrow \text{GS.KeyGen}(\text{pp})$, any message M , and any identity $\text{id}_i \in \mathcal{I}$ indexed by $i \in [N]$, if $\text{gsig} \leftarrow \text{GS.Sign}(\text{gsk}[i], M, \text{gpk}, \text{pp})$ then

$$\text{GS.Verify}(\text{gpk}, M, \text{gsig}, \text{gpk}, \text{pp}) = 1 \text{ and } \text{GS.Open}(\text{gmsk}, M, \text{gsig}, \text{gpk}, \text{pp}) = \text{id}_i.$$

The full-anonymity property ensures that a group signature gsig^* remains anonymous for anyone but the group manager. The original BMW model requires that this property holds even when the adversary has access to an oracle $\mathcal{O}_{\text{Open}}$ that can lift anonymity of any other group signature. The similarities with the decryption oracle of IND-CCA encryption schemes have sometimes led to refer to this property as CCA-anonymity, in contrast to a weaker CPA-anonymity notion introduced by Boneh, Boyen, and Shacham [BBS04], where such oracle queries are forbidden.

In this paper, we merely use CPA-anonymity for sake of clarity and comparison with prior works, as an intermediate step towards our CCA-anonymous group signature. In particular, since the overhead of the latter is very small in our case, we see no reason for considering the CPA-anonymous variant for real-world applications.

Definition 2.4. A static group signature $\mathcal{GS} = (\text{GS.Setup}, \text{GS.KeyGen}, \text{GS.Sign}, \text{GS.Verify}, \text{GS.Open})$ is said ε_{an} -fully-anonymous if for a group size N polynomial in the security parameter λ , and all PPT adversaries $\mathcal{A}$, the advantage of $\mathcal{A}$ in the game of Figure 2.1 defined by $\text{Adv}_{an}[\mathcal{A}] = |\mathbb{P}[\text{Game}_{\mathcal{A}}^{an}(1^\lambda, 1^N) = 1] - 1/2|$ is upper-bounded by ε_{an} .

```

Game $_{\mathcal{A}}^{an}(1^\lambda, 1^N)$ :
1  $\text{pp} \leftarrow \text{GS.Setup}(1^\lambda, 1^N)$ 
2  $(\text{gpk}, \text{gmsk}, \text{gsk}) \leftarrow \text{GS.KeyGen}(\text{pp})$ 
3  $(\text{st}, i_0, i_1, M) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Open}}}(\text{choose}, \text{pp}, \text{gpk}, \text{gsk})$ 
4  $b \leftarrow U(\{0, 1\})$ 
5  $\text{gsig}^* \leftarrow \text{GS.Sign}(\text{gsk}[i_b], M, \text{gpk}, \text{pp})$ 
6  $b' \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Open}}}(\text{guess}, \text{st}, \text{pp}, \text{gpk}, \text{gsk}, \text{gsig}^*)$ 
7 if  $\llbracket i_0 = i_1 \rrbracket$  or  $(M, \text{gsig}^*)$  was queried to  $\mathcal{O}_{\text{Open}}$ , return 0
8 return  $\llbracket b = b' \rrbracket$ 

```

Fig. 2.1. CCA-anonymity game for static group signatures. The variable st refers to the state an adversary may want to keep for the second stage of its attacks.

Finally, the full-traceability property ensures that no colluding set $\mathcal{C}$ of group members can produce a signature that verifies but doesn't open (traceability) or opens to an identity outside the coalition (non-frameability), even knowing the group manager secret key. The adversary is allowed to adaptatively corrupt

group members and query group signatures from any group member. This is modelled by two oracles: (1) $\mathcal{O}_{\text{Cor}}$ is queried with an index $i \in [N]$ corresponding to an identity $\text{id} \in \mathcal{I}$ and returns $\text{gsk}[i]$ while updating $\mathcal{C} = \mathcal{C} \cup \{\text{id}\}$. (2) $\mathcal{O}_{\text{Sign}}$ is queried with an index $i \in [N]$ (corresponding to an identity $\text{id} \in \mathcal{I}$) and a message M , and outputs $\text{gsig} = \text{GS.Sign}(\text{gsk}[i], M, \text{gpk}, \text{pp})$, while storing $(\text{id}, M, \text{gsig})$ in a set $\mathcal{S}$. Following [BCC⁺16, Fig. 2], we slightly strengthen the BMW model by allowing the adversary to win even if it reuses a pair (i, M) queried to $\mathcal{O}_{\text{Sign}}$, provided that the returned group signature gsig^* differs from the oracle output (similarly to the strong unforgeability notion for a standard signature scheme).

Definition 2.5. A static group signature $\mathcal{GS} = (\text{GS.Setup}, \text{GS.KeyGen}, \text{GS.Sign}, \text{GS.Verify}, \text{GS.Open})$ is said ε_{tr} -fully-traceable if for a group size N polynomial in the security parameter λ , and all PPT adversaries $\mathcal{A}$, the advantage of $\mathcal{A}$ in the game of Figure 2.2 defined by $\text{Adv}_{\text{tr}}[\mathcal{A}] = \mathbb{P}[\text{Game}_{\mathcal{A}}^{\text{tr}}(1^\lambda, 1^N) = 1]$ is upper-bounded by ε_{tr} .

```

Game $\mathcal{A}$ tr( $1^\lambda, 1^N$ ):
1  $\mathcal{C}, \mathcal{S} \leftarrow \emptyset$ 
2  $\text{pp} \leftarrow \text{GS.Setup}(1^\lambda, 1^N)$ 
3  $(\text{gpk}, \text{gmsk}, \text{gsk}) \leftarrow \text{GS.KeyGen}(\text{pp})$ 
4  $(M, \text{gsig}^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Cor}}, \mathcal{O}_{\text{Sign}}}(\text{pp}, \text{gpk}, \text{gmsk})$ 
5  $\text{id} \leftarrow \text{GS.Open}(\text{gmsk}, M, \text{gsig}^*, \text{gpk}, \text{pp})$ 
7 return  $\llbracket \text{GS.Verify}(\text{gpk}, M, \text{gsig}^*, \text{pp}) = 1 \rrbracket \wedge \llbracket \text{id} \notin \mathcal{C} \rrbracket \wedge \llbracket (\text{id}, M, \text{gsig}^*) \notin \mathcal{S} \rrbracket$ 

```

Fig. 2.2. Traceability game for static group signatures

3 Module-LWE with Hints over Multiple Secrets

To simplify the security analysis of our group signature while improving its efficiency, we leverage a variant of the Hint Module Learning With Errors problem introduced by Kim et al. [KLSS23]. Concretely, in addition to giving the regular M-LWE instance $(\mathbf{A}, \mathbf{b})$ with $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$ or $\mathbf{b}$ uniform, the problem gives out Q_{hint} hints of the form $\mathbf{z}_i = \mathbf{y}_i + \gamma_i[\mathbf{s}^T \mathbf{e}^T]^T$ for some public (possibly adversarially controlled) γ_i , and secret masks $\mathbf{y}_i$. The authors show that distinguishing $(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e}, (\mathbf{z}_i)_i)$ from $(\mathbf{A}, \mathbf{b}, (\mathbf{z}_i)_i)$ with $\mathbf{b}$ uniform reduces from the standard version of M-LWE (i.e., without hints) when $\mathbf{s}, \mathbf{e}, \mathbf{y}_i$ are discrete Gaussians with carefully chosen parameters. Before that, the presence of such hints was generally handled using statistical noise flooding (requiring superpolynomial parameters), Rényi divergence flooding (for search problems), or rejection sampling. The Hint-M-LWE assumption leads to smaller parameters while efficiently reducing from a well-known problem.

This assumption has since then been used and generalized in different contexts, allowing for more efficient cryptographic schemes. In particular, in [ENP24],

the authors generalized the assumption so as to handle hints of the form $\mathbf{z}_i = \mathbf{y}_i + \mathbf{M}_i[\mathbf{s}^T | \mathbf{e}^T]^T$ for square matrices $\mathbf{M}_i$. It naturally encompasses [KLSS23] by taking $\mathbf{M}_i = \gamma_i \mathbf{I}$. We note that the works of [KLSS23, ENP24] consider Q_{hint} of this form for the same secret-error vector. In our group signature construction, we depart from it by considering a variant with multiple secrets but a single hint per secret. In particular, each hint uses the same matrix $\mathbf{M}$. Additionally, we generalize [ENP24] to handle matrices of arbitrary dimensions as our matrix is later a single row vector. Despite our specific use, we present our assumption and its reduction from M-LWE in the most general way, i.e., with arbitrary $\mathbf{M}$, and Q_{hint} hints per independent secret. For conciseness and clarity, for $\mathbf{X} = [\mathbf{x}_1 | \dots | \mathbf{x}_k] \in \mathcal{R}^{\alpha \times k}$, we define $\tilde{\mathbf{X}} = [\tau(\mathbf{x}_1) | \dots | \tau(\mathbf{x}_k)] \in \mathbb{Z}^{\alpha n \times k}$. Note that when $Q_{\text{hint}} = 0$, this corresponds to the multiple secret version of regular M-LWE.

Definition 3.1 (Matrix-Hint-M-LWE). Let $\mathcal{R}$ be the ring of integers of a number field of degree n . Let m, d, q, k be in $\mathbb{N}^\times$ and $Q_{\text{hint}} \in \mathbb{N}$. Let $\mathcal{D}_s$ be a distribution over $\mathcal{R}^{d+m}$, and for $i \in [Q_{\text{hint}}]$, let $\mathcal{M}_i$ be a distribution over $\mathbb{Z}^{h_i \times n(m+d)}$ for some $h_i \in \mathbb{N}^\times$, and $\mathcal{D}_{y,i}$ a distribution over $\mathbb{Z}^{h_i}$. The Matrix-Hint Module Learning With Errors problem, denoted $\text{MH-M-LWE}_{n,d,m,q,\mathcal{D}_s,(\mathcal{D}_{y,i})_i,(\mathcal{M}_i)_i,Q_{\text{hint}}}^k$, asks to distinguish the following distributions.

$\mathcal{P}_0$	Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{X} \leftarrow \mathcal{D}_s^k$, and for $i \in [Q_{\text{hint}}]$, $\mathbf{M}_i \leftarrow \mathcal{M}_i$, $\mathbf{Y}_i \leftarrow \mathcal{D}_{y,i}^k$. Sample $\mathbf{B} \leftarrow U(\mathcal{R}_q^{m \times k})$, and compute $\mathbf{Z}_i = \mathbf{Y}_i + \mathbf{M}_i \tilde{\mathbf{X}}$ for all $i \in [Q_{\text{hint}}]$. Output $(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i)$.
$\mathcal{P}_1$	Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{X} \leftarrow \mathcal{D}_s^k$, and for $i \in [Q_{\text{hint}}]$, $\mathbf{M}_i \leftarrow \mathcal{M}_i$, $\mathbf{Y}_i \leftarrow \mathcal{D}_{y,i}^k$. Compute $\mathbf{B} = [\mathbf{A} \mathbf{I}_m] \mathbf{X} \bmod q\mathcal{R}$, and $\mathbf{Z}_i = \mathbf{Y}_i + \mathbf{M}_i \tilde{\mathbf{X}}$ for all $i \in [Q_{\text{hint}}]$. Output $(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i)$.

We define $\text{Adv}_{\text{MH-M-LWE}}[\mathcal{A}] = |\mathbb{P}_{(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i) \sim \mathcal{P}_0}[\mathcal{A}(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i) = 1] - \mathbb{P}_{(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i) \sim \mathcal{P}_1}[\mathcal{A}(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i) = 1]|$ as the advantage of a PPT adversary in solving MH-M-LWE. We call $\varepsilon_{\text{MH-M-LWE}} = \sup_{\mathcal{A}} \text{PPT Adv}_{\text{MH-M-LWE}}[\mathcal{A}]$ the hardness bound of MH-M-LWE, the parameters being clear from the context. The search version corresponds to finding $\mathbf{X}$ given an instance drawn from $\mathcal{P}_1$.

We now show an efficient reduction from M-LWE to MH-M-LWE^k when $\mathcal{D}_s$ and $\mathcal{D}_{y,i}$ are discrete Gaussian distributions and when the matrices $\mathbf{M}_i$ are appropriately bounded. The reduction follows the blueprint of [KLSS23, ENP24], but is generalized to tackle non-square matrix hints and multiple secrets. The reduction first proceeds by a hybrid argument to reduce to a single secret instance, and then follows [KLSS23, ENP24] generalized to non-square matrix hints. The proof is given in Appendix A for completeness.

Theorem 3.1. Let $\mathcal{R}$ be the ring of integers of a number field of degree n . Let $m, d, q, k, Q_{\text{hint}}$ be in $\mathbb{N}^\times$ and polynomial in the security parameter. For $i \in [Q_{\text{hint}}]$, let $\mathcal{M}_i$ be a distribution over $\mathbb{Z}^{h_i \times n(m+d)}$ for some $h_i \in \mathbb{N}^\times$ such that $\sup_{\mathbf{M} \in \text{Supp}(\mathcal{M}_i)} \|\mathbf{M}\|_2 \leq \beta_i$. Then, let $\mathbf{S}$ be in $\mathbb{R}^{n(d+m) \times n(d+m)}$ and $(\mathbf{S}_i)_{i \in [Q_{\text{hint}}]}$ in

$\times_{i \in [Q_{\text{hint}}]} \mathbb{R}^{h_i \times h_i}$ be positive definite matrices, such that

$$s' := \frac{1}{\sqrt{2}} \left(\|\mathbf{S}^{-1}\|_2 + \sum_{i \in [Q_{\text{hint}}]} \beta_i^2 \|\mathbf{S}_i^{-1}\|_2 \right)^{-1/2}$$

verifies $s' \geq \sqrt{2}\eta_\varepsilon(\mathcal{R}^{d+m})$ for some $\varepsilon \in (0, 1)$. There is a PPT reduction from $\text{M-LWE}_{n,d,m,q,\mathcal{D}_{\mathcal{R}^{d+m},s'}}$ to $\text{MH-M-LWE}_{n,d,m,q,\mathcal{D}_{\mathcal{R}^{d+m},\sqrt{\mathbf{S}}},(\mathcal{D}_{\mathbb{Z}^{h_i},\sqrt{\mathbf{S}_i^{-1}}})_i,(\mathcal{M}_i)_i,Q_{\text{hint}}}$ that shows

$$\varepsilon_{\text{MH-M-LWE}^k} \leq k \left(\varepsilon_{\text{M-LWE}} + \frac{\varepsilon}{1 - \varepsilon} \right),$$

where $\varepsilon_{\text{M-LWE}}, \varepsilon_{\text{MH-M-LWE}^k}$ are the respective hardness bounds. The same holds for the search versions, without the factor k .

We note that as in [ENP24], our hints are over the integers but can be used to define hints over the ring by choosing distributions $\mathcal{M}_i$ over multiplication matrices (e.g., negacirculant matrices when $\mathcal{R}$ is a power-of-two cyclotomic ring). In our construction, we use hints of the form $\mathbf{Z}_i = \mathbf{Y}_i + \mathbf{M}_i \mathbf{X}$ over the ring $\mathcal{R}$. We then may index the problem with distributions over $\mathcal{R}$ instead of $\mathbb{Z}$. In our case, we use this new assumption with $Q_{\text{hint}} = 1$, $m = d$, $h_1 = 1$ (or n when embedded into $\mathbb{Z}$), and where $\mathcal{M}_1$ is the distribution $\mathcal{W}_{d,w_{\text{op}}/2}^2$ as a row vector, truncated in spectral norm at β_{op} . More precisely, taking $\mathbf{x}_{\text{op}}^T = [\mathbf{s}_{\text{op}}^T | \mathbf{e}_{\text{op}}^T] \in \mathcal{R}^{1 \times 2d}$, we have

$$\|M_\tau(\mathbf{x}_{\text{op}}^T)\|_2^2 = \lambda_{\max}(M_\tau(\mathbf{x}_{\text{op}}^T)M_\tau(\mathbf{x}_{\text{op}}^T)^T) = \|M_\tau(\mathbf{x}_{\text{op}}^* \mathbf{x}_{\text{op}})\|_2 = \|\sigma(\mathbf{x}_{\text{op}}^* \mathbf{x}_{\text{op}})\|_\infty,$$

with σ the canonical embedding of $\mathcal{K}$, and where the second equality uses the ring homomorphism properties of M_τ , and the last equality holds by [BJRW23, Lem. 2.3]. Thence, we can then easily verify and enforce a bound $\|M_\tau(\mathbf{x}_{\text{op}}^T)\|_2 \leq \beta_{\text{op}}$ when generating the opening key.

In our scheme, we also choose a spherical Gaussian for $\mathcal{D}_{y,1}$, i.e., $\mathbf{S}_1 = s_{e1}^2 \mathbf{I}_n$, and a nearly spherical one for $\mathcal{D}_s$, i.e., $\mathbf{S} = \text{diag}(s_{se}^2 \mathbf{I}_{2nd}, s_{e1}^2 \mathbf{I}_n)$. Roughly speaking, we will have $s_{e1} \approx s_{se} \beta_{\text{op}}$, which means $\|\mathbf{S}^{-1}\|_2 = \max(1/s_{se}^2, 1/s_{e1}^2) = 1/s_{se}^2$. We then have $s' = (s_{se}^{-2} + \beta_{\text{op}}^2 s_{e1}^{-2})^{-1/2} / \sqrt{2}$. We then need to set s_{se} and s_{e1} so that $s' \geq \sqrt{2}\eta_\varepsilon(\mathbb{Z}^{2nd})$. For a free parameter $\alpha > \sqrt{2}$, we can set $s_{se} = \alpha \sqrt{2}\eta_\varepsilon(\mathbb{Z}^{2nd})$ and $s_{e1} = \alpha \sqrt{2} / \sqrt{\alpha^2 - 2} \cdot \sqrt{2}\eta_\varepsilon(\mathbb{Z}^{2nd}) \cdot \beta_{\text{op}}$.

4 Group Signature

4.1 High-Level Description

Informally, group members must carry out two tasks when producing signatures: (1) to prove that they are indeed members of a group managed by the group manager and (2) to provide a way for the group manager to recover their identity. Obviously, this must be done while ensuring anonymity, which rules out most basic solutions addressing (1) and (2).

As explained in Section 1.1.1, regarding (1), our strategy departs from previous constructions where a group member i would receive a unique certificate $(t_i, \mathbf{v}_i)$ such that $\mathbf{A}_{t_i} \mathbf{v}_i = \mathbf{u}$ (for some public $\mathbf{u}$ and where $\mathbf{A}_{t_i}$ is derived from the group manager's public key), and prove knowledge of the latter in a zero-knowledge way. In our case, user i is given a whole short basis $\mathbf{S}_i$ of $\mathcal{L}_q^\perp(\mathbf{A}_{t_i})$ which is generated using the recent sampler **TSampler** of [JS26] and turned into a basis using the **ToBasis** algorithm from Lemma 2.1. In more details, $\mathbf{A}_{t_i}$ is defined as $[\mathbf{I}_d | \mathbf{A} | q_L t_i \mathbf{G}_H - \mathbf{B} | \mathbf{a}_3] \bmod q\mathcal{R}$, where $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{a}_3$ are parts of the group manager's public key, $\mathbf{G}_H$ is a gadget matrix (see its definition in **GS.Setup**), $q_L | q$ and t_i is some a short element of $\mathcal{R}$ defining the identity of user i (and then differs for each group member). For each t_i , the group manager can generate sufficiently many short preimages of $\mathbf{0}$ for $\mathbf{A}_{t_i}$ to have a full-rank set $\mathbf{U}_i$ of lattice points, which is enough to obtain a basis $\mathbf{S}_i$ of the same quality with **ToBasis**. Obviously, this approach increases complexity compared to a mere certificate generation but this later allows for significantly decreasing the complexity of the group signature itself, as we shall see.

Once group member i has received $\mathbf{S}_i$, it can use it to sample short Gaussian preimages $\mathbf{v}$ of any syndrome $\mathbf{u}$ using **KleinSamplePre**. Note that i and the group manager are the only ones⁵ that have the ability to compute such preimages $\mathbf{v}$ for $\mathbf{A}_{t_i}$. This ability then acts as a proof of membership for user i , thereby addressing point (1) mentioned above. Moreover, the user generates a fresh vector $\mathbf{v}$ for each group signature, which means $\mathbf{v}$ does not need to be hidden in a NIZK proof and can directly be included in the signature, which is the key to the efficiency of our system. Compared to the previous approach which required hiding $(t_i, \mathbf{v}_i)$ in a NIZK, we now only have to hide t_i and prove it verifies $\mathbf{A}_{t_i} \mathbf{v} = \mathbf{u}$.

The subtlety, mentioned in the introduction, lies in the choice of the syndrome $\mathbf{u}$. Indeed, as we explained in Section 1.1.1, while publishing $\mathbf{u}$ gives an insecure scheme, letting the user choose it without constraints is just as bad. In our construction, $\mathbf{u}$ is defined as an M-LWE instance, namely $\mathbf{A}\mathbf{s} + \mathbf{e} \bmod q\mathcal{R}$, where the short $\mathbf{s}$ and $\mathbf{e}$ are chosen by the user and whose knowledge is proven in the NIZK. It means that we now hide $(t_i, \mathbf{s}, \mathbf{e})$ and prove the relation $\mathbf{A}_{t_i} \mathbf{v} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q\mathcal{R}$ with $\mathbf{s}, \mathbf{e}$ short. If we take a step back, we essentially traded in the NIZK witness a fairly large and high-dimensional Gaussian vector $\mathbf{v}_i$ for a much shorter and lower-dimensional vector $[\mathbf{s} | \mathbf{e}]$ that only needs to be distributed as in a M-LWE secret-error pair. Given the framework of [LNP22] is very sensitive to the dimension and the norm of the witnesses, this results in significant performance improvements.

It now remains to address (2), namely to provide a way for the opener to lift anonymity of any group signature. This is generally done by adding a verifiable encryption of some data related to the member's identity, which corresponds to a ciphertext (typically a version of Kyber [BDK⁺18]) along with a

⁵ In particular, user $j \neq i$ (or a coalition of users) holding a trapdoor $\mathbf{S}_j$ for $\mathbf{A}_{t_j}$ with $t_j \neq t_i$ cannot produce short preimages for $\mathbf{A}_{t_i}$. This non-frameability property is taken into account in the full-traceability of Definition 2.5, which we later show for our construction based on standard assumptions (M-SIS and M-LWE).

NIZK proof that it decrypts to the correct message. The latter NIZK can be combined with the one we already have to limit the impact of the ciphertext, but the ciphertext randomness needs to be included in the witness nonetheless, which further worsens the group signature complexity. In our case, we avoid this extra cost by leveraging an interesting feature of the matrix structure of [JS26]. Indeed, while $\mathbf{A}_{t_i} \bmod q\mathcal{R}$ depends on t_i , we note that $\mathbf{A}_{t_i} \bmod q_L\mathcal{R} = [\mathbf{I}_d|\mathbf{A}] - \mathbf{B}|\mathbf{a}_3] \bmod q_L\mathcal{R}$, which is independent of t_i . This means that anyone, given $\mathbf{v}$, can recover $\mathbf{A}\mathbf{s} + \mathbf{e} \bmod q_L\mathcal{R} = [\mathbf{I}_d|\mathbf{A}] - \mathbf{B}|\mathbf{a}_3]\mathbf{v} \bmod q_L\mathcal{R}$, which corresponds to the first part of a Dual-Regev/Kyber ciphertext. All we need to complete this ciphertext, in a CPA setting, is $\mathbf{ct}_1 = \mathbf{b}_{\text{op}}^T \mathbf{s} + e_1 + \lfloor q_L/2 \rfloor t_i \bmod q_L\mathcal{R}$, where $\mathbf{b}_{\text{op}} = \mathbf{A}^T \mathbf{s}_{\text{op}} + \mathbf{e}_{\text{op}} \bmod q_L\mathcal{R}$ is the encryption public key. Note that $\mathbf{ct}_1$ is just one ring element and makes use of $\mathbf{s}$ which is already part of the NIZK witnesses. We thus only need to include e_1 in the latter, which has a marginal impact on performance.

As hinted in Section 1.1.2, recycling the same randomness for both $\mathbf{u}$ and the ciphertext component creates some challenges in the traceability proof. The goal of our traceability proof is to use a successful adversary to produce a short $\mathbf{x}$ such that $\mathbf{A}_{t_h} \mathbf{x} = \mathbf{0}$ for t_h identifying the honest user targetted in the traceability. This can then be turned into an M-SIS solution for the matrix $[\mathbf{I}_d|\mathbf{A}|\mathbf{a}_3]$ following the proof strategy of [JS26]. To carry the latter, one essentially needs to make the gadget $\mathbf{G}_H$ vanish for the tag t_h . This is done by tweaking the public key matrix $\mathbf{B} = [\mathbf{I}_d|\mathbf{A}]\mathbf{R}$ (where $\mathbf{R}$ is the **TSampler** trapdoor known only to the group manager to generate user keys) and generating it as $\mathbf{B} = [\mathbf{I}_d|\mathbf{A}]\mathbf{R} + q_L t_h \mathbf{G}_H$ instead. As a result, from the $\mathbf{x}$ above, we have $\mathbf{0} = \mathbf{A}_{t_h} \mathbf{x} = [\mathbf{I}_d|\mathbf{A}] - [\mathbf{I}_d|\mathbf{A}]\mathbf{R}|\mathbf{a}_3]\mathbf{x}$, which rearranges to $[\mathbf{I}_d|\mathbf{A}|\mathbf{a}_3]\mathbf{x}' = \mathbf{0}$ for $\mathbf{x}' = [(\mathbf{x}_1 - \mathbf{R}\mathbf{x}_2)^T | x_3]^T$. However, setting $\mathbf{B}$ in this way prevents the group manager from generating a basis of $\mathcal{L}_q^\perp(\mathbf{A}_{t_h})$ with **TSampler**. We must then simulate all the signing queries to the user identified by t_h without resorting to a basis nor **KleinSamplePre**.

One could be tempted to use a sampler simulation result by sampling $\mathbf{v}$ without the basis and programming the syndrome $\mathbf{u} = \mathbf{A}_{t_i} \mathbf{v}$ afterwards. But because the adversary is given the opening key, we must do so while ensuring the simulated group signatures still open to t_h . As a result, $(\mathbf{u}, \mathbf{ct}_1)$ needs to satisfy a specific structure to enable opening. By carefully changing $\mathbf{ct}_1$, we show that we can simulate $\mathbf{u}$ (and in turn $\mathbf{v}$ without the basis as above) under the Matrix-Hint-M-LWE assumption introduced in Section 3. In our case, the hint is of the form

$$z = e_1 + [\mathbf{e}_{\text{op}}^T | -\mathbf{s}_{\text{op}}^T] \begin{bmatrix} \mathbf{s} \\ \mathbf{e} \end{bmatrix}$$

where e_1 acts as the mask $\mathbf{Y}_1$ from Definition 3.1, and $[\mathbf{e}_{\text{op}}^T | -\mathbf{s}_{\text{op}}^T]$ as the matrix $\mathbf{M}_1$. The reduction from regular M-LWE of Theorem 3.1 then dictates how $\mathbf{s}, \mathbf{e}, e_1$ are chosen with respect to each other.

In particular, they should be distributed as discrete Gaussians, while having e_1 sufficiently larger than $[\mathbf{e}_{\text{op}}^T | -\mathbf{s}_{\text{op}}^T][\mathbf{s}^T | \mathbf{e}^T]^T$. It prevents us from using very compact encryption parameters as in Kyber [BDK⁺18], thus requiring a larger encryption modulus q_L for correct decryption. This is unfortunately not opti-

mal, as we also desire q_L to be as small as possible to have shorter bases from the sampler of [JS26]. To separate these requirements, we resort to the following trick. We additionally provide $\text{ct}'_0 = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q'\mathcal{R}$, where q' is a (very) small integer⁶, coprime with q . Using the Chinese Remainder isomorphism, one can use $\mathbf{A}\mathbf{s} + \mathbf{e} \bmod q_L\mathcal{R} = [\mathbf{I}_d|\mathbf{A}] - \mathbf{B}[\mathbf{a}_3]\mathbf{v} \bmod q_L\mathcal{R}$, and $\text{ct}'_0 = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q'\mathcal{R}$ to obtain $\text{ct}_0 = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q_Lq'\mathcal{R}$. One can thus implicitly increase the ciphertext modulus to q_Lq' and thus ensure decryption correctness (ct_1 is now provided modulo q_Lq'). Compared to the classical approach with an independent ciphertext, this strategy is still much more efficient because ct'_0 is provided modulo an integer (q') much smaller than the one necessary for an independent ciphertext, and we do not introduce additional witnesses in the NIZK proof as ct'_0 is built using $\mathbf{s}$ and $\mathbf{e}$.

Finally, we extend our scheme to achieve CCA-anonymity by following the Naor-Yung approach [NY90]. However, as we aim to recycle the randomness $\mathbf{s}$ and $\mathbf{e}$ (and thus the whole $\text{ct}_0 = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q_Lq'\mathcal{R}$ vector) for the two ciphertexts resulting from this approach, we are closer in essence to the Naor-Yung with shared randomness paradigm [BMV16]. More precisely, group signatures now include an extra ciphertext element $\text{ct}_2 = \mathbf{b}_{\text{op},2}^T \mathbf{s} + e_2 + \lfloor q_Lq'/2 \rfloor t_i \bmod q_Lq'\mathcal{R}$, where $\mathbf{b}_{\text{op},2}$ is a second encryption key, and the noise e_2 is also included in the witness of the NIZK. The overhead going from the CPA version to the CCA one is therefore very small. Recycling the randomness $(\mathbf{s}, \mathbf{e})$ for the two ciphertexts elements ct_1 and ct_2 however creates a few pitfalls in the proof of CCA-anonymity. Fortunately, the situation is very similar to the one we described for the traceability proof above: as we need to keep at least one opening key $\mathbf{s}_{\text{op},1}$ or $\mathbf{s}_{\text{op},2}$ at all times, we cannot simulate ct_1, ct_2 simultaneously from the same M-LWE instance. But doing it in turn, e.g., simulating ct_2 first, requires tackling the shared component $(\mathbf{u}, \text{ct}'_0)$ and programming ct_1 using a hint-type assumption. Using the exact same MH-M-LWE assumption, we are able to carry the CCA-anonymity proof in this setting. We insist that, as shown by Theorem 3.1, the latter reduces from regular M-LWE for our parameters, meaning our final construction is too based on well-known lattice assumptions, i.e., M-LWE and M-SIS.

4.2 Construction

We give here the formal specifications of **GS.Setup**, **GS.KeyGen**, **GS.Sign**, **GS.Verify** and **GS.Open** of our group signature. We note it corresponds to the CCA-anonymous version of our scheme, but we highlight the parts that are specific to making it so. Without these highlighted portions, the group signature is only CPA-anonymous, which allows for a more relevant comparison with prior CPA-anonymous schemes.

⁶ In our parameter selection of Section 6.1, $q' = 4$ is sufficient to ensure perfect decryption correctness, without hindering security.

Algorithm 4.1: GS.Setup($1^\lambda, 1^N$)

1. Choose $n \geq \kappa$ to be powers of two
2. Choose $d \in \mathbb{N}^\times$
- Parameters of TSampler.**
3. Choose $b_H \in \mathbb{N}^\times$ a small prime s.t. $b_H = 2\kappa + 1 \pmod{4\kappa}$ and $b_H > \sqrt{\kappa}^\kappa$
4. Choose $k_H, b_L, k_L \in \mathbb{N}^\times$ s.t. $b_L^{k_L} \wedge b_H^{k_H} = 1$
5. $q_L \leftarrow b_L^{k_L}$
6. $q_H \leftarrow b_H^{k_H}$
7. $q \leftarrow q_L q_H$
8. Choose $w \in \mathbb{N}^\times$ s.t. $\binom{n}{w} \geq N$ and set $N \leftarrow \binom{n}{w}$
9. $\mathcal{I} \leftarrow \{t \in T_1 : \|t\|_1 = w\}$
10. $B_R \leftarrow \frac{3}{4}(\sqrt{nd} + \sqrt{ndk_H} + 6)$
11. For $x \in \{L, H\}$, define $\mathbf{G}_x = [1 \dots |b_x^{k_x} - 1|] \otimes \mathbf{I}_d \in \mathcal{R}^{d \times dk_x}$
12. Choose $\varepsilon \in (0, 1)$ ▷ typically $\varepsilon \approx 2^{-40}$
13. $s_L \leftarrow b_L \cdot \sqrt{\ln(2ndk_L/\varepsilon)/\pi}$ ▷ $s_L \approx b_L \eta_\varepsilon(\mathbb{Z}^{ndk_L}) \geq b_L \eta_{\varepsilon/dk_L}(\mathbb{Z}^n)$ by Lemma 2.3
14. $s_H \leftarrow b_H \cdot \sqrt{\ln(2ndk_H/\varepsilon)/\pi}$ ▷ $s_H \approx b_H \eta_\varepsilon(\mathbb{Z}^{ndk_H}) \geq b_H \eta_{\varepsilon/dk_H}(\mathbb{Z}^n)$ by Lemma 2.3
15. $\eta \leftarrow \sqrt{\ln(2nd(k_L + k_H)/\varepsilon)/\pi}$ ▷ $\eta \approx \eta_\varepsilon(\mathbb{Z}^{n(dk_L + k_H)})$ by Lemma 2.3
16. $s_1 \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$
17. $s_2 \leftarrow s_L^2/\sqrt{s_L^2 - \eta^2}$
18. $s_{11} \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) \cdot (b_L^{2k_L} - 1)/(b_L^2 - 1) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$
19. $s_3 \leftarrow \sqrt{3s_H^2 B_R/\sqrt{s_H^2 - \eta^2}}$
20. $s_4 \leftarrow \sqrt{3s_H^2/\sqrt{s_H^2 - \eta^2}}$
21. $B_{11} \leftarrow c_{nd} s_{11} \sqrt{nd}$ ▷ set by Lemma 2.5
22. $B_{12} \leftarrow c_{nd} s_3 \sqrt{nd}$ ▷ set by Lemma 2.5
23. $B_2 \leftarrow c_{n(dk_H+1)} s_4 \sqrt{n(dk_H+1)}$ ▷ set by Lemma 2.5
24. $\beta \leftarrow \sqrt{B_{11}^2 + B_{12}^2 + B_2^2}$
- Parameters of KleinSamplePre.**
25. $m \leftarrow 2d + dk_H + 1$
26. Choose $\varepsilon_v \in (0, 1)$ ▷ typically $\varepsilon_v \leq 2^{-\lambda}$
27. $\eta_v \leftarrow \sqrt{\ln(2nm/\varepsilon_v)/\pi}$ ▷ $\eta_v \approx \eta_{\varepsilon_v}(\mathbb{Z}^{nm})$ by Lemma 2.3
28. $s_v \leftarrow \eta_v \beta$
- Opening Parameters.**
29. Choose w_{op} ▷ Hamming weight of opening key
30. $B_{\text{op}} \leftarrow \sqrt{w_{\text{op}}}$
31. Choose β_{op} ▷ Spectral norm bound of opening key
32. Choose $\varepsilon_{\text{op}} \in (0, 1)$ ▷ typically $\varepsilon_{\text{op}} \leq 2^{-\lambda}/Q_{\text{Sign}}$
33. $\eta_{\text{op}} \leftarrow \sqrt{\ln(4nd/\varepsilon_{\text{op}})/\pi}$ ▷ $\eta_{\text{op}} \approx \eta_{\varepsilon_{\text{op}}}(\mathbb{Z}^{2nd})$ by Lemma 2.3
34. Choose $\alpha > \sqrt{2}$
35. $s_{se} \leftarrow \alpha \cdot \sqrt{2} \eta_{\text{op}}$
36. $s_{e1} \leftarrow \frac{\alpha \sqrt{2}}{\sqrt{\alpha^2 - 2}} \cdot \sqrt{2} \eta_{\text{op}} \cdot \beta_{\text{op}}$
37. $B_{se} \leftarrow c_{2nd} s_{se} \sqrt{2nd}$
38. $B_{e1} \leftarrow c_{ns_{e1}} \sqrt{n}$
39. Choose $q' \in \mathbb{N}^\times$ s.t. $q \wedge q' = 1$ and $q_L q' > 4(B_{e1} + B_{se} B_{\text{op}})$
40. **params** $\leftarrow (N, n, d, b_H, k_H, q_H, b_L, k_L, q_L, q, w, B_R, \varepsilon, s_L, s_H, \eta, s_1, s_2, s_3, s_4, s_{1,1}, B_{11}, B_{12}, B_2, \beta, m, \varepsilon_u, \eta_u, s_u, B_{\text{op}}, \beta_{\text{op}}, \varepsilon_{\text{op}}, \eta_{\text{op}}, s_{se}, s_{e1}, B_{se}, B_{e1}, q')$
41. $\mathbf{A} \leftarrow U(\mathcal{R}_{qq'}^{d \times d})$
42. $\mathbf{a}_3 \leftarrow U(\mathcal{R}_q^d)$
- Output:** $\text{pp} = (\text{params}, \mathbf{A}, \mathbf{a}_3)$

Algorithm 4.2: GS.KeyGen(pp)

1. $\mathbf{x}_{\text{op},1} = [\mathbf{s}_{\text{op},1}^T | \mathbf{e}_{\text{op},1}^T]^T \leftarrow \mathcal{W}_{d, w_{\text{op}}/2}^2$ conditioned on $\|\sigma(\mathbf{x}_{\text{op},1}^* \mathbf{x}_{\text{op},1})\|_\infty \leq \beta_{\text{op}}^2$
2. $\mathbf{b}_{\text{op},1} = \mathbf{A}^T \mathbf{s}_{\text{op},1} + \mathbf{e}_{\text{op},1} \pmod{q_L q' \mathcal{R}}$

```

3.  $\mathbf{x}_{\text{op},2} = [\mathbf{s}_{\text{op},2}^T | \mathbf{e}_{\text{op},2}^T]^T \leftarrow \mathcal{W}_{d,w_{\text{op}}/2}^2$  conditioned on  $\|\sigma(\mathbf{x}_{\text{op},2}^*, \mathbf{x}_{\text{op},2})\|_\infty \leq \beta_{\text{op}}^2$ 
4.  $\mathbf{b}_{\text{op},2} = \mathbf{A}^T \mathbf{s}_{\text{op},2} + \mathbf{e}_{\text{op},2} \bmod q_L q' \mathcal{R}$ 
5.  $\mathbf{R}_1, \mathbf{R}_2 \leftarrow \mathcal{B}_1^{d \times dk_H}$  conditioned on  $\|\mathbf{R}_i\|_2 \leq B_R$ 
6.  $\mathbf{B} \leftarrow \mathbf{R}_1 + \mathbf{A} \mathbf{R}_2 \bmod q \mathcal{R} \in \mathcal{R}_q^{d \times dk_H}$ 
7. For  $t_i \in \mathcal{I}$  ( $i = 1, \dots, N$ ) do
8.    $\mathbf{U}_i \leftarrow \emptyset$ 
9.   While  $\mathbf{U}_i$  does not have  $m$  linearly independent columns do
10.     $x_{3,i} \leftarrow \mathcal{D}_{\mathcal{R},s_4}$ 
11.     $\mathbf{x}'_i \leftarrow \text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, -x_{3,i} \mathbf{a}_3 \bmod q \mathcal{R}, t_i, s_1, s_2, s_3, s_4, s_L, s_H)$ 
12.     $\mathbf{x}_i \leftarrow [\mathbf{x}'_i | x_{3,i}]^T$ 
13.    If  $\|\mathbf{x}_i\|_2 \leq \beta$  and  $\mathbf{x}_i$  linearly independent with  $\mathbf{U}_i$ ,  $\mathbf{U}_i \leftarrow [\mathbf{U}_i | \mathbf{x}_i]$ 
14.     $\mathbf{S}_i \leftarrow \text{ToBasis}(\mathbf{U}_i, \mathcal{L}_q^\perp(\mathbf{A}_{t_i})) \quad \triangleright \mathbf{A}_{t_i} = [\mathbf{I}_d | \mathbf{A}]_{q_L t_i} \mathbf{G}_H - \mathbf{B} | \mathbf{a}_3 \bmod q \mathcal{R}$ 

```

Output: $\text{gpk} = (\mathbf{B}, \mathbf{b}_{\text{op},1}, \mathbf{b}_{\text{op},2})$, $\text{gmsk} = \mathbf{s}_{\text{op},1}$, $\text{gsk} = (t_i, \mathbf{S}_i)_{i \in [N]}$

Algorithm 4.3: GS.Sign($\text{gsk}[i], \mathbf{M}, \text{gpk}, \text{pp}$)

```

1.  $\mathbf{x} = [\mathbf{s}^T | \mathbf{e}^T]^T \leftarrow \mathcal{D}_{\mathcal{R}^{2d}, s_{se}}$  conditioned on  $\|\mathbf{x}\|_2 \leq B_{se}$ 
2.  $e_1 \leftarrow \mathcal{D}_{\mathcal{R}, s_{e1}}$  conditioned on  $\|e_1\|_2 \leq B_{e1}$ 
3.  $e_2 \leftarrow \mathcal{D}_{\mathcal{R}, s_{e1}}$  conditioned on  $\|e_2\|_2 \leq B_{e1}$ 
4.  $\text{ct}'_0 \leftarrow \mathbf{A} \mathbf{s} + \mathbf{e} \bmod q' \mathcal{R}$ 
5.  $\text{ct}_1 \leftarrow \mathbf{b}_{\text{op},1}^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t_i \bmod q_L q' \mathcal{R}$ 
6.  $\text{ct}_2 \leftarrow \mathbf{b}_{\text{op},2}^T \mathbf{s} + e_2 + \lfloor q_L q' / 2 \rfloor t_i \bmod q_L q' \mathcal{R}$ 
7.  $\mathbf{u} \leftarrow \mathbf{A} \mathbf{s} + \mathbf{e} \bmod q \mathcal{R}$ 
8.  $\mathbf{v} \leftarrow \text{KleinSamplePre}(\mathbf{S}_i, \mathbf{u}, s_v)$ 
9. Parse  $\mathbf{v}$  as  $\mathbf{v} = [\mathbf{v}_1^T | \mathbf{v}_2^T | v_3]^T$  with  $\mathbf{v}_1 \in \mathcal{R}^{2d}, \mathbf{v}_2 \in \mathcal{R}^{dk_H}$ 
10. If  $\llbracket \|\mathbf{v}_1\|_2 > B_{v1} \rrbracket \vee \llbracket \|\mathbf{v}_2\|_2 > B_{v2} \rrbracket \vee \llbracket \|\mathbf{v}_3\|_2 > B_{v3} \rrbracket \vee \llbracket v_3 = 0 \rrbracket$ , go to 8.
11.  $\pi \leftarrow \text{Prove}((t_i, \mathbf{s}, \mathbf{e}, e_1, e_2); \text{aux} = \mathbf{M})$ 

```

Output: $\text{gsig} = (\mathbf{v}, \text{ct}'_0, \text{ct}_1, \text{ct}_2, \pi)$

Algorithm 4.4: GS.Verify($\text{gpk}, \mathbf{M}, \text{gsig}, \text{pp}$)

```

1. Parse  $\mathbf{v}$  as  $\mathbf{v} = [\mathbf{v}_1^T | \mathbf{v}_2^T | v_3]^T$  with  $\mathbf{v}_1 \in \mathcal{R}^{2d}, \mathbf{v}_2 \in \mathcal{R}^{dk_H}$ 
2.  $b_1 \leftarrow \llbracket \|\mathbf{v}_1\|_2 \leq B_{v1} \rrbracket \quad \triangleright B_{vi} = c_{\dim(\mathbf{v}_i)} s_v \sqrt{\dim(\mathbf{v}_i)}$  set by Lemma 2.5
3.  $b_2 \leftarrow \llbracket \|\mathbf{v}_2\|_2 \leq B_{v2} \rrbracket$ 
4.  $b_3 \leftarrow \llbracket 0 < \|v_3\|_2 \leq B_{v3} \rrbracket$ 
5.  $b_4 \leftarrow \llbracket \text{Verify}(\pi; \text{aux} = \mathbf{M}) = 1 \rrbracket$ 

```

Output: $b_1 \wedge b_2 \wedge b_3 \wedge b_4$ (1 if valid, 0 otherwise)

Algorithm 4.5: GS.Open($\text{gmsk}, \mathbf{M}, \text{gsig}, \text{gpk}, \text{pp}$)

```

1.  $\text{ct}_{0,L} = [\mathbf{I}_d | \mathbf{A}] - \mathbf{B} | \mathbf{a}_3 \mathbf{v} \bmod q_L \mathcal{R}$ 
2.  $\text{ct}_0 = q' [q'^{-1}]_{q_L} \cdot \text{ct}_{0,L} + q_L [q_L^{-1}]_{q'} \cdot \text{ct}'_0 \bmod q_L q' \mathcal{R}$ 
3.  $a_1 = \text{ct}_1 - \mathbf{s}_{\text{op},1}^T \text{ct}_0 \bmod q_L q' \mathcal{R}$ 
4. For  $i = 0, \dots, n-1$  do
5.    $\tau_i(t) \leftarrow 0$  if  $|\tau_i(a_1)| < |\tau_i(a_1) \pm q_L q' / 2|$  else  $\tau_i(t) \leftarrow 1$ 
6. If  $\llbracket t \in \mathcal{I} \rrbracket \wedge \llbracket \text{GS.Verify}(\text{gpk}, \mathbf{M}, \text{gsig}, \text{pp}) = 1 \rrbracket$ ,  $\text{id} \leftarrow t$  else  $\text{id} \leftarrow \perp$ 

```

Output: $\text{id} \in \mathcal{I} \cup \{\perp\}$

Remark 4.1. Following [BMW03], the group manager generates all the group member keys at once, in the `GS.KeyGen` algorithm. This allows for simplifying the syntax as there is no need to store the group manager's trapdoor in such a case. However, in practice, the group manager may proceed differently. It could for example generate member keys by smaller batches (e.g., when the previous batch nears depletion), or individually, upon joining request. This way, the number of generated keys would scale with the number of actual group members and not with the size of the identity space $\mathcal{I}$.

4.3 Zero-Knowledge Relation

As mentioned in Sections 1 and 4.1, our core contributions aim at reducing the impact of the zero-knowledge proof, in particular by drastically reducing the witness dimension. The relation we prove in zero-knowledge then becomes rather simple compared to prior works as all the equations are linear in the witness, except for norm constraints. The proof system (Prove, Verify) is then a non-interactive zero-knowledge argument proving knowledge of $(t, \mathbf{s}, \mathbf{e}, e_1, e_2) \in \mathcal{R}^{2d+2+1}$ such that

$$\begin{aligned} \mathbf{A}\mathbf{s} + \mathbf{e} - q_L t \mathbf{G}_H \mathbf{v}_2 &= [\mathbf{I}_d | \mathbf{A}] - \mathbf{B} | \mathbf{a}_3] \mathbf{v} \bmod q\mathcal{R} \\ \mathbf{A}\mathbf{s} + \mathbf{e} &= \text{ct}'_0 \bmod q'\mathcal{R} \\ \mathbf{b}_{\text{op},1}^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t &= \text{ct}_1 \bmod q_L q' \mathcal{R} \\ \mathbf{b}_{\text{op},2}^T \mathbf{s} + e_2 + \lfloor q_L q' / 2 \rfloor t &= \text{ct}_2 \bmod q_L q' \mathcal{R} \\ \|\mathbf{s}^T | \mathbf{e}^T\|^T\|_2 &\leq B_{se}, \quad \|\mathbf{e}_1\|_2 \leq B_{e1}, \quad \|\mathbf{e}_2\|_2 \leq B_{e1}, \quad t \in \mathcal{I} \end{aligned} \tag{1}$$

The statement `statement` then contains all the public elements of the above relation, those not in red. This relation can be proven using the framework of [LNP22, LN22], as detailed in Appendix B. In particular, we note that the set of identities $\mathcal{I}$ corresponds to the set of tags denoted $\mathcal{T}_w$ in previous works [AGJ⁺24, JS25a, JS25b, JS26], whose membership can be efficiently proven using [LNP22]. We also provide optimizations to the framework of [LNP22] in Appendix B so as to tailor it to our relation in a more efficient way. Some of these optimizations may be of independent interest.

We specify that the NIZK is obtained by applying the Fiat-Shamir transform to the interactive proof system of [LNP22], and the signed message $\mathbf{M}$ is included as an auxiliary input to the random oracles when deriving the different challenges, as in prior works.

5 Security Analysis

We now analyze the security of the group signature scheme presented in Section 4, showing it is CCA-anonymous and fully-traceable. As we explained, the MH-M-LWE assumption is used to simplify the proof, but we recall that the latter accepts a reduction from M-LWE (Theorem 3.1) and thus by no means weakens our result. In the security statement, we use $\mathcal{D}_{\text{op}}, \mathcal{D}_{se}, \mathcal{D}_{e1}$ to respectively denote $\mathcal{W}_{d, w_{\text{op}}/2}^2$ truncated in spectral norm at β_{op} (as specified in GS.KeyGen), $\mathcal{D}_{\mathcal{R}^{2d}, s_{se}}$ truncated in ℓ^2 norm at B_{se} , and $\mathcal{D}_{\mathcal{R}, s_{e1}}$ truncated in ℓ^2 norm at B_{e1} (as specified in GS.Sign).

Theorem 5.1. *The group signature of Section 4 is CCA-anonymous based on the zero-knowledge and simulation soundness properties of (Prove, Verify), and the MH-M-LWE¹ and M-LWE assumptions. More precisely, the scheme is ε_{an} -CCA-anonymous with*

$$\varepsilon_{\text{an}} = \varepsilon_{\text{snd}} + \varepsilon_{zk} + 3\varepsilon_{\text{M-LWE}}^{(\text{op})} + 2\varepsilon_{\text{MH-M-LWE}} + \varepsilon_{\text{M-LWE}}^{(\text{ct})} + \delta_v/2,$$

where $\varepsilon_{snd}, \varepsilon_{zk}$ are respectively the soundness error and zero-knowledge security of (Prove, Verify), $\varepsilon_{\text{M-LWE}}^{(\text{op})}, \varepsilon_{\text{M-LWE}}^{(\text{ct})}, \varepsilon_{\text{MH-M-LWE}}$ are the respective hardness bounds of $\text{M-LWE}_{n,d,d,q_L q', \mathcal{D}_{\text{op}}}$, $\text{M-LWE}_{n,d,d+1,q q', \mathcal{D}_{se} \otimes \mathcal{D}_{e1}}$, and the matrix hint version $\text{MH-M-LWE}_{n,d,d+1,q q', \mathcal{D}_{se} \otimes \mathcal{D}_{e1}, \mathcal{D}_{e1}, \mathcal{D}_{\text{op}} \otimes \{0\}, 1}$. Finally, the parameter δ_v is defined by $\delta_v = \frac{1+\varepsilon_v}{1-\varepsilon_v} \left(\frac{1+\varepsilon_v/nm}{1-\varepsilon_v/nm} \right)^{nm} - 1 \sim_{\varepsilon_v \rightarrow 0^+} 4\varepsilon_v$.

Proof. We now prove our scheme is CCA-anonymous. More concretely, we proceed with a game-based proof that, at a high-level, aims at simulating the challenge group signature gsig^* without using the identity i_b (and in turn the bit b). The NIZK proof π can be simulated based on the zero-knowledge property, so we really need to focus on the other elements. The dependency in i_b comes from the preimage $\mathbf{v}$, as it may depend on the trapdoor $\mathbf{S}_{i_b}$, and the ciphertexts ct_1, ct_2 . The former can be tackled by the security of [KleinSamplePre](#) by sampling $\mathbf{v}$ without $\mathbf{S}_{i_b}$ and reprogramming the syndrome $\mathbf{u}$, but it requires $\mathbf{u}$ to be uniform. It means we must first simulate $\mathbf{u}, \text{ct}'_0, \text{ct}_1, \text{ct}_2$, which share the same randomness. Although it seems we can do so with a single M-LWE instance, we need to account for opening queries. They prevent us from turning both opening keys $\mathbf{b}_{\text{op},1}, \mathbf{b}_{\text{op},2}$ into uniform vectors simultaneously. As explained in Section 4.1, we are able to carry out the Naor-Yung proof, with shared randomness, while dealing with opening queries, but this requires us to rely on a hint version of M-LWE. More precisely, we rely on our new assumption MH-M-LWE with $Q_{\text{hint}} = 1$ and $k = 1$. Let us now formalize these arguments. We call Game_0 the original CCA-anonymity game of Figure 2.1, and then define a sequence of modifications in games Game_i . We denote by Adv_i the advantage of a PPT adversary $\mathcal{A}$ in Game_i .

Game₁. We start by ensuring all the opening queries are well-formed. Typically, an opening query can submit a group signature which does not verify, in which case [GS.Open](#) detects it and returns $\perp$. In case it does verify, to carry out the Naor-Yung strategy, we must ensure that the ciphertexts ct_1, ct_2 indeed encrypt the same tag t_i . As the NIZK proof verifies, we can argue that an adversary who managed to produce ciphertexts ct_1, ct_2 which encrypt two different tags broke the soundness of (Prove, Verify). Concretely, in Game_1 , the challenger holds on to $\mathbf{s}_{\text{op},2}$ when generating the keys in [GS.KeyGen](#). Then, for each query to $\mathcal{O}_{\text{Open}}$, the challenger changes the call to [GS.Open](#) as follows. It first carries steps 1 to 5 of [GS.Open](#) as usual, and then computes $a_2 = \text{ct}_2 - \mathbf{s}_{\text{op},2}^T \text{ct}_0 \bmod q_L q' \mathcal{R}$. It decodes a_2 similarly to a_1 to get some t' in $\mathcal{R}$. Then, if $\llbracket t = t' \rrbracket \wedge \llbracket t \in \mathcal{I} \rrbracket \wedge \llbracket \text{GS.Verify}(\text{gpk}, \text{M}, \text{gsig}, \text{pp}) = 1 \rrbracket$, it returns t , and $\perp$ otherwise.

The only condition that changes is that both ciphertexts should decrypt to the same tag. By definition of π , the opening queries are correctly simulated unless the adversary $\mathcal{A}$ managed to provide (gsig, M) such that ct_1, ct_2 decrypt to two different tags while having a valid proof π . In that case, we can readily use $\mathcal{A}$ into an adversary against the soundness of (Prove, Verify) by using one such cheating opening query. As a result, Lemma B.1 gives

$$|\text{Adv}_0 - \text{Adv}_1| \leq \varepsilon_{snd}$$

where ε_{snd} is the soundness error defined in Lemma B.1. From now on, we can assume that all opening queries with valid group signatures contain two ciphertexts ct_1, ct_2 that are consistent in the sense that they decrypt to the same tag. This later allows us to use either one of the keys to answer opening queries.

Game₂. We now proceed by simulating the zero-knowledge proof π within the challenge group signature gsig^* computed by the challenger. More precisely, instead of faithfully generating π in the call to **GS.Sign**, we use the simulator from Lemma B.1. As a result, a distinguisher between the views of $\mathcal{A}$ in **Game₁** and **Game₂** can readily be used to construct an attacker for the zero-knowledge property of (**Prove**, **Verify**). Hence, we have

$$|\text{Adv}_1 - \text{Adv}_2| \leq \varepsilon_{zk},$$

where ε_{zk} is defined in Lemma B.1.

Game₃. Note that as of **Game₁**, we are ensured that in opening queries with group signatures passing verification, the ciphertexts ct_1, ct_2 decrypt to the same tag. We can thus decrypt either and correctly simulate queries. We do so with $\mathbf{s}_{\text{op},1}$, which allows us to proceed with simulating ct_2 in the challenge group signature gsig^* following a typical IND-CPA argument. We thus start by simulating the second opening key $\mathbf{b}_{\text{op},2}$. Concretely, **Game₃** corresponds to **Game₂** at the exception that $\mathbf{b}_{\text{op},2}$ is sampled uniformly in $\mathcal{R}_{q_L q'}^d$ (and $\mathbf{x}_{\text{op},2}$ is not sampled) in **GS.KeyGen**. As we do not use $\mathbf{s}_{\text{op},2}$ in the opening queries as explained above, this change is licit and we now argue this is indistinguishable under M-LWE $_{n,d,d,q_L q', \mathcal{D}_{\text{op}}}$, where $\mathcal{D}_{\text{op}}$ corresponds to $\mathcal{W}_{d,w_{\text{op}}/2}^2$ truncated in spectral norm at β_{op} .

Indeed, let $\mathcal{D}$ be a distinguisher between the views of $\mathcal{A}$ in **Game₂** and **Game₃**. We construct a distinguisher $\mathcal{B}$ against M-LWE which, upon reception of the instance $(\mathbf{A}_{\text{op}}, \mathbf{b}_{\text{op},2})$ in $\mathcal{R}_{q_L q'}^{d \times d} \times \mathcal{R}_{q_L q'}^d$, proceeds as follows. It first samples $\mathbf{A}_H$ from $U(\mathcal{R}_{q_H}^{d \times d})$ and defines $\mathbf{A}$ in **GS.Setup** as the inverse CRT of $(\mathbf{A}_H, \mathbf{A}_{\text{op}}^T)$, i.e.,

$$\mathbf{A} = q_L q' [(q_L q')^{-1}]_{q_H} \cdot \mathbf{A}_H + q_H [q_H^{-1}]_{q_L q'} \cdot \mathbf{A}_{\text{op}}^T \bmod q q' \mathcal{R}$$

It then defines all other elements correctly using the constructed $\mathbf{A}$ except $\mathbf{b}_{\text{op},2}$ which it takes directly from the instance. From this, $\mathcal{B}$ calls the distinguisher $\mathcal{D}$ and outputs 1 (M-LWE) if $\mathcal{D}$ returns **Game₂**, and 0 (uniform) if $\mathcal{D}$ returns **Game₃**. Let us show that $\mathcal{B}$ simulates the challenger correctly.

First, as $\mathbf{A}_{\text{op}}$ is uniform modulo $q_L q'$, $\mathbf{A}_H$ is uniform modulo q_H and $q_H \wedge q_L q' = 1$ (as $q_H \wedge q_L = q_H \wedge q' = q_L \wedge q' = 1$), the chinese remainder theorem gives that $\mathbf{A}$ is indeed uniform modulo $q q'$ as required, and that $\mathbf{A} \bmod q_L q' \mathcal{R} = \mathbf{A}_{\text{op}}^T$. Then, if $\mathbf{b}_{\text{op},2} = \mathbf{A}_{\text{op}} \mathbf{s}_{\text{op},2} + \mathbf{e}_{\text{op},2} \bmod q_L q' \mathcal{R}$ for some $\mathbf{x}_{\text{op},2} = [\mathbf{s}_{\text{op},2}^T | \mathbf{e}_{\text{op},2}^T]^T \sim \mathcal{D}_{\text{op}}$, then we indeed have $\mathbf{b}_{\text{op},2} = \mathbf{A}^T \mathbf{s}_{\text{op},2} + \mathbf{e}_{\text{op},2} \bmod q_L q' \mathcal{R}$ as in **Game₂**. And if $\mathbf{b}_{\text{op},2}$ is uniform, then it correctly simulates **Game₃**. It then yields

$$|\text{Adv}_2 - \text{Adv}_3| \leq \varepsilon_{\text{M-LWE}}^{(\text{op})},$$

where $\varepsilon_{\text{M-LWE}}^{(\text{op})}$ is the hardness bound of M-LWE $_{n,d,d,q_L q', \mathcal{D}_{\text{op}}}$.

Game₄. We now wish to rely on a IND-CPA argument to simulate the second ciphertext ct_2 . However, because we share some of the same randomness between the two ciphertexts ct_1 and ct_2 , we cannot rely directly on M-LWE. To prepare for this simulation of ct_2 , we slightly change the way ct_1 is generated in gsig^* . Concretely, the challenger generates $\mathbf{s}, \mathbf{e}, e_1, e_2$ as usual, and computes $\bar{\mathbf{u}} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod qq'\mathcal{R}$. It then defines $\mathbf{u} = \bar{\mathbf{u}} \bmod q\mathcal{R}$ and $\text{ct}'_0 = \bar{\mathbf{u}} \bmod q'\mathcal{R}$. So far, this is perfectly identical to Game₃ by a simple rewriting. It also defines $\text{ct}_2 = \mathbf{b}_{\text{op},2}^T \mathbf{s} + e_2 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$ as usual. For ct_1 however, it defines it as

$$\text{ct}_1 = \mathbf{s}_{\text{op},1}^T (\bar{\mathbf{u}} \bmod q_L q' \mathcal{R}) + e_1 + [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0] \begin{bmatrix} \mathbf{s} \\ \mathbf{e} \\ e_2 \end{bmatrix} + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}.$$

We now argue that this is also perfectly equivalent to Game₃. In the latter, we have

$$\begin{aligned} \text{ct}_1 &= (\mathbf{A}^T \mathbf{s}_{\text{op},1} + \mathbf{e}_{\text{op},1})^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R} \\ &= \mathbf{s}_{\text{op},1}^T \mathbf{A} \mathbf{s} + \mathbf{e}_{\text{op},1}^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R} \\ &= \mathbf{s}_{\text{op},1}^T (\mathbf{A} \mathbf{s} + \mathbf{e}) - \mathbf{s}_{\text{op},1}^T \mathbf{e} + \mathbf{e}_{\text{op},1}^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R} \\ &= \mathbf{s}_{\text{op},1}^T (\mathbf{A} \mathbf{s} + \mathbf{e} \bmod q_L q' \mathcal{R}) + e_1 + (\mathbf{e}_{\text{op},1}^T \mathbf{s} - \mathbf{s}_{\text{op},1}^T \mathbf{e}) + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R} \end{aligned}$$

Yet, as $q_L q' | qq'$, we have $\mathbf{A} \mathbf{s} + \mathbf{e} \bmod q_L q' \mathcal{R} = (\mathbf{A} \mathbf{s} + \mathbf{e} \bmod qq' \mathcal{R}) \bmod q_L q' \mathcal{R}$. Hence, ct_1 is exactly the same in both games. As a result, both games are identical and we obtain

$$\text{Adv}_3 = \text{Adv}_4.$$

Game₅. We now simulate $\mathbf{u}, \text{ct}'_0, \text{ct}_1, \text{ct}_2$ in the challenge group signature gsig^* . As it resembles an M-LWE instance, we would want to simulate $\bar{\mathbf{u}}$ under M-LWE and define $\mathbf{u}, \text{ct}'_0, \text{ct}_1, \text{ct}_2$ from $\bar{\mathbf{u}}$ as in the previous game. Indeed, by the previous rewriting, $\mathbf{u}, \text{ct}'_0, \text{ct}_1$ can be generated from the same $\bar{\mathbf{u}}$ modulo qq' , and ct_2 from an additional row of the M-LWE matrix. However, it also requires the additional element $e_1 + (\mathbf{e}_{\text{op},1}^T \mathbf{s} - \mathbf{s}_{\text{op},1}^T \mathbf{e})$ due to the randomness shared across ct_1 and ct_2 . Fortunately, this additional element has the form of a noisy linear hint on the M-LWE secret-error vector $(\mathbf{s}, \mathbf{e}, e_2)$. Hence, we can rely on the MH-M-LWE assumption we introduced in Section 3 for specific parameters. In particular, we choose parameters that are compatible with the reduction from M-LWE of Theorem 3.1. We note here that we need a single hint on a single secret, i.e., $Q_{\text{hint}} = k = 1$. Later on in the traceability proof, we will need the multiple secret version with one hint per independent secret. Here, the hint we need is of the form

$$z = e_1 + [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0] \begin{bmatrix} \mathbf{s} \\ \mathbf{e} \\ e_2 \end{bmatrix}$$

Concretely, the challenger changes the call to GS.Sign and samples $(\mathbf{s}, \mathbf{e}, e_1, e_2)$ as usual but now samples $\bar{\mathbf{u}} \leftarrow U(\mathcal{R}_{qq'}^d)$ and $u_2 \leftarrow U(\mathcal{R}_{qq'})$. It then sets everything as in Game₄, that is $\mathbf{u} = \bar{\mathbf{u}} \bmod q\mathcal{R}$, $\text{ct}'_0 = \bar{\mathbf{u}} \bmod q'\mathcal{R}$ and $\text{ct}_1 = \mathbf{s}_{\text{op}}^T (\bar{\mathbf{u}} \bmod q_L q' \mathcal{R}) + e_1 + [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0] \begin{bmatrix} \mathbf{s} \\ \mathbf{e} \\ e_2 \end{bmatrix} + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$.

$q_L q' \mathcal{R}) + e_1 + \mathbf{e}_{\text{op}}^T \mathbf{s} - \mathbf{s}_{\text{op}}^T \mathbf{e} + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$, except ct_2 which is defined as $\text{ct}_2 = (u_2 \bmod q_L q' \mathcal{R}) + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$. We now argue this is indistinguishable under $\text{MH-M-LWE}_{n,d,d+1,qq',\mathcal{D}_{se} \otimes \mathcal{D}_{e1},\mathcal{D}_{e1},\mathcal{D}_{\text{op}} \otimes \{0\},1}^1$, where $\mathcal{D}_{se} = \mathcal{D}_{\mathcal{R}^{2d},s_{se}}$ truncated at B_{se} , $\mathcal{D}_{e1} = \mathcal{D}_{\mathcal{R},s_{e1}}$ truncated at B_{e1} , and $\mathcal{D}_{\text{op}} = \mathcal{W}_{d,w_{\text{op}}/2}^2$ truncated in spectral norm at β_{op} .

Indeed, let $\mathcal{D}$ be a distinguisher between the views of $\mathcal{A}$ in Game_4 and Game_5 . We construct a distinguisher $\mathcal{B}$ against MH-M-LWE which, upon reception of the instance $(\tilde{\mathbf{A}}, \tilde{\mathbf{u}}, \mathbf{m}^T, z)$ in $\mathcal{R}_{qq'}^{(d+1) \times d} \times \mathcal{R}_{qq'}^{(d+1) \times 1} \times \mathcal{R}^{1 \times (2d+1)} \times \mathcal{R}^{1 \times 1}$, proceeds as follows. It first parses the instance as

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} \\ \tilde{\mathbf{b}}_{\text{op},2}^T \end{bmatrix}, \quad \tilde{\mathbf{u}} = \begin{bmatrix} \bar{\mathbf{u}} \\ u_2 \end{bmatrix}, \quad \mathbf{m}^T = [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0].$$

It defines $\mathbf{b}_{\text{op},2} = \tilde{\mathbf{b}}_{\text{op},2} \bmod q_L q' \mathcal{R}$, and then uses $\mathbf{A}$ in the definition of pp in [GS.Setup](#) and $\mathbf{b}_{\text{op},2}, \mathbf{s}_{\text{op},1}, \mathbf{e}_{\text{op},1}$ in the definition of the opening keys in [GS.KeyGen](#). Then, when generating gsig^* , the distinguisher $\mathcal{B}$ uses $\bar{\mathbf{u}}$ to define $\mathbf{u} = \bar{\mathbf{u}} \bmod q \mathcal{R}$, $\text{ct}'_0 = \bar{\mathbf{u}} \bmod q' \mathcal{R}$, and $\text{ct}_1 = \mathbf{s}_{\text{op},1}^T \bar{\mathbf{u}} + z + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$. It also defines $\text{ct}_2 = (u_2 \bmod q_L q' \mathcal{R}) + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$. Finally, it calls $\mathcal{D}$ on the total view of $\mathcal{A}$ in this simulated environment and outputs 1 (MH-M-LWE) if $\mathcal{D}$ returns Game_4 , and 0 (uniform) otherwise.

We now check $\mathcal{B}$ correctly simulates both games. First, notice that whether $\tilde{\mathbf{u}}$ is uniform or not, $\mathbf{A}$ is uniform modulo qq' as required. Similarly, we have that $\tilde{\mathbf{b}}_{\text{op},2}$ is uniform modulo qq' which means $\mathbf{b}_{\text{op},2}$ is indeed uniform modulo $q_L q'$ as $q_L q' | qq'$. Additionally, the opening key is correctly distributed. Indeed, as $-\mathcal{W}_{d,w_{\text{op}}/2} = \mathcal{W}_{d,w_{\text{op}}/2}$ and $\|M_\tau([\mathbf{s}_{\text{op},1}^T | \mathbf{e}_{\text{op},1}^T])\|_2 = \|M_\tau([\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0])\|_2 \leq \beta_{\text{op}}$, it holds that $(\mathbf{s}_{\text{op},1}, \mathbf{e}_{\text{op},1})$ follows $\mathcal{D}_{\text{op}}$ as well, as required. It also holds that z is of the form $z = e_1 + \mathbf{m}^T [\mathbf{s}^T | \mathbf{e}^T | e_2]^T = e_1 + \mathbf{e}_{\text{op},1}^T \mathbf{s} - \mathbf{s}_{\text{op},1}^T \mathbf{e}$ for some $e_1 \sim \mathcal{D}_{e1}$ and $(\mathbf{s}, \mathbf{e}, e_2) \sim \mathcal{D}_{se} \otimes \mathcal{D}_{e1}$.

If $\tilde{\mathbf{u}}$ is uniform modulo qq' , so is $\bar{\mathbf{u}}$. Also, as $q_L q'$ divides qq' , $u_2 \bmod q_L q' \mathcal{R}$ is uniform in $\mathcal{R}_{q_L q'}$, then correctly simulating Game_5 . If on the other hand $\tilde{\mathbf{u}} = \tilde{\mathbf{A}} \mathbf{s} + [\mathbf{e}^T | e_2]^T \bmod qq' \mathcal{R}$ (for the same $\mathbf{s}, \mathbf{e}, e_2$ that is used in z), we have

$$\begin{aligned} \mathbf{u} &= \bar{\mathbf{u}} \bmod q \mathcal{R} = \mathbf{A} \mathbf{s} + \mathbf{e} \bmod q \mathcal{R} \\ \text{ct}'_0 &= \bar{\mathbf{u}} \bmod q' \mathcal{R} = \mathbf{A} \mathbf{s} + \mathbf{e} \bmod q' \mathcal{R} \\ \text{ct}_1 &= \mathbf{s}_{\text{op},1}^T (\bar{\mathbf{u}} \bmod q_L q' \mathcal{R}) + e_1 + [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0] \begin{bmatrix} \mathbf{s} \\ \mathbf{e} \\ e_2 \end{bmatrix} + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R} \\ \text{ct}_2 &= u_2 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R} = \mathbf{b}_{\text{op},2}^T \mathbf{s} + e_2 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}, \end{aligned}$$

which then perfectly simulate Game_4 . It then yields

$$|\text{Adv}_4 - \text{Adv}_5| \leq \varepsilon_{\text{MH-M-LWE}}$$

where the parameter $\varepsilon_{\text{MH-M-LWE}}$ denotes the hardness bound of the problem $\text{MH-M-LWE}_{n,d,d+1,qq',\mathcal{D}_{se} \otimes \mathcal{D}_{e1},\mathcal{D}_{e1},\mathcal{D}_{\text{op}} \otimes \{0\},1}^1$. Then, by Theorem 3.1, it holds that

$\varepsilon_{\text{MH-M-LWE}} \leq 1 \cdot (\varepsilon_{\text{M-LWE}} + \varepsilon_{\text{op}} / (1 - \varepsilon_{\text{op}}))$ for the appropriate M-LWE parameters. We later use this methodology to assess the value of the $\varepsilon_{\text{MH-M-LWE}}$ above, as described in Section 6.1.

Game₆. We further change the definition of ct_2 by simulating it without using t_{i_b} . More precisely, the challenger samples $\text{ct}_2 \leftarrow U(\mathcal{R}_{q_L q'})$ when calling **GS.Sign** to generate gsig^* . In **Game₅**, $u_2 \bmod q_L q' \mathcal{R}$ is uniform, used nowhere else, and independent of $\lfloor q_L q' / 2 \rfloor t_{i_b}$, it then holds that ct_2 is exactly uniform in **Game₅**. Hence, we directly have

$$\text{Adv}_5 = \text{Adv}_6$$

Game₇. Now that ct_2 is perfectly simulated, we can revert the changes of **Game₃** to **Game₅** in reverse order without changing ct_2 . First, the challenger changes the way it generates $\bar{\mathbf{u}}$ and thus $\mathbf{u}, \text{ct}'_0, \text{ct}_1$. When generating the challenge signature by calling (the modified version of) **GS.Sign**, it samples $(\mathbf{s}, \mathbf{e}, e_1, e_2)$ as usual and defines again $\bar{\mathbf{u}} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q q' \mathcal{R}$. It then defines $\mathbf{u} = \bar{\mathbf{u}} \bmod q \mathcal{R} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q \mathcal{R}$, $\text{ct}'_0 = \bar{\mathbf{u}} \bmod q' \mathcal{R}$, and $\text{ct}_1 = \mathbf{s}_{\text{op},1}^T (\bar{\mathbf{u}} \bmod q_L q' \mathcal{R}) + e_1 + [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0] [\mathbf{s}^T | \mathbf{e}^T | e_2]^T + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$. But note that it leaves ct_2 completely random. Using the same argument as in **Game₅** (except that the instance has one fewer row due to leaving ct_2 untouched), it holds that

$$|\text{Adv}_6 - \text{Adv}_7| \leq \varepsilon'_{\text{MH-M-LWE}} \leq \varepsilon_{\text{MH-M-LWE}}$$

where $\varepsilon'_{\text{MH-M-LWE}}$ is the hardness bound of $\text{MH-M-LWE}_{n,d,d,qq',\mathcal{D}_{se},\mathcal{D}_{e1},\mathcal{D}_{\text{op}},1}^1$. The last inequality is due to a trivial reduction consisting in discarding the $(d+1)$ -th sample in the assumption from **Game₅**. This is possible because the hint features a 0 for the error e_2 (corresponding to the error of the $(d+1)$ -th sample).

Game₈. We now revert the change from **Game₄** by changing the generation of ct_1 back to its original computation in the challenge signature. Precisely, the challenger modifies the call to **GS.Sign** by generating $(\mathbf{s}, \mathbf{e}, e_1, e_2)$ as usual, but generates $\mathbf{u} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q \mathcal{R}$, $\text{ct}'_0 = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q' \mathcal{R}$ directly, and $\text{ct}_1 = \mathbf{b}_{\text{op},1}^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$. As shown in **Game₄**, both games are identically distributed and thus

$$\text{Adv}_7 = \text{Adv}_8$$

Game₉. We now reintroduce the second opening key $\mathbf{s}_{\text{op},2}$ by reverting the change from **Game₃**. Concretely, the challenger goes back to running **GS.KeyGen** legitimately and sampling $\mathbf{x}_{\text{op},2} \leftarrow \mathcal{W}_{d,w_{\text{op}}/2}^2$ conditioned on $\|\sigma(\mathbf{x}_{\text{op},2}^* \mathbf{x}_{\text{op},2})\|_{\infty} \leq \beta_{\text{op}}$ and defining $\mathbf{b}_{\text{op},2} = [\mathbf{A}^T | \mathbf{I}_d] \mathbf{x}_{\text{op},2} \bmod q_L q' \mathcal{R}$. Using the exact same argument as in **Game₃** in reverse, we obtain

$$|\text{Adv}_8 - \text{Adv}_9| \leq \varepsilon_{\text{M-LWE}}^{(\text{op})}.$$

At this stage, we are essentially in the same situation we were in **Game₂**, except that we now sample ct_2 uniformly at random. Moving forward, we will now switch the key used to answer opening queries, simulate $\mathbf{b}_{\text{op},1}$, and finally simulate $\mathbf{u}, \text{ct}'_0, \text{ct}_1$. After that, we can simulate $\mathbf{v}$ which concludes the proof.

Game₁₀. We now switch keys when answering the queries to $\mathcal{O}_{\text{Open}}$ so as to use $\mathbf{s}_{\text{op},2}$ instead of $\mathbf{s}_{\text{op},1}$. More precisely, the challenger changes the calls to **GS.Open** in the opening queries as follows. It still computes ct_0 , but then computes $a_2 = \text{ct}_2 - \mathbf{s}_{\text{op},2}^T \text{ct}_0 \bmod q_L q' \mathcal{R}$. It then decodes a_2 to get t and returns t if $\llbracket t \in \mathcal{I} \rrbracket \wedge \llbracket \text{GS.Verify}(\text{gpk}, \mathbf{M}, \text{gsig}, \text{pp}) = 1 \rrbracket$, and $\perp$ otherwise. Recall that as of **Game₁**, we have ensured that for queries with valid group signatures, the two ciphertexts were generated by encrypting the same tag t . Hence, the opening queries in **Game₁₀** are perfectly simulated. It means

$$\text{Adv}_9 = \text{Adv}_{10}$$

Game₁₁. We now simulate the first opening key. As $\mathbf{s}_{\text{op},1}$ is not used, we can thus sample $\mathbf{b}_{\text{op},1}$ uniformly. Hence, when generating the keys with **GS.KeyGen**, the challenger samples $\mathbf{b}_{\text{op},1} \leftarrow U(\mathcal{R}_{q_L q'}^d)$ without sampling $\mathbf{x}_{\text{op},1}$. Using the same argument as in **Game₃** and **Game₉**, it holds that

$$|\text{Adv}_{10} - \text{Adv}_{11}| \leq \varepsilon_{\text{M-LWE}}^{(\text{op})}.$$

Game₁₂. Now, notice that because ct_2 is sampled uniformly independently of all the rest, we can simplify the arguments of **Game₄** and **Game₅** when simulating ct_1 in the challenge signature. Indeed, we no longer have correlated randomness at this point which allows us to simulate $\mathbf{u}, \text{ct}'_0, \text{ct}_1$ from a regular M-LWE instance. Concretely, when generating the challenge signature with the modified **GS.Sign**, the challenger samples $\mathbf{u} \leftarrow U(\mathcal{R}_q^d), \text{ct}'_0 \leftarrow U(\mathcal{R}_{q'}^d)$ and $u_1 \leftarrow \mathcal{R}_{q_L q'}$ to define $\text{ct}_1 = u_1 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$. We now argue that this is indistinguishable under $\text{M-LWE}_{n,d,d+1,qq',\mathcal{D}_{se} \otimes \mathcal{D}_{e1}}$. For a distinguisher $\mathcal{D}$, we construct the adversary $\mathcal{B}$ for M-LWE as follows. Given the instance $(\tilde{\mathbf{A}}, \tilde{\mathbf{b}}) \in \mathcal{R}_{qq'}^{(d+1) \times d} \times \mathcal{R}_{qq'}^{d+1}$, it parses it as

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} \\ \tilde{\mathbf{b}}_{\text{op},1}^T \end{bmatrix}, \text{ and } \tilde{\mathbf{b}} = \begin{bmatrix} \tilde{\mathbf{u}} \\ \tilde{u}_1 \end{bmatrix}.$$

It then uses $\mathbf{A}$ when setting the public parameters in **GS.Setup**, and defines $\mathbf{b}_{\text{op},1} = \tilde{\mathbf{b}}_{\text{op},1} \bmod q_L q' \mathcal{R}$ in **GS.KeyGen**. Then, when generating the group signature in its call to **GS.Sign**, it defines $\mathbf{u} = \tilde{\mathbf{u}} \bmod q \mathcal{R}$, $\text{ct}'_0 = \tilde{\mathbf{u}} \bmod q' \mathcal{R}$, $u_1 = \tilde{u}_1 \bmod q_L q' \mathcal{R}$, and $\text{ct}_1 = u_1 + \lfloor q_L q' / 2 \rfloor t_{i_b} \bmod q_L q' \mathcal{R}$. Finally, if $\mathcal{D}$ returns **Game₁₁**, $\mathcal{B}$ returns 1 (M-LWE) and 0 otherwise. We now verify it correctly simulates the games so that the output of $\mathcal{B}$ is as expected.

First, notice that as $\tilde{\mathbf{A}}$ is uniform modulo qq' , it directly holds that $\mathbf{A}$ is uniform in $\mathcal{R}_{qq'}^{d \times d}$ and that $\mathbf{b}_{\text{op},1}$ is uniform in $\mathcal{R}_{q_L q'}^d$ as $q_L q' | qq'$. Now, assume $\tilde{\mathbf{b}} = \tilde{\mathbf{A}} \mathbf{s} + \tilde{\mathbf{e}} \bmod qq' \mathcal{R}$ for some $[\mathbf{s}^T | \tilde{\mathbf{e}}^T]^T$ drawn from $\mathcal{D}_{se} \otimes \mathcal{D}_{e1}$. Parsing $\tilde{\mathbf{e}}^T$ as $[\mathbf{e}^T | e_1]$ gives that $\mathbf{u} = \mathbf{A} \mathbf{s} + \mathbf{e} \bmod q \mathcal{R}$, $\text{ct}'_0 = \mathbf{A} \mathbf{s} + \mathbf{e} \bmod q' \mathcal{R}$, and $u_1 = \mathbf{b}_{\text{op},1}^T \mathbf{s} + e_1 \bmod q_L q' \mathcal{R}$, thus correctly simulating **Game₁₁**. Then, if $\tilde{\mathbf{b}}$ is uniform modulo qq' , the Chinese Remainder Theorem gives that $\mathbf{u}$ and ct'_0 are independent and uniform modulo q and q' respectively as $q \wedge q' = 1$. We also directly get that u_1 is uniform modulo $q_L q'$ as $q_L q' | qq'$, hence correctly simulating **Game₁₂**. We obtain that

$$|\text{Adv}_{11} - \text{Adv}_{12}| \leq \varepsilon_{\text{M-LWE}}^{(\text{ct})},$$

where $\varepsilon_{\text{M-LWE}}^{(\text{ct})}$ is the hardness bound of M-LWE $_{n,d,d+1,qq',\mathcal{D}_{se} \otimes \mathcal{D}_{e1}}$. Note that in the secret-error distribution, the term e_1 has a larger parameter than $\mathbf{s}, \mathbf{e}$. Because s_{e1} is larger than s_{se} , adding an independent Gaussian noise with parameter $\sqrt{s_{e1}^2 - s_{se}^2} \geq \sqrt{2}\eta_{\varepsilon_{\text{op}}}(\mathbb{Z}^n)$ to the last sample gives a trivial reduction from the same M-LWE assumption but with a secret-error distribution that is spherical with parameter s_{se} . We use this method to evaluate $\varepsilon_{\text{M-LWE}}^{(\text{ct})}$ in Section 6.1.

Game₁₃. Here, we simply generate ct_1 to be perfectly uniform without using t_{i_b} . Hence, the challenger samples $\text{ct}_1 \leftarrow U(\mathcal{R}_{q_L q'})$ when calling **GS.Sign** in **Game₁₃**. As u_1 in **Game₁₂** is uniform and independent of $\lfloor q_L q' / 2 \rfloor t_{i_b}$, it holds that ct_1 is exactly uniform in **Game₁₂**. Hence, we directly have

$$\text{Adv}_{12} = \text{Adv}_{13}$$

Game₁₄. At this point, the only element that might still depend on the identity i_b is the preimage $\mathbf{v}$. We thus simulate it without the identity, while (virtually) reprogramming the syndrome $\mathbf{u}$. Concretely, when calling **GS.Sign** in **Game₁₄**, the challenger samples $\mathbf{v}$ from $\mathcal{D}_{\mathcal{R}^m, s_v}$ (truncated at the bounds B_{v1}, B_{v2}, B_{v3}) and (virtually) defines $\mathbf{u} = \mathbf{A}_{t_{i_b}} \mathbf{v} \bmod q\mathcal{R}$. Although we do not give $\mathbf{u}$ to the adversary and it cannot query **gsig*** for opening, Lemma 2.6 shows that the pair $(\mathbf{v}, \mathbf{u})$ are statistically close in both games. In particular, as we are only concerned with the marginal distribution of $\mathbf{v}$, the data processing inequality of the statistical distance gives $\Delta(\mathbf{v}_{\text{Game}_{13}}, \mathbf{v}_{\text{Game}_{14}}) \leq \Delta((\mathbf{v}, \mathbf{u})_{\text{Game}_{13}}, (\mathbf{v}, \mathbf{u})_{\text{Game}_{14}}) \leq \delta_v/2$, where the last inequality stems from Lemma 2.6 and

$$\delta_v = \frac{1 + \varepsilon_v}{1 - \varepsilon_v} \left(\frac{1 + \varepsilon_v/nm}{1 - \varepsilon_v/nm} \right)^{nm} - 1$$

Indeed, for the distributions $\mathcal{P}_0, \mathcal{P}_1$ over $S = \mathcal{R}^m \times \mathcal{R}_q^d$ of Lemma 2.6, we have

$$\Delta(\mathcal{P}_0, \mathcal{P}_1) = \frac{1}{2} \sum_{(\mathbf{v}, \mathbf{u}) \in S} |\mathcal{P}_0(\mathbf{v}, \mathbf{u}) - \mathcal{P}_1(\mathbf{v}, \mathbf{u})| \leq \frac{1}{2} \sum_{(\mathbf{v}, \mathbf{u}) \in S} \mathcal{P}_1(\mathbf{v}, \mathbf{u}) \delta_v = \frac{\delta_v}{2}.$$

As a result, the data processing inequality of the statistical distance yields

$$|\text{Adv}_{13} - \text{Adv}_{14}| \leq \Delta(\mathbf{v}_{\text{Game}_{13}}, \mathbf{v}_{\text{Game}_{14}}) \leq \delta_v/2$$

Combining everything. In **Game₁₄**, the view of the adversary is completely independent of the identity i_b and in turn the bit b , which yields $\text{Adv}_{14} = 0$. Combining the previous hybrids with the triangle inequality, we then obtain

$$\text{Adv}_0 \leq \varepsilon_{\text{snd}} + \varepsilon_{\text{zk}} + 3\varepsilon_{\text{M-LWE}}^{(\text{op})} + 2\varepsilon_{\text{MH-M-LWE}} + \varepsilon_{\text{M-LWE}}^{(\text{ct})} + \delta_v/2,$$

which concludes the proof. $\square$

Remark 5.1. Albeit proven in the CCA-anonymity model, our construction and proof can be simplified if one only aims for CPA-anonymity. As mentioned in

Section 4, one only needs to remove the highlighted parts in the group signature specifications. Because the CPA model does not provide the adversary with opening queries, the challenger has a total freedom on the encryption key. In particular, it can simulate the opening public key $\mathbf{b}_{\text{op},1}$ without knowing the underlying $\mathbf{s}_{\text{op},1}$ while still being able to carry the anonymity game.

The proof is then greatly simplified: we simply have to (1) simulate the zero-knowledge proof in $\mathbf{gsig}^*$, (2) simulate the opening key $\mathbf{b}_{\text{op},1}$ replacing it by a uniform value, (3) simulate $\mathbf{u}, \text{ct}'_0, \text{ct}_1$ from the same M-LWE instance with matrix $[\mathbf{A}^T | \mathbf{b}_{\text{op},1}^T]^T$, and (4) simulate $\mathbf{v}$ without the trapdoor $\mathbf{S}_{i_b}$ and reprogram $\mathbf{u}$ accordingly. In the end, using the same quantities as in Theorem 5.1, the updated construction is ε'_{an} -CPA-anonymous with $\varepsilon'_{an} = \varepsilon_{zk} + \varepsilon_{\text{M-LWE}}^{(\text{op})} + \varepsilon_{\text{M-LWE}}^{(\text{ct})} + \delta_v/2$.

Theorem 5.2. *The group signature of Section 4 is fully-traceable based on the zero-knowledge and soundness properties of (Prove, Verify), and the M-LWE, M-SIS, and MH-M-LWE $^{Q_{\text{sign}}}$ assumptions, where Q_{sign} is the maximal number of emitted group signatures per group member. More precisely, the scheme is ε_{tr} -fully-traceable with*

$$\varepsilon_{tr} = \varepsilon_{snd} + \varepsilon_{\text{M-LWE}}^{(\text{op})} + N \left(Q_{\text{sign}} \varepsilon_{zk} + \varepsilon_{\text{MH-M-LWE}^{Q_{\text{sign}}}} \right. \\ \left. + \left(1 + \frac{\lambda(2\lambda - 1)\delta_v^2}{(1 - \delta_v)^{2\lambda+1}} \right)^{Q_{\text{sign}}/2\lambda} h^{\circ dk_H} (2\varepsilon_{\text{M-SIS}})^{1 - \frac{1}{2\lambda}} \right)$$

where $\varepsilon_{snd}, \varepsilon_{zk}$ are respectively the soundness error and zero-knowledge security of (Prove, Verify), $\varepsilon_{\text{M-LWE}}^{(\text{op})}, \varepsilon_{\text{MH-M-LWE}^{Q_{\text{sign}}}}, \varepsilon_{\text{M-SIS}}$ are the respective hardness bounds of $\text{M-LWE}_{n,d,d,q_L,q',\mathcal{D}_{\text{op}}}$, $\text{MH-M-LWE}_{n,d,d+1,q,q',\mathcal{D}_{se} \otimes \mathcal{D}_{e1}, \mathcal{D}_{e1}, \mathcal{D}_{\text{op}} \otimes \{0\},1}$, and $\text{M-SIS}_{n,d,2d+1,q,B}$ with $B^2 = (B_{v1} + c\sqrt{nd}B_{v2} + B_{se})^2 + B_{v3}^2$ and $c \approx 1.061$. The function $h^{\circ dk_H}$ is to the dk_H -th composition power of the function⁷ h defined by

$$h(x) = \varepsilon_{\text{M-LWE}}^{(\text{gpk})} + \mu \left(2\varepsilon_{\text{M-LWE}}^{(\text{gpk})} + \mu \left(\varepsilon_{\text{M-LWE}}^{(\text{gpk})} + x \right)^{1 - \frac{1}{2\lambda}} \right)^{1 - \frac{1}{2\lambda}},$$

with $\varepsilon_{\text{M-LWE}}^{(\text{gpk})}$ the hardness bound of $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1^{2d}}$ and

$$\mu = \left(1 + \frac{\lambda(2\lambda - 1)\delta_{\text{gpk}}^2}{(1 - \delta_{\text{gpk}})^{2\lambda+1}} \right)^{(N-1)m/2\lambda} \\ \delta_{\text{gpk}} = \left(\frac{1 + \frac{\varepsilon}{dk_L}}{1 - \frac{\varepsilon}{dk_L}} \right)^{2dk_L} \left(\frac{1 + \frac{\varepsilon}{dk_H}}{1 - \frac{\varepsilon}{dk_H}} \right)^{2dk_H} \left(\frac{1 + \varepsilon}{1 - \varepsilon} \right)^{12d(n-1)+6} - 1.$$

Finally, the parameter δ_v is defined by $\delta_v = \frac{1+\varepsilon_v}{1-\varepsilon_v} \left(\frac{1+\varepsilon_v/nm}{1-\varepsilon_v/nm} \right)^{nm} - 1$.

⁷ The function h has been introduced in [AGJ⁺24] and its impact on the bit-security of its input is studied in [AGJ⁺24, App. A.1]. In particular, dk_H compositions reduces the bit security by $O(dk_H)$ bits, which we account for in our security assessment.

Proof. We now prove the traceability of our scheme. At a high-level, we show that an adversary making the opening fail (i.e., **GS.Open** returns $\perp$) readily breaks the soundness of (**Prove**, **Verify**), while an adversary framing an honest group member can be used to forge a signature for the scheme of [JS26]. On this second point, we actually do not rely directly on the unforgeability but rather carry out the reduction from M-SIS by using the vector $\mathbf{v}$ provided by the adversary. Following the proof strategy of [JS26], reducing to M-SIS requires making the gadget vanish for the tag t^* used in the group signature of the adversary. As such, we guess which honest user i^+ is targeted (in the hope that the tag t^* outputted by **GS.Open** at the end matches t_{i^+}). Because we make the gadget vanish for t_{i^+} , we cannot produce a secret key for user i^+ and must then, beforehand, simulate all of its group signature queries without resorting to its trapdoor $\mathbf{S}_{i^+}$. Because the adversary is given the opening key $\mathbf{gmsk}$, we must do so while ensuring these signatures still open coherently, i.e., ensuring **GS.Open** indeed returns t_{i^+} . Let us now formalize these arguments. We call **Game**₀ the original traceability game from Figure 2.2, and then define a sequence of modifications in games **Game** _{i} . We denote by Adv_i the advantage of a PPT adversary $\mathcal{A}$ in **Game** _{i} .

Game₁. We start by extracting the zero-knowledge proof π within the group signature $\mathbf{gsig}^*$ outputted by the adversary, so that we can later exploit it to construct an M-SIS solution. More precisely, at the end of **Game**₁, the challenger extracts a witness $(t^*, \mathbf{s}^*, \mathbf{e}^*, e_1^*, e_2^*)$ with the extractor from Lemma B.1, which verifies

$$\begin{aligned} \mathbf{A}\mathbf{s}^* + \mathbf{e}^* - q_L t^* \mathbf{G}_H \mathbf{v}_2 &= [\mathbf{I}_d |\mathbf{A}| - \mathbf{B}|\mathbf{a}_3] \mathbf{v} \bmod q\mathcal{R} \\ \mathbf{A}\mathbf{s}^* + \mathbf{e}^* &= \mathbf{ct}'_0 \bmod q'\mathcal{R} \\ \mathbf{b}_{\text{op},1}^T \mathbf{s}^* + e_1^* + \lfloor q_L q' / 2 \rfloor t^* &= \text{ct}_1 \bmod q_L q' \mathcal{R} \\ \mathbf{b}_{\text{op},2}^T \mathbf{s}^* + e_2^* + \lfloor q_L q' / 2 \rfloor t^* &= \text{ct}_2 \bmod q_L q' \mathcal{R} \\ [\mathbf{s}^{*T} | \mathbf{e}^{*T}]^T &\leq B_{se}, \|e_1^*\|_2 \leq B_{e1}, \|e_2^*\|_2 \leq B_{e1}, t^* \in \mathcal{I}. \end{aligned}$$

By Lemma B.1, it then holds that

$$\text{Adv}_0 \leq \text{Adv}_1 + \varepsilon_{\text{snd}}$$

At this stage, the output of **GS.Open** is necessarily t^* . Indeed, **GS.Open** computes $\mathbf{ct}_{0,L} = [\mathbf{I}_d |\mathbf{A}| - \mathbf{B}|\mathbf{a}_3] \mathbf{v} \bmod q_L \mathcal{R} = \mathbf{A}\mathbf{s}^* + \mathbf{e}^* \bmod q_L \mathcal{R}$. Because $\mathbf{ct}'_0 = \mathbf{A}\mathbf{s}^* + \mathbf{e}^* \bmod q' \mathcal{R}$, the inverse CRT computation of $\mathbf{ct}_0$ gives $\mathbf{ct}_0 = \mathbf{A}\mathbf{s}^* + \mathbf{e}^* \bmod q_L q' \mathcal{R}$ (as $q' \wedge q_L = 1$). As a result, after simplification we have

$$a_1 = e_1^* + \underbrace{[\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T]}_{f^*} \begin{bmatrix} \mathbf{s}^* \\ \mathbf{e}^* \end{bmatrix} + \lfloor q_L q' / 2 \rfloor t^* \bmod q_L q' \mathcal{R}.$$

The ciphertext noise f^* can then be bounded in infinity norm by $\|e_1^*\|_\infty + \|(\mathbf{e}_{\text{op},1}, \mathbf{s}_{\text{op},1})\|_2 \|(\mathbf{s}^*, \mathbf{e}^*)\|_2 \leq B_{e1} + B_{se} B_{\text{op}}$, as $\|(\mathbf{e}_{\text{op},1}, \mathbf{s}_{\text{op},1})\|_2 = \sqrt{w_{\text{op}}} = B_{\text{op}}$.

Based on how the parameters are set in **GS.Setup**, it holds that $B_{e1} + B_{se}B_{op} < q_L q' / 4$. As a result, the decoding process in **GS.Open** constructs $t = t^*$. As $t^* \in \mathcal{I}$, it indeed holds that **GS.Open** returns t^* .

It thus means that the adversary cannot win **Game₁** (and the subsequent games we later introduce) by producing a group signature making the opening fail. As such, it must hold that t^* corresponds to the tag of an honest group member⁸. With that in mind, we proceed with the rest of the traceability proof, showing we can use an adversary to solve an M-SIS instance.

Game₂. We start by simulating the second opening key $\mathbf{b}_{op,2}$. Note that although we give $\mathbf{gmsk}$ to the adversary, $\mathbf{gmsk}$ only contains the secret to the first opening key $\mathbf{b}_{op,1}$. Hence, the second key $\mathbf{s}_{op,2}$ is not only unknown to the adversary, but it is also never used on the challenger's side. As a result, the challenger samples $\mathbf{b}_{op,2} \leftarrow U(\mathcal{R}_{q_L q'}^d)$ when generating the keys in **GS.KeyGen**. Using the exact argument as for **Game₃** in the CCA-anonymity proof (Theorem 5.1), it holds that

$$\text{Adv}_1 \leq \text{Adv}_2 + \varepsilon_{\text{M-LWE}}^{(\text{op})}$$

where $\varepsilon_{\text{M-LWE}}^{(\text{op})}$ is the hardness bound of M-LWE $_{n,d,d,q_L q', \mathcal{D}_{op}}$, where $\mathcal{D}_{op} = \mathcal{W}_{d, w_{op}/2}^2$ truncated in spectral norm at β_{op} .

Game₃. We now introduce a guess on the honest group member that is targetted by the adversary. At the outset of the game, the challenger samples $i^+ \leftarrow U([N])$ and defines $t^+ = t_{i^+}$. At the end of the game, it aborts if the output of **GS.Open** on $\mathbf{gsig}^*$ does not match t^+ . Because we have shown **GS.Open** outputs t^* (which matches the extracted tag), the guess being correct comes down to checking $t^+ = t^*$. As i^+ (and in turn t^+) is uniform and independent of $\mathcal{A}$'s view, we have $\mathbb{P}[\mathcal{A} \text{ wins Game}_3] = \mathbb{P}[\mathcal{A} \text{ wins Game}_2 \wedge t^+ = t^*] = \mathbb{P}[\mathcal{A} \text{ wins Game}_2] \mathbb{P}[t^+ = t^*]$ which directly yields

$$\text{Adv}_2 = N \cdot \text{Adv}_3 = \binom{n}{w} \text{Adv}_3.$$

Note that from **Game₃** onwards, we can assume i^+ is never queried to $\mathcal{O}_{\text{Cor}}$ (i.e., never corrupted), otherwise $\mathcal{A}$ would lose the game. More precisely, the challenger would abort if queried with $\mathcal{O}_{\text{Cor}}(i^+)$. It would not change the advantage of $\mathcal{A}$ as querying $\mathcal{O}_{\text{Cor}}(i^+)$ would automatically mean that t^+ , and in turn t^* , would designate a corrupt user. We now aim at simulating signing queries to user i^+ without resorting to its secret key $\mathbf{S}_{i^+}$ to eventually perform a reduction from M-SIS. To do so, we need to simulate the $\mathbf{v}$'s without the key, which imposes reprogramming the syndromes $\mathbf{u}$. This can only be done when $\mathbf{u}$ has already been replaced by a perfectly uniform syndromes, which then requires simulating the NIZK proofs.

Game₄. We thus start by simulating all the NIZK proofs π in each query to user i^+ . Concretely, when the adversary queries $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$ for some message $\mathbf{M}$, the

⁸ In the model of [BSZ05], it means that we showed the scheme was traceable. We must now show the scheme is non-frameable.

challenger proceeds as in **GS.Sign**($\text{gsk}[i^+], \mathbf{M}, \text{gpk}, \text{pp}$) except in the generation of π where it uses the simulator from Lemma B.1. As a result, a distinguisher between the views of $\mathcal{A}$ in **Game**₃ and **Game**₄ can readily be used to construct an attacker for the zero-knowledge property of (**Prove**, **Verify**). As we must simulate at most Q_{sign} such proofs, where Q_{sign} is the maximal number of group signature queries per user, it yields

$$\text{Adv}_3 \leq \text{Adv}_4 + Q_{\text{sign}} \cdot \varepsilon_{zk},$$

where ε_{zk} is defined in Lemma B.1.

Game₅. To prepare the simulation of $\mathbf{u}$ while maintaining a consistent opening, we slightly change the way $\mathbf{u}, \text{ct}'_0, \text{ct}_1$ are generated. We recall that as the adversary is given the opening key $\text{gmsk} = \mathbf{s}_{\text{op},1}$, it has the ability to run **GS.Open**. Hence, when answering a query to $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$, we need to ensure that the produced group signature opens to identity i^+ , i.e., **GS.Open** returns t^+ . We thus define **Game**₅ as follows. When queried for $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$, the challenger changes the call to **GS.Sign** by sampling $\mathbf{s}, \mathbf{e}, e_1, e_2$ as usual but by computing $\bar{\mathbf{u}} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod qq'\mathcal{R}$. It then computes $\mathbf{u} = \bar{\mathbf{u}} \bmod q\mathcal{R}$, and $\text{ct}'_0 = \bar{\mathbf{u}} \bmod q'\mathcal{R}$. Note that this is perfectly equivalent to what is done in **Game**₄, this is only a mere rewriting. Then, for ct_1 , the challenger defines it as

$$\text{ct}_1 = \mathbf{s}_{\text{op},1}^T (\bar{\mathbf{u}} \bmod q_L q' \mathcal{R}) + e_1 + [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0] \begin{bmatrix} \mathbf{s} \\ \mathbf{e} \\ e_2 \end{bmatrix} + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R}.$$

We now argue that this is also perfectly equivalent to **Game**₄. In the latter, we have

$$\begin{aligned} \text{ct}_1 &= (\mathbf{A}^T \mathbf{s}_{\text{op},1} + \mathbf{e}_{\text{op},1})^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R} \\ &= \mathbf{s}_{\text{op},1}^T \mathbf{A}\mathbf{s} + \mathbf{e}_{\text{op},1}^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R} \\ &= \mathbf{s}_{\text{op},1}^T (\mathbf{A}\mathbf{s} + \mathbf{e}) - \mathbf{s}_{\text{op},1}^T \mathbf{e} + \mathbf{e}_{\text{op},1}^T \mathbf{s} + e_1 + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R} \\ &= \mathbf{s}_{\text{op},1}^T (\mathbf{A}\mathbf{s} + \mathbf{e} \bmod q_L q' \mathcal{R}) + e_1 + (\mathbf{e}_{\text{op},1}^T \mathbf{s} - \mathbf{s}_{\text{op},1}^T \mathbf{e}) + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R} \end{aligned}$$

Yet, as $q_L q' | qq'$, we have $\mathbf{A}\mathbf{s} + \mathbf{e} \bmod q_L q' \mathcal{R} = (\mathbf{A}\mathbf{s} + \mathbf{e} \bmod qq' \mathcal{R}) \bmod q_L q' \mathcal{R}$. Hence, ct_1 is exactly the same in both games. As a result, both games are identical and we obtain

$$\text{Adv}_4 = \text{Adv}_5.$$

Game₆. We now simulate $\mathbf{u}$ by replacing it by a uniform syndrome, in each signing query to user i^+ . As $\text{ct}'_0, \text{ct}_1, \text{ct}_2$ share some of the randomness used in $\mathbf{u}$, we cannot change $\mathbf{u}$ independently of the rest, hence the changes from previous games. We thus end up changing all $(\mathbf{u}, \text{ct}'_0, \text{ct}_1, \text{ct}_2)$ in each query to $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$. This follows a similar argument to **Game**₅ in the CCA-anonymity proof (Theorem 5.1), except that we must now simulate Q_{sign} instances instead of just one. This corresponds to the multiple secret variant of MH-M-LWE. Indeed, by the previous rewriting, $\mathbf{u}, \text{ct}'_0, \text{ct}_1$ can be generated from the same $\bar{\mathbf{u}}$ modulo qq' , while ct_2

can be generated from an extra row to the MH-M-LWE instance. However, it also requires the additional element $e_1 + (\mathbf{e}_{\text{op}}^T \mathbf{s} - \mathbf{s}_{\text{op}}^T \mathbf{e})$ that essentially allows the correct decryption (so that the opening returns t^+ as needed). Fortunately, although $(\mathbf{s}_{\text{op},1}, \mathbf{e}_{\text{op},1})$ is given to the adversary, this additional element has the form of a noisy linear hint on the M-LWE secret-error vector $(\mathbf{s}, \mathbf{e}, e_2)$. Hence, we can rely on the MH-M-LWE assumption we introduced in Section 3 for specific parameters. In particular, we choose parameters that are compatible with the reduction from M-LWE of Theorem 3.1. We also note that as we need to simulate each of the at most Q_{sign} queries $\mathcal{O}_{\text{Sign}}(i^+, \cdot)$, we need to consider MH-M-LWE k with $k = Q_{\text{sign}}$.

Concretely, on receipt of a query $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$, the challenger further changes the call to **GS.Sign**. It samples $(\mathbf{s}, \mathbf{e}, e_1, e_2)$ as usual, but now samples $\bar{\mathbf{u}} \leftarrow U(\mathcal{R}_{qq'}^d)$. Following **Game₅**, it sets $\mathbf{u} = \bar{\mathbf{u}} \bmod q\mathcal{R}$, $\text{ct}_0' = \bar{\mathbf{u}} \bmod q'\mathcal{R}$ and $\text{ct}_1 = \mathbf{s}_{\text{op},1}^T (\bar{\mathbf{u}} \bmod q_L q' \mathcal{R}) + e_1 + \mathbf{e}_{\text{op},1}^T \mathbf{s} - \mathbf{s}_{\text{op},1}^T \mathbf{e} + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R}$. It then also samples $u_2 \leftarrow U(\mathcal{R}_{q_L q'})$ and sets $\text{ct}_2 = u_2 + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R}$. We now argue this is indistinguishable under MH-M-LWE $_{n,d,d+1,qq',\mathcal{D}_{se} \otimes \mathcal{D}_{e1}, \mathcal{D}_{e1}, \mathcal{D}_{\text{op}} \otimes \{0\}, 1}^{Q_{\text{sign}}}$ where $\mathcal{D}_{se} = \mathcal{D}_{\mathcal{R}^2 d, s_{se}}$ truncated at B_{se} , $\mathcal{D}_{e1} = \mathcal{D}_{\mathcal{R}, s_{e1}}$ truncated at B_{e1} , and $\mathcal{D}_{\text{op}} = \mathcal{W}_{d, w_{\text{op}}/2}^2$ truncated in spectral norm at β_{op} .

Indeed, let $\mathcal{D}$ be a distinguisher between the views of $\mathcal{A}$ in **Game₅** and **Game₆**. We construct a distinguisher $\mathcal{B}$ against MH-M-LWE which, upon reception of the instance $(\tilde{\mathbf{A}}, \tilde{\mathbf{U}}, \mathbf{m}^T, \mathbf{z}^T)$ in $\mathcal{R}_{qq'}^{(d+1) \times d} \times \mathcal{R}_{qq'}^{(d+1) \times Q_{\text{sign}}} \times \mathcal{R}^{1 \times (2d+1)} \times \mathcal{R}^{1 \times Q_{\text{sign}}}$, proceeds as follows. It first parses the instance as

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} \\ \tilde{\mathbf{b}}_{\text{op},2}^T \end{bmatrix}, \quad \tilde{\mathbf{U}} = \begin{bmatrix} \mathbf{U} \\ \mathbf{u}_2^T \end{bmatrix}, \quad \mathbf{m}^T = [\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0].$$

It defines $\mathbf{b}_{\text{op},2} = \tilde{\mathbf{b}}_{\text{op},2} \bmod q_L q' \mathcal{R}$, and then uses $\mathbf{A}$ in the definition of **pp** in **GS.Setup** and $\mathbf{b}_{\text{op},2}, \mathbf{s}_{\text{op},1}, \mathbf{e}_{\text{op},1}$ in the definition of the opening keys in **GS.KeyGen**. Then, in the j -th query to $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$, the distinguisher $\mathcal{B}$ defines $\bar{\mathbf{u}} = \mathbf{U}[j]$, $\mathbf{u} = \bar{\mathbf{u}} \bmod q\mathcal{R}$, $\text{ct}_0' = \bar{\mathbf{u}} \bmod q'\mathcal{R}$, and $\text{ct}_1 = \mathbf{s}_{\text{op},1}^T \bar{\mathbf{u}} + \mathbf{z}^T[j] + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R}$. It also defines $\text{ct}_2 = (\mathbf{u}_2^T[j] \bmod q_L q' \mathcal{R}) + \lfloor q_L q' / 2 \rfloor t^+ \bmod q_L q' \mathcal{R}$. Finally, it calls $\mathcal{D}$ on the total view of $\mathcal{A}$ in this simulated environment and outputs 1 (MH-M-LWE) if $\mathcal{D}$ returns **Game₅**, and 0 (uniform) otherwise.

We now check $\mathcal{B}$ correctly simulates both games. First, notice that whether $\tilde{\mathbf{U}}$ is uniform or not, $\mathbf{A}$ is uniform modulo qq' as required. We also have that $\tilde{\mathbf{b}}_{\text{op},2}$ is uniform modulo qq' which means $\mathbf{b}_{\text{op},2}$ is indeed uniform modulo $q_L q'$ as $q_L q' | qq'$. Additionally, the opening key is correctly distributed. Indeed, as $-\mathcal{W}_{d, w_{\text{op}}/2} = \mathcal{W}_{d, w_{\text{op}}/2}$ and because $\|M_\tau([\mathbf{s}_{\text{op},1}^T | \mathbf{e}_{\text{op},1}^T])\|_2 = \|M_\tau([\mathbf{e}_{\text{op},1}^T | -\mathbf{s}_{\text{op},1}^T | 0])\|_2 \leq \beta_{\text{op}}$, it holds that $(\mathbf{s}_{\text{op},1}, \mathbf{e}_{\text{op},1})$ follows $\mathcal{D}_{\text{op}}$ as well, as required. It also holds that the j -th entry of $\mathbf{z}^T$ is of the form $e_{1,j} + \mathbf{m}^T[\mathbf{s}_j^T | \mathbf{e}_j^T | e_{2,j}]^T = e_{1,j} + \mathbf{e}_{\text{op},1}^T \mathbf{s}_j - \mathbf{s}_{\text{op},1}^T \mathbf{e}_j$ for some $e_{1,j} \sim \mathcal{D}_{e1}$ and $(\mathbf{s}_j, \mathbf{e}_j, e_{2,j}) \sim \mathcal{D}_{se} \otimes \mathcal{D}_{e1}$.

Then, if $\tilde{\mathbf{U}}$ is uniform modulo qq' , each $\bar{\mathbf{u}} = \mathbf{U}[j]$ is uniform modulo qq' . Also, as $q_L q' | qq'$, each $u_2 = \mathbf{u}_2^T[j] \bmod q_L q' \mathcal{R}$ is also uniform in $\mathcal{R}_{q_L q'}$ as desired, which then simulates **Game₆** correctly. If on the other hand $\tilde{\mathbf{U}} = [\mathbf{A} | \mathbf{I}_{d+1}] \mathbf{X} \bmod q_L q' \mathcal{R}$ where the j -th column of $\mathbf{X}$ is $[\mathbf{s}_j^T | \mathbf{e}_j^T | e_{2,j}]^T$ (for the same $\mathbf{s}_j, \mathbf{e}_j, e_{2,j}$ that

are used in $\mathbf{z}^T[j]$), we have for each query

$$\begin{aligned}\mathbf{u} &= \bar{\mathbf{u}} \bmod q\mathcal{R} = \mathbf{A}\mathbf{s}_j + \mathbf{e}_j \bmod q\mathcal{R} \\ \mathbf{ct}'_0 &= \bar{\mathbf{u}} \bmod q'\mathcal{R} = \mathbf{A}\mathbf{s}_j + \mathbf{e}_j \bmod q'\mathcal{R} \\ \mathbf{ct}_1 &= \mathbf{s}_{\text{op},1}^T(\bar{\mathbf{u}} \bmod q_L q'\mathcal{R}) + e_{1,j} + \mathbf{e}_{\text{op},1}^T \mathbf{s}_j - \mathbf{s}_{\text{op},1}^T \mathbf{e}_j + \lfloor q_L q'/2 \rfloor t^+ \bmod q_L q'\mathcal{R} \\ \mathbf{ct}_2 &= u_2 + \lfloor q_L q'/2 \rfloor t^+ \bmod q_L q'\mathcal{R} = \mathbf{b}_{\text{op},2}^T \mathbf{s}_j + e_{2,j} + \lfloor q_L q'/2 \rfloor t^+ \bmod q_L q'\mathcal{R},\end{aligned}$$

which then perfectly simulate **Game**₅. It then yields

$$\text{Adv}_5 \leq \text{Adv}_6 + \varepsilon_{\text{MH-M-LWE}^{Q_{\text{sign}}}},$$

where the parameter $\varepsilon_{\text{MH-M-LWE}^{Q_{\text{sign}}}}$ denotes the hardness bound of the problem $\text{MH-M-LWE}_{n,d,d+1,qq',\mathcal{D}_{sc} \otimes \mathcal{D}_{e1},\mathcal{D}_{e1},\mathcal{D}_{\text{op}} \otimes \{0\},1}^{Q_{\text{sign}}}$.

Game₇. We now again separate the sampling of $\bar{\mathbf{u}} \bmod q\mathcal{R}$ and $\bar{\mathbf{u}} \bmod q'\mathcal{R}$ so that they are clearly independent. For each query $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$, instead of sampling $\bar{\mathbf{u}}$ uniformly in $\mathcal{R}_{qq'}^d$ in the call to **GS.Sign**, it samples $\mathbf{u} \leftarrow U(\mathcal{R}_q^d)$ and $\mathbf{ct}'_0 \leftarrow U(\mathcal{R}_{q'}^d)$, and computes $\bar{\mathbf{u}}$ as the inverse CRT of $(\mathbf{u}, \mathbf{ct}'_0)$ as $\bar{\mathbf{u}} = q'[q'^{-1}]_q \mathbf{u} + q[q^{-1}]_{q'} \mathbf{ct}'_0 \bmod qq'\mathcal{R}$. As a result, $\mathbf{u} = \bar{\mathbf{u}} \bmod q\mathcal{R}$, and $\mathbf{ct}'_0 = \bar{\mathbf{u}} \bmod q'\mathcal{R}$, while $\bar{\mathbf{u}} \bmod q_L q'\mathcal{R}$ is a combination of $\mathbf{u}$ and $\mathbf{ct}'_0$. By the Chinese Remainder Theorem and the fact that $q \wedge q' = 1$, it holds that the distribution of $\bar{\mathbf{u}}$ is indeed uniform in $\mathcal{R}_{qq'}^d$. As such, the games are identically distributed. Thence, we get

$$\text{Adv}_6 = \text{Adv}_7.$$

Game₈. We are now finally able to simulate the $\mathbf{v}$'s without the trapdoor of user i^+ in the signing queries to said user. This is a necessary step for the reduction from M-SIS as we aim at making the gadget vanish for the tag t^+ , preventing us from sampling the secret key $\mathbf{S}_{i^+}$ for user i^+ . We thus change the queries to $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$. Note that as of **Game**₇, $\mathbf{u}$ is uniform and independent from all other random variables (except $\mathbf{ct}_1$ which is constructed from $\mathbf{u}$, or rather $\bar{\mathbf{u}}$). Concretely, when calling **GS.Sign** in a query to $\mathcal{O}_{\text{Sign}}(i^+, \mathbf{M})$, the challenger samples $\mathbf{v}$ from $\mathcal{D}_{\mathcal{R}^m, s_v}$ (truncated at the bounds B_{v1}, B_{v2}, B_{v3}) and defines $\mathbf{u} = \mathbf{A}_{t^+} \mathbf{v} \bmod q\mathcal{R}$. This $\mathbf{u}$ is then used to define $\bar{\mathbf{u}}$ and construct $\mathbf{ct}_1$. Lemma 2.6 shows $\mathcal{P}_7 \in [1 - \delta_v, 1 + \delta_v] \mathcal{P}_8$ with

$$\delta_v = \frac{1 + \varepsilon_v}{1 - \varepsilon_v} \left(\frac{1 + \varepsilon_v/nm}{1 - \varepsilon_v/nm} \right)^{nm} - 1,$$

and where $\mathcal{P}_7, \mathcal{P}_8$ are defined as below.

$\mathcal{P}_7$	Sample $\mathbf{u} \leftarrow U(\mathcal{R}_q^d)$, and compute $\mathbf{v} \leftarrow \text{KleinSamplePre}(\mathbf{S}_{i^+}, \mathbf{u}, s_v)$. Output $(\mathbf{v}, \mathbf{u})$.
$\mathcal{P}_8$	Sample $\mathbf{v} \leftarrow \mathcal{D}_{\mathcal{R}^m, s_v}$, and compute $\mathbf{u} = \mathbf{A}_{t^+} \mathbf{v} \bmod q\mathcal{R}$. Output $(\mathbf{v}, \mathbf{u})$.

Truncating the $\mathbf{v}$ to the verification bounds does not change the marginal distance between the distributions. As **Game**₇ can be simulated from distribution

$\mathcal{P}_7^{Q_{\text{sign}}}$ and Game_8 from distribution $\mathcal{P}_8^{Q_{\text{sign}}}$ with the same randomized function, Lemma 2.2 with $a = 2\lambda$ yields

$$\text{Adv}_7 \leq \left(1 + \frac{\lambda(2\lambda - 1)\delta_v^2}{(1 - \delta_v)^{2\lambda+1}}\right)^{Q_{\text{sign}}/2\lambda} \cdot \text{Adv}_8^{1-1/2\lambda}.$$

Now none of the queries to $\mathcal{O}_{\text{Sign}}(i^+, \cdot)$ use $\mathbf{S}_{i^+}$, which means the challenger does not have to generate it.

Game₉. We now proceed with the reduction from M-SIS following the blueprint of [JS26]. The idea is to hide the tag guess t^+ within gpk , while still being able to answer signing queries. As the challenger does not need to produce $\mathbf{S}_{i^+}$, it means we can carry the security proof of [JS26] without producing signatures for the tag t^+ . To avoid repeating their exact security argument, we directly define **Game₉** after their whole hybrid argument. More precisely, in **Game₉**, the challenger changes the construction of $\mathbf{B}$ in **GS.KeyGen** by instead defining it as

$$\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 + q_L t^+ \mathbf{G}_H \bmod q\mathcal{R}.$$

Using the hybrid trapdoor switching argument of [JS26] corresponding to games $G_{0,0}, \dots, G_{0,9}, G_{1,0}, \dots, G_{1,9}, \dots, G_{dk_H-1,0}, \dots, G_{dk_H-1,9}$ in their proof of [JS26, Thm. 4.1] (note that we are in what they call branch **1**), we obtain

$$\text{Adv}_8 \leq h^{\text{od}k_H}(\text{Adv}_9),$$

where $h^{\text{od}k_H}$ is the dk_H -th composition power of the function h defined by

$$h(x) = \varepsilon_{\text{M-LWE}}^{(\text{gpk})} + \mu \left(2\varepsilon_{\text{M-LWE}}^{(\text{gpk})} + \mu \left(\varepsilon_{\text{M-LWE}}^{(\text{gpk})} + x \right)^{1-\frac{1}{2\lambda}} \right)^{1-\frac{1}{2\lambda}},$$

with $\varepsilon_{\text{M-LWE}}^{(\text{gpk})}$ the hardness bound of $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1^{2d}}$ and

$$\begin{aligned} \mu &= \left(1 + \frac{\lambda(2\lambda - 1)\delta_{\text{gpk}}^2}{(1 - \delta_{\text{gpk}})^{2\lambda+1}} \right)^{(N-1)m/2\lambda} \\ \delta_{\text{gpk}} &= \left(\frac{1 + \frac{\varepsilon}{dk_L}}{1 - \frac{\varepsilon}{dk_L}} \right)^{2dk_L} \left(\frac{1 + \frac{\varepsilon}{dk_H}}{1 - \frac{\varepsilon}{dk_H}} \right)^{2dk_H} \left(\frac{1 + \varepsilon}{1 - \varepsilon} \right)^{12d(n-1)+6} - 1. \end{aligned}$$

We refer to [JS26] for the details of each hybrid. We simply note that δ_{gpk} comes from their trapdoor switching lemma [JS26, Lem. 3.4], and that as opposed to their proof where they use the trapdoor switching to Q preimage generation, we use it for $(N-1)m$ preimages. This is because we apply it to m preimages per group member (except for user i^+). This explains the exponent $(N-1)m$ in the definition of μ . The definition of h and the inequality featuring $h^{\text{od}k_H}$ then rigorously follow by applying their games $G_{0,0}$ to $G_{dk_H-1,9}$.

Game₁₀. We now enforce a bound on the $\mathbf{v}$ in the final signature outputted by the adversary. Namely, the challenger aborts at the end of the game if

$$\left\| \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \end{bmatrix} \mathbf{v}_2 \right\|_2 > c\sqrt{nd}\|\mathbf{v}_2\|_2$$

By the (slightly updated) concentration bound on $\|\cdot\|_{2,\mathbb{Z}}$ from [AGJ⁺24,JS25a], the bound is satisfied with probability at least 1/2 for a small constant c close to 1. We discuss it further in Section 6.1, but we empirically verify that $c \approx 1.061$ yields such a concentration probability. We then have

$$\text{Adv}_9 \leq 2 \cdot \text{Adv}_{10},$$

Bounding Adv_{10} . We finally bound Adv_{10} by giving a reduction from M-SIS. Let $\mathcal{A}$ be a PPT adversary for Game_{10} . On input the M-SIS instance $[\mathbf{I}_d | \mathbf{A}'' | \mathbf{a}_3]$, the challenger samples $\mathbf{A}' \leftarrow U(\mathcal{R}_{q'}^{d \times d})$. It then defines $\mathbf{A}$ as the inverse CRT of $(\mathbf{A}'', \mathbf{A}')$, i.e., $\mathbf{A} = q'[q'^{-1}]_q \mathbf{A}'' + q[q^{-1}]_{q'} \mathbf{A}' \bmod qq'\mathcal{R}$. Notice that because $q \wedge q' = 1$, the Chinese Remainder Theorem ensures that $\mathbf{A}$ is indeed uniform in $\mathcal{R}_{qq'}^{d \times d}$ as required per Game_{10} . It also clearly holds by construction that $\mathbf{A} \bmod q\mathcal{R} = \mathbf{A}''$.

The challenger then uses $\mathbf{A}$ to define pp in [GS.Setup](#) and carries Game_{10} with $\mathcal{A}$ normally. In the end, $\mathcal{A}$ outputs $\text{gsig}^* = (\mathbf{v}, \text{ct}'_0, \text{ct}_1, \text{ct}_2, \pi)$ and the challenger extracted π to get $(t^*, \mathbf{s}^*, \mathbf{e}^*, e_1^*, e_2^*)$ (as per the change in Game_1). Because $\mathcal{A}$ wins Game_{10} , it must hold (by the change in Game_3) that $t^+ = t^*$. Hence $q_L t^* \mathbf{G}_H - \mathbf{B} = -\mathbf{R}_1 - \mathbf{A}\mathbf{R}_2 \bmod q\mathcal{R}$. Based on the extraction equations we then have

$$[\mathbf{I}_d | \mathbf{A} | q_L t^* \mathbf{G}_H - \mathbf{B} | \mathbf{a}_3] \mathbf{v} = \mathbf{A} \mathbf{s}^* + \mathbf{e}^* \bmod q\mathcal{R},$$

which can be rewritten as

$$[\mathbf{I}_d | \mathbf{A}'' | \mathbf{a}_3] \cdot \underbrace{\begin{bmatrix} \mathbf{v}_1 - \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \end{bmatrix} \mathbf{v}_2 - \begin{bmatrix} \mathbf{e}^* \\ \mathbf{s}^* \end{bmatrix} \\ v_3 \end{bmatrix}}_{\mathbf{x}} = \mathbf{0} \bmod q\mathcal{R}.$$

Because the group signature passes verification, we must have $v_3 \neq 0$, and thus $\mathbf{x} \neq \mathbf{0}$. Also, based on the bounds verified by $\mathbf{v}_1, \mathbf{v}_2, v_3, (\mathbf{s}^*, \mathbf{e}^*)$ and the change in Game_{10} , we have

$$\|\mathbf{x}\|_2 \leq \sqrt{(B_{v1} + c\sqrt{nd}B_{v2} + B_{se})^2 + B_{v3}^2} =: B.$$

Hence $\mathbf{x}$ is a solution to $\text{M-SIS}_{n,d,2d+1,q,B}$ which readily proves

$$\text{Adv}_{10} \leq \varepsilon_{\text{M-SIS}}$$

where $\varepsilon_{\text{M-SIS}}$ is the hardness bound of $\text{M-SIS}_{n,d,2d+1,q,B}$.

Combining everything. Putting everything together then gives

$$\begin{aligned} \text{Adv}_0 \leq \varepsilon_{\text{snd}} + \varepsilon_{\text{M-LWE}}^{(\text{op})} + N \left(Q_{\text{sign}} \varepsilon_{zk} + \varepsilon_{\text{MH-M-LWE}}^{Q_{\text{sign}}} \right. \\ \left. + \left(1 + \frac{\lambda(2\lambda - 1)\delta_v^2}{(1 - \delta_v)^{2\lambda+1}} \right)^{Q_{\text{sign}}/2\lambda} h^{\text{od}k_H} (2\varepsilon_{\text{M-SIS}})^{1 - \frac{1}{2\lambda}} \right) \end{aligned}$$

as claimed. $\square$

6 Performance Estimation

6.1 Parameter Selection and Security Estimation Methodology

We instantiate the group signature scheme of Section 4 with concrete parameters targetting a classical security of $\lambda = 128$ and a group size of $N \approx 2^{32}$, and $Q_{\text{sign}} = 2^{32}$. The parameters are set according to [GS.Setup](#) so as to provide sufficient security for the multiple M-LWE and M-SIS assumptions. In particular, we note that we need to take larger parameters than in [\[JS26\]](#) when instantiating their sampler as our M-SIS bound from Theorem 5.2 follows from that of a preimage sampled with a delegated trapdoor. Additionally, the choice of q' in [GS.Setup](#) is so that our IND-CCA verifiable encryption is perfectly correct, while minimizing the size of the ciphertext components and maximizing the M-LWE security.

Selecting Truncation Bounds. The different Gaussian tail bounds $B_{1,1}$, $B_{1,2}$, B_2 , B_{se} , B_{e1} , B_{v1} , B_{v2} , B_{v3} are selected based on Lemma 2.5, with a tailcut c_D and tailcut probability $(c_D \sqrt{2\pi e} e^{-\pi c_D^2})^D$ set so as to avoid too many rejections when sampling such elements.

We also address the other concentration bounds we need in the specifications or security proofs of our scheme. First, the matrices $\mathbf{R}_i$ sampled in [GS.KeyGen](#) are truncated in spectral norm at $B_R = \frac{3}{4}(\sqrt{nd} + \sqrt{ndk_H} + 6)$. This bound is proven for unstructured binomial matrices using random matrix theory [\[Ver12\]](#), and has been empirically verified for structured matrices over $\mathcal{R}$ and used in many prior works, e.g., [\[GMPW20, LNPS21, JRS23, LNP22, AGJ⁺24, JS25a\]](#). In particular, we observe that it is verified with a (empirical) probability of at least 0.99, which avoids rejecting too many keys in [GS.KeyGen](#) and is sufficient in our case.

To obtain a tighter M-SIS bound at the end of our proof of traceability, we also bound $\|\mathbf{R}\mathbf{v}_2\|_2$ by $c\sqrt{nd}\|\mathbf{v}_2\|_2$ for a small constant c . This is obtained by using a Johnson-Lindenstrauss-like bound, which was formalized in [\[JS25a\]](#) as bounding the *integer* spectral norm $\|\mathbf{R}\|_{2,\mathbb{Z}} = \max_{\mathbf{x} \in \mathbb{Z}^D \setminus \{\mathbf{0}\}} \|\mathbf{R}\mathbf{x}\|_2 / \|\mathbf{x}\|_2$. Again, although bounds can be proven for unstructured matrices, works using structured ones instead rely on empirically verified bounds. In our case, we observe that $c \approx 1.061$ yields a concentration probability of at least 1/2, for the matrix $\mathbf{R} = [\mathbf{R}_1^T | \mathbf{R}_2^T]^T$ drawn from a binomial and conditioned on the spectral norm of $\mathbf{R}_1, \mathbf{R}_2$ being bounded by B_R .

Finally, for the reduction from M-LWE to MH-M-LWE, we also need to truncate the spectral norm of $M_\tau(\mathbf{x}_{\text{op}}^T)$ for $\mathbf{x}_{\text{op}}$ drawn from $\mathcal{W}_{d, w_{\text{op}}/2}^2$. As described in Section 3, this spectral norm is equal to $\|\sigma(\mathbf{x}_{\text{op}}^* \mathbf{x}_{\text{op}})\|_\infty$, where σ is the canonical embedding of $\mathcal{K}$. Here again, we can truncate the spectral norm at key generation by requiring it to be upper-bounded by some β_{op} . The latter is chosen to have a concentration probability of at least 0.99, which is sufficient in our case. Through statistical estimation and interpolation, we empirically observe that β_{op} scales with $\sqrt{w_{\text{op}}}$, and decreases with d for fixed n, w_{op} . For our parameters (n, d, w_{op}) , we are able to heuristically estimate β_{op} when setting our parameters.

Assessing Matrix-Hint M-LWE. The security analysis of our scheme relies on the MH-M-LWE¹ and MH-M-LWE^{Q_{sign}} assumptions, which, by Theorem 3.1, reduce from M-LWE for the parameters we instantiate our scheme with. Following this reduction, one can thus estimate the hardness bound of MH-M-LWE through a carefully parameterized M-LWE assumption. When setting the parameters s_{se} and s_{e1} , we choose the free parameter α in GS.Setup so as to reduce constraints on other parameters. We observe that choosing α closer to $\sqrt{2}$ (while avoiding $1/\sqrt{\alpha^2 - 2}$ blowing up too much) gives the best results. In particular, we have $s_{e1} \gg s_{se}$. Applying Theorem 3.1 with $\mathbf{S} = \text{diag}(s_{se}^2 \mathbf{I}_{2nd}, s_{e1}^2 \mathbf{I}_n)$ as described in Section 3, we obtain

$$\varepsilon_{\text{MH-M-LWE}^k} \leq k \left(\varepsilon_{\text{M-LWE}} + \frac{\varepsilon_{\text{op}}}{1 - \varepsilon_{\text{op}}} \right)$$

where $\varepsilon_{\text{M-LWE}}$ is the hardness bound of M-LWE _{$n, d, d+1, qq', \mathcal{D}_{\mathcal{R}^{2d+1}, \sqrt{2}\eta_{\text{op}}}$} , and where $\varepsilon_{\text{MH-M-LWE}^k}$ is that of MH-M-LWE _{$n, d, d+1, qq', \mathcal{D}_{se} \otimes \mathcal{D}_{e1}, \mathcal{D}_{e1}, \mathcal{D}_{\text{op}} \otimes \{0\}, 1$} . Applying it for $k = 1$ and $k = Q_{\text{sign}}$ then gives the hardness bounds needed in Theorems 5.1 and 5.2.

Assessing M-LWE and M-SIS. The security assessment of the M-SIS and M-LWE assumptions underlying our scheme (and the proof system as described in Lemma B.1 in the expression of ε_{snd} and ε_{zk}) is carried with the latest version of lattice estimator [APS15] on the unstructured search assumptions. Although we consider all attacks, the results show that lattice reduction attacks outperform algebraic and combinatorial ones. For lattice reduction attacks, we use the more realistic lattice sieving cost model to estimate the hardness bounds as $2^{-(0.292B+16.4)}$, where B is the minimal BKZ blocksize outputted by the lattice estimator. We note that as opposed to other constructions in the Core-SVP model (i.e., with $2^{-0.292B}$) which aim for $\lambda = 110$, we aim for $\lambda = 128$ which then provides a similar security level.

We simply note that the hardness bound $\varepsilon_{\text{M-LWE}}^{(\text{ct})}$ is upper bounded by that of M-LWE _{$n, d, d+1, qq', \mathcal{D}_{\mathcal{R}^{2d+1}, s_{se}}$} . Indeed, in Theorem 5.1, $\varepsilon_{\text{M-LWE}}^{(\text{ct})}$ is related to the elliptical secret-error distribution $\mathcal{D}_{\mathcal{R}^{2d}, s_{se}} \otimes \mathcal{D}_{\mathcal{R}, s_{e1}}$. Given an instance for the spherical distribution $\mathcal{D}_{\mathcal{R}^{2d+1}, s_{se}}$, because $s_{e1} \gg s_{se}$, we can add an independent Gaussian error e' of width $\sqrt{s_{e1}^2 - s_{se}^2}$ to the last sample. It would turn the distribution $\mathcal{D}_{\mathcal{R}^{2d+1}, s_{se}}$ into $\mathcal{D}_{\mathcal{R}^{2d}, s_{se}} \otimes \mathcal{D}_{\mathcal{R}, s_{e1}}$ by simple convolution of discrete Gaussians (as $\sqrt{s_{e1}^2 - s_{se}^2} \gg \sqrt{2}\eta_{\varepsilon}(\mathbb{Z}^n)$). Adding this independent noise would also preserve the uniform distribution when the given instance is uniform. It then shows a simple reduction from M-LWE _{$n, d, d+1, qq', \mathcal{D}_{\mathcal{R}^{2d+1}, s_{se}}$} to M-LWE _{$n, d, d+1, qq', \mathcal{D}_{\mathcal{R}^{2d}, s_{se}} \otimes \mathcal{D}_{\mathcal{R}, s_{e1}}$} , allowing us to assess the hardness of the latter in a simpler way with the lattice estimator.

Final Security. The final security of our scheme then follows from the hardness bounds and parameter selection above by plugging them in the formulae from Theorems 5.1 and 5.2. We opt for the common practice of not accounting for the guessing loss N in the traceability proof, but we still keep N to be polynomially bounded. In the end, we reach 127 bits of classical security for

the CCA-anonymity and 126 bits for the full-traceability. When relaxing our construction to only reach CPA-anonymity, the security proof of Theorem 5.1 is greatly simplified as noted in Remark 5.1. In this case, we reach 146 bits of classical security for the CPA-anonymity, and still 126 bits for the full-traceability.

6.2 Performance Comparison

We report in Table 6.1 the size of our group signature, and associated security, for the CCA-anonymous version as well as the CPA-anonymous version for a fairer comparison with prior works. The full parameter set and derived hardness bounds for the CCA version are given in Appendix C in Tables C.1, C.2 and C.3.

	Group Signature Size						Security	
	$ \mathbf{v} $	$ \mathbf{ct}'_0 $	$ \mathbf{ct}_1 $	$ \mathbf{ct}_2 $	$ \pi $	Total	λ_{an}	λ_{tr}
CPA	26.59 KB	0.44 KB	0.50 KB	-	24.96 KB	52.49 KB	146	126
CCA	26.59 KB	0.44 KB	0.50 KB	0.50 KB	25.84 KB	53.87 KB	127	126

Table 6.1. Size and security estimates for the group signature (CPA and CCA variants). All sizes are expressed in kilobytes (KB). The anonymity security λ_{an} and traceability security λ_{tr} are expressed in bit security (i.e., $\lambda_x = \lfloor -\log_2 \epsilon_x \rfloor$). The security for the CPA variant results from a much simpler anonymity proof.

The comparison with prior constructions [LNPS21, LN22, LNP22, JS25b] is then given in Table 6.2 for similar group sizes. We observe that revealing $\mathbf{v}$ in clear limits the impact of the zero-knowledge proof, leading to performance gains over the most compact runtime-efficient construction to date, i.e., that of [LNP22] with the sampler of [JS25b]. Our technique of recycling randomness to avoid transmitting a full ciphertext also contributes to reducing our group signature size. We also observe that said technique only entails a very small overhead of less than 1.5 KB going from CPA to CCA-anonymity. This is not the case in prior constructions where the CCA version is either non-trivial to obtain as in [LNPS21, LN22], or incurs a larger overhead (e.g., 6.5 KB in [LNP22, JS25b]).

Additionally, compared to [GKV10] and [LN22], our group signature performance is independent of N in both size and computational complexity. This is mainly due to the fact that the NIZK relation from Section 4.3 does not depend on N , which is not the case for OR-proofs used in [GKV10, LN22].

Besides the performance improvement, we believe that one of the main contributions of this work is to highlight that the seminal blueprint of BMW may not be the best suited for lattices, where NIZK proofs have relatively high size or computational complexity. In particular, it shows that the delegated basis approach has a clear potential, all the more so as it has been arguably far less scrutinized than the BMW one. We therefore hope that our construction will foster new ones that will exploit this paradigm shift in an even more efficient way.

	Approach	NIZK Compl.	Size	CCA Overh.
[LNPS21]	PoK of Certificate	Low	203.00 KB	unspecified
[LNP22]	PoK of Certificate	Low	98.02 KB	6.47 KB
[LN22]	OR Proof	High	19.40 KB	unspecified
[JS25b]	PoK of Certificate	Low	75.71 KB	6.47 KB
Ours	Delegated Basis	Low	52.49 KB	1.38 KB

Table 6.2. Comparison of state-of-the-art lattice-based static group signatures. The “NIZK Compl.” column assesses the NIZK computational complexity. The “Size” column reports the group signature size in kilobytes (KB) of the CPA-anonymous construction. The “CCA Overh.” column details the overhead on the group signature size of turning the construction from CPA-anonymous to CCA-anonymous.

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A Reduction for Matrix-Hint M-LWE (Theorem 3.1)

Proof. We first proceed by showing MH-M-LWE^k reduces from MH-M-LWE^1 via a hybrid argument as k is polynomial, and then proving a reduction from M-LWE to MH-M-LWE^1 as a generalized version of [ENP24]. We note that we could alternatively show a reduction from M-LWE^k to MH-M-LWE^k , but the loss in advantage would become $((\frac{1+\varepsilon}{1-\varepsilon})^k - 1)/2$ which is larger than $k\varepsilon/(1-\varepsilon)$, albeit asymptotically equivalent. Our reduction is therefore tighter.

Reducing from Single Secret. We prove a general reduction in the following lemma, which holds for parameters polynomial in the security parameter and for any efficiently sampleable distributions.

Lemma A.1. *Let $\mathcal{R}$ be the ring of integers of a number field of degree n . Let m, d, q, k be in $\mathbb{N}^\times$ and $Q_{\text{hint}} \in \mathbb{N}$, all polynomial in the security parameter. Let $\mathcal{D}_s$ be a distribution over $\mathcal{R}^{d+m}$, and for $i \in [Q_{\text{hint}}]$, let $\mathcal{M}_i$ be a distribution over $\mathbb{Z}^{h_i \times n(m+d)}$ for some $h_i \in \mathbb{N}^\times$, and $\mathcal{D}_{y,i}$ a distribution over $\mathbb{Z}^{h_i}$. There is an efficient probabilistic reduction from $\text{MH-M-LWE}_{n,d,m,q,\mathcal{D}_s,(\mathcal{D}_{y,i})_i,(\mathcal{M}_i)_i,Q_{\text{hint}}}^1$*

to $\text{MH-M-LWE}_{n,d,m,q,\mathcal{D}_s,(\mathcal{D}_{y,i})_i,(\mathcal{M}_i)_i,Q_{\text{hint}}}^k$ which shows the respective hardness bounds satisfy $\varepsilon_{\text{MH-M-LWE}^k} \leq k \cdot \varepsilon_{\text{MH-M-LWE}^1}$. The same holds for the search versions, without the factor k .

Proof (Lemma A.1). Let $\mathcal{A}$ be a PPT adversary against MH-M-LWE^k . We show that $\text{Adv}_{\text{MH-M-LWE}^k}[\mathcal{A}] \leq k\varepsilon_{\text{MH-M-LWE}^1}$ by providing the following reduction. For each $j \in \{0, \dots, k\}$, we define the distribution $\mathcal{H}_j$ as

$\mathcal{H}_j$ Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{X} \leftarrow \mathcal{D}_s^k$, and for $i \in [Q_{\text{hint}}]$, $\mathbf{M}_i \leftarrow \mathcal{M}_i$, $\mathbf{Y}_i \leftarrow \mathcal{D}_{y,i}^k$ and compute $\mathbf{Z}_i = \mathbf{Y}_i + \mathbf{M}_i \tilde{\mathbf{X}}$. For $\ell \in [j]$, compute $\mathbf{b}_\ell = [\mathbf{A}|\mathbf{I}_m]\mathbf{X}[\ell] \bmod q\mathcal{R}$. Then, for $\ell \in [j+1, k]$, sample $\mathbf{b}_\ell \leftarrow U(\mathcal{R}_q^m)$, and finally compute $\mathbf{B} = [\mathbf{b}_1 | \dots | \mathbf{b}_k]$. Output $(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i)$.

We now use $\mathcal{A}$ to construct an adversary $\mathcal{B}$ against MH-M-LWE^1 which, on input an instance $(\mathbf{A}, \mathbf{b}, (\mathbf{M}_i)_i, (\mathbf{z}_i)_i)$ of $\text{MH-M-LWE}_{n,d,m,q,\mathcal{D}_s,(\mathcal{D}_{y,i})_i,(\mathcal{M}_i)_i,Q_{\text{hint}}}^1$, proceeds as follows.

$\mathcal{B}(\mathbf{A}, \mathbf{b}, (\mathbf{M}_i)_i, (\mathbf{z}_i)_i)$:

- 1 $j^* \leftarrow U([k])$
- 2 **for all** $j \in [k] \setminus \{j^*\}$ **do**
- 3 $\mathbf{x}_j \leftarrow \mathcal{D}_s$
- 4 **for all** $i \in [Q_{\text{hint}}]$ **do**
- 5 $\mathbf{y}_{j,i} \leftarrow \mathcal{D}_{y,i}$
- 6 $\mathbf{z}_{j,i} \leftarrow \mathbf{y}_{j,i} + \mathbf{M}_i \tau(\mathbf{x}_j)$
- 7 **if** $j < j^*$ **then** $\mathbf{b}_j \leftarrow [\mathbf{A}|\mathbf{I}_m]\mathbf{x}_j \bmod q\mathcal{R}$
- 8 **if** $j > j^*$ **then** $\mathbf{b}_j \leftarrow U(\mathcal{R}_q^m)$
- 9 $\mathbf{B} \leftarrow [\mathbf{b}_1 | \dots | \mathbf{b}_{j^*-1} | \mathbf{b} | \mathbf{b}_{j^*+1} | \dots | \mathbf{b}_k]$
- 10 **for all** $i \in [Q_{\text{hint}}]$ **do**
- 11 $\mathbf{Z}_i \leftarrow [\mathbf{z}_{1,i} | \dots | \mathbf{z}_{j^*-1,i} | \mathbf{z}_i | \mathbf{z}_{j^*+1,i} | \dots | \mathbf{z}_{k,i}]$
- 12 **return** $\mathcal{A}(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i)$

As all the parameters are polynomials and the distributions are efficiently samplable, it holds that $\mathcal{B}$ runs in the same time as $\mathcal{A}$ with some additive polynomial time overhead. Let us now analyze the reduction by conditioning on $j^* = j$. If $\mathbf{b}$ is uniform, then the distribution provided as input to $\mathcal{A}$ exactly follows distribution $\mathcal{H}_{j-1}$, as we still have $\mathbf{z}_i = \mathbf{y}_i + \mathbf{M}_i \mathbf{x}$ for some $\mathbf{x} \sim \mathcal{D}_s$ and $\mathbf{y}_i \sim \mathcal{D}_{y,i}$, which means $\mathbf{Z}_i$ is correctly formed as $\mathbf{Y}_i + \mathbf{M}_i \tilde{\mathbf{X}}$. If on the other hand $\mathbf{b} = [\mathbf{A}|\mathbf{I}_m]\mathbf{x} \bmod q\mathcal{R}$, the input to $\mathcal{A}$ then follows $\mathcal{H}_j$. We then have that the advantage $\text{Adv}_{\text{MH-M-LWE}^1}[\mathcal{B}]$ is equal to

$$\begin{aligned}
& |\mathbb{P}_{\mathbf{b} \text{ unif}}[\mathcal{B}(\mathbf{A}, \mathbf{b}, (\mathbf{M}_i)_i, (\mathbf{z}_i)_i) = 1] - \mathbb{P}_{\mathbf{b} \text{ M-LWE}}[\mathcal{B}(\mathbf{A}, \mathbf{b}, (\mathbf{M}_i)_i, (\mathbf{z}_i)_i) = 1]| \\
&= \left| \sum_{j \in [k]} \mathbb{P}[j^* = j] \mathbb{P}_{\mathbf{b} \text{ unif}}[\mathcal{B}(\mathbf{A}, \mathbf{b}, (\mathbf{M}_i)_i, (\mathbf{z}_i)_i) = 1 \mid j^* = j] \right. \\
&\quad \left. - \sum_{j \in [k]} \mathbb{P}[j^* = j] \mathbb{P}_{\mathbf{b} \text{ M-LWE}}[\mathcal{B}(\mathbf{A}, \mathbf{b}, (\mathbf{M}_i)_i, (\mathbf{z}_i)_i) = 1 \mid j^* = j] \right|
\end{aligned}$$

$$\begin{aligned}
&= \left| \sum_{j \in [k]} \frac{1}{k} (\mathbb{P}_{X \sim \mathcal{H}_{j-1}}[\mathcal{A}(X) = 1] - \mathbb{P}_{X \sim \mathcal{H}_j}[\mathcal{A}(X) = 1]) \right| \\
&= \frac{1}{k} |\mathbb{P}_{X \sim \mathcal{H}_0}[\mathcal{A}(X) = 1] - \mathbb{P}_{X \sim \mathcal{H}_k}[\mathcal{A}(X) = 1]| \\
&= \frac{1}{k} \text{Adv}_{\text{MH-M-LWE}^k}[\mathcal{A}]
\end{aligned}$$

where the second equality comes from the total probability formula, the third from the above observation and the fact that j^* is uniform, and the last equality by the fact that $\mathcal{H}_0 = \mathcal{P}_0$ and $\mathcal{H}_k = \mathcal{P}_1$, where $\mathcal{P}_0, \mathcal{P}_1$ are the distributions in the definition of MH-M-LWE^k in Definition 3.1. By definition of $\varepsilon_{\text{MH-M-LWE}^1}$, we have $\text{Adv}_{\text{MH-M-LWE}^1}[\mathcal{B}] \leq \varepsilon_{\text{MH-M-LWE}^1}$. We then showed that for any PPT adversary $\mathcal{A}$, $\text{Adv}_{\text{MH-M-LWE}^k}[\mathcal{A}] \leq k\varepsilon_{\text{MH-M-LWE}^1}$, which indeed concludes that $\varepsilon_{\text{MH-M-LWE}^k} \leq k\varepsilon_{\text{MH-M-LWE}^1}$ by taking the supremum.

For the search versions, the reduction $\mathcal{B}$ simply samples $k-1$ extra columns with independent secrets. That is, $\mathcal{B}$ samples $\mathbf{x}_2, \dots, \mathbf{x}_k \leftarrow \mathcal{D}_s, (\mathbf{y}_{2,i})_i, \dots, (\mathbf{y}_{k,i})_i$ from $(\mathcal{D}_{y,i})_i$. It then defines $\mathbf{B} = [\mathbf{b} | [\mathbf{A} | \mathbf{I}_m] \mathbf{x}_2 | \dots | [\mathbf{A} | \mathbf{I}_m] \mathbf{x}_k] \bmod q\mathcal{R}$, and uses the $\mathbf{M}_i$ from the instance to define $\mathbf{Z}_i = [\mathbf{z}_i | \mathbf{y}_{2,i} + \mathbf{M}_i \tau(\mathbf{x}_2) | \dots | \mathbf{y}_{k,i} + \mathbf{M}_i \tau(\mathbf{x}_k)]$ for all $i \in [Q_{\text{hint}}]$. It then returns the first column of $\mathcal{A}(\mathbf{A}, \mathbf{B}, (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i)$, i.e., the secret-error vector corresponding to $\mathbf{b}$. Hence the advantage of $\mathcal{B}$ is that of $\mathcal{A}$. In that case, we directly get $\text{Adv}[\mathcal{A}] \leq \varepsilon_{\text{search-MH-M-LWE}^1}$ and thus $\varepsilon_{\text{search-MH-M-LWE}^k} \leq \varepsilon_{\text{search-MH-M-LWE}^1}$. $\square$

Reducing Single Secret from M-LWE. We now bound $\varepsilon_{\text{MH-M-LWE}^1}$ by generalizing the reduction of [ENP24] to non-square matrices. Let $\mathcal{A}$ be a PPT adversary for $\text{MH-M-LWE}^1_{n,d,m,q,\mathcal{D}_{\mathcal{R}^{d+m},s},(\mathcal{D}_{\mathbb{Z}^{h_i},s_i})_i,(\mathcal{M}_i)_i,Q_{\text{hint}}}$. We now show that $\text{Adv}_{\text{MH-M-LWE}^1}[\mathcal{A}] \leq \varepsilon_{\text{M-LWE}} + \varepsilon/(1-\varepsilon)$ by constructing an adversary $\mathcal{B}$ against $\text{M-LWE}_{n,d,m,q,\mathcal{D}_{\mathcal{R}^{d+m},s'}}$ as follows.

$\mathcal{B}(\mathbf{A}, \mathbf{b})$:

- 1 **for all** $i \in [Q_{\text{hint}}]$ **do** $\mathbf{M}_i \leftarrow \mathcal{M}_i$
- 2 $\mathbf{S}_0 \leftarrow (\mathbf{S}^{-1} + \sum_{i \in [Q_{\text{hint}}]} \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{M}_i)^{-1}$
- 3 $\mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R}^{d+m}, \sqrt{s}}$
- 4 **for all** $i \in [Q_{\text{hint}}]$ **do**
- 5 $\mathbf{y}_i \leftarrow \mathcal{D}_{\mathbb{Z}^{h_i}, \sqrt{s_i}}$
- 6 $\mathbf{z}_i \leftarrow \mathbf{y}_i + \mathbf{M}_i \tau(\mathbf{x})$
- 7 $\mathbf{c} \leftarrow \mathbf{S}_0 \sum_{i \in [Q_{\text{hint}}]} \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{z}_i$
- 8 $\mathbf{t} \leftarrow \tau^{-1}(\mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}_0 - s'^2 \mathbf{I}_{n(m+d)}}}, \mathbf{c})$
- 9 $\mathbf{b}' \leftarrow \mathbf{b} + [\mathbf{A} | \mathbf{I}_m] \mathbf{t} \bmod q\mathcal{R}$
- 10 **return** $\mathcal{A}(\mathbf{A}, \mathbf{b}', (\mathbf{M}_i)_i, (\mathbf{z}_i)_i)$

Well-definition and Runtime. We first check that the reduction we described is well-defined and efficient. As all the parameters and dimensions are polynomial, the reduction is efficient and well-defined as long as the sampling of $\mathbf{t}$ is. It is the case if and only if $\mathbf{S}'_0 := \mathbf{S}_0 - s'^2 \mathbf{I}_{n(m+d)}$ is positive definite. We must then

show that the smallest eigenvalue of $\mathbf{S}_0$ exceeds s'^2 . It holds that

$$\begin{aligned}
\lambda_{\min}(\mathbf{S}_0) &= \lambda_{\max}(\mathbf{S}_0^{-1})^{-1} = \left\| \mathbf{S}^{-1} + \sum_{i \in [Q_{\text{hint}}]} \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{M}_i \right\|_2^{-1} \\
&\geq \left(\|\mathbf{S}^{-1}\|_2 + \sum_{i \in [Q_{\text{hint}}]} \|\mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{M}_i\|_2 \right)^{-1} \\
&\geq \left(\|\mathbf{S}^{-1}\|_2 + \sum_{i \in [Q_{\text{hint}}]} \|\mathbf{M}_i\|_2^2 \|\mathbf{S}_i^{-1}\|_2 \right)^{-1} \\
&\geq \left(\|\mathbf{S}^{-1}\|_2 + \sum_{i \in [Q_{\text{hint}}]} \beta_i^2 \|\mathbf{S}_i^{-1}\|_2 \right)^{-1} \\
&= 2s'^2
\end{aligned}$$

This proves that $\mathbf{S}'_0$ is positive definite but also larger than $s'^2 \mathbf{I}_{n(m+d)}$ for the Loewner order. Hence, the reduction is well-defined and polynomial-time.

Correctness. We now check the reduction is correct in that the input to $\mathcal{A}$ is close to a well-distributed MH-M-LWE¹ instance. First, if $\mathbf{b}$ is uniform, and as $\mathbf{t}$ is sampled independently of $\mathbf{b}$, it holds that $\mathbf{b}'$ is also uniform in $\mathcal{R}_q^m$. Hence the input to $\mathcal{A}$ is distributed as in $\mathcal{P}_0$ of Definition 3.1.

Next, assume $\mathbf{b} = [\mathbf{A}|\mathbf{I}_m]\mathbf{x}' \bmod q\mathcal{R}$ for some $\mathbf{x}' \leftarrow \mathcal{D}_{\mathcal{R}^{d+m}, s'}$. Then, $\mathbf{b}' = [\mathbf{A}|\mathbf{I}_m]\mathbf{r} \bmod q\mathcal{R}$ for $\mathbf{r} = \mathbf{x}' + \mathbf{t}$. Let us look at the conditional distribution of $\tau(\mathbf{r})$ given $(\mathbf{M}_i, \mathbf{z}_i)_i$. It holds that the conditional distribution follows

$$\tau(\mathbf{r}) \sim \mathcal{D}_{\mathbb{Z}^{n(m+d)}, s'} + \mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}'_0}, \mathbf{c}}.$$

Because $\mathbf{S}'_1 = s'^2 \mathbf{I}_{n(m+d)}$ and $\mathbf{S}'_0$ are positive definite, we can consider their inverse and define $\mathbf{T}^{-1} = \mathbf{S}'_0^{-1} + \mathbf{S}'_1^{-1} = \mathbf{S}'_0^{-1} + \frac{1}{s'^2} \mathbf{I}_{n(m+d)}$. We then have $\lambda_{\max}(\mathbf{T}^{-1}) = \|\mathbf{T}^{-1}\|_2 = \frac{1}{s'^2} + \|\mathbf{S}'_0^{-1}\|_2$. Yet, we have shown that $\|\mathbf{S}'_0^{-1}\|_2 = 1/(\lambda_{\min}(\mathbf{S}_0) - s'^2)$ is at most $1/s'^2$. So $\lambda_{\max}(\mathbf{T}^{-1}) \leq 2/s'^2$. This yields $\lambda_{\min}(\mathbf{T}) \geq s'^2/2 \geq \eta_\varepsilon(\mathbb{Z}^{n(m+d)})^2$, and therefore $\sqrt{\mathbf{T}} \geq \eta_\varepsilon(\mathbb{Z}^{n(m+d)})$.

By the convolution lemma of Lemma 2.4, we then obtain that conditioned on $(\mathbf{M}_i, \mathbf{z}_i)_i$, $\mathcal{D}_{\mathbb{Z}^{n(m+d)}, s'} + \mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}'_0}, \mathbf{c}} \in [\frac{1-\varepsilon}{1+\varepsilon}, \frac{1+\varepsilon}{1-\varepsilon}] \mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}_0}, \mathbf{c}}$. By definition of the conditional distribution and taking the statistical distance, it thus holds that $(\mathbf{A}, \mathbf{b}', (\mathbf{M}_i)_i, (\mathbf{Z}_i)_i)$ is within statistical distance $\frac{1}{2}(\frac{1+\varepsilon}{1-\varepsilon} - 1) = \frac{\varepsilon}{1-\varepsilon}$ of

$\mathcal{P}_2$ Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R}^{d+m}, \sqrt{\mathbf{S}}}$, and for $i \in [Q_{\text{hint}}]$, $\mathbf{M}_i \leftarrow \mathcal{M}_i$, $\mathbf{y}_i \leftarrow \mathcal{D}_{\mathbb{Z}^{h_i}, \sqrt{\mathbf{S}_i}}$. Compute $\mathbf{z}_i = \mathbf{y}_i + \mathbf{M}_i \tau(\mathbf{x})$ for all $i \in [Q_{\text{hint}}]$, and $\mathbf{c} = \mathbf{S}_0 \sum_{i \in [Q_{\text{hint}}]} \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{z}_i$. Then, sample $\mathbf{r} \leftarrow \tau^{-1}(\mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}_0}, \mathbf{c}})$. Finally, compute $\mathbf{b} = [\mathbf{A}|\mathbf{I}_m]\mathbf{r} \bmod q\mathcal{R}$, and output $(\mathbf{A}, \mathbf{b}, (\mathbf{M}_i)_i, (\mathbf{z}_i)_i)$.

We then show through the following lemma that $\mathcal{P}_2$ is identical to $\mathcal{P}_1$ from Definition 3.1.

Lemma A.2. *The following distributions $\mathcal{P}'_1, \mathcal{P}'_2$ are identical.*

$\mathcal{P}'_1$	Sample $\mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}}}$, and for $i \in [Q_{\text{hint}}]$, $\mathbf{y}_i \leftarrow \mathcal{D}_{\mathbb{Z}^{h_i}, \sqrt{\mathbf{S}_i}}$. Compute $\mathbf{z}_i = \mathbf{y}_i + \mathbf{M}_i \mathbf{x}$ for all $i \in [Q_{\text{hint}}]$. Output $(\mathbf{x}, (\mathbf{z}_i)_i)$.
$\mathcal{P}'_2$	Sample $\mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}}}$, and for $i \in [Q_{\text{hint}}]$, $\mathbf{y}_i \leftarrow \mathcal{D}_{\mathbb{Z}^{h_i}, \sqrt{\mathbf{S}_i}}$. Compute $\mathbf{z}_i = \mathbf{y}_i + \mathbf{M}_i \mathbf{x}$ for all $i \in [Q_{\text{hint}}]$, and $\mathbf{c} = \mathbf{S}_0 \sum_{i \in [Q_{\text{hint}}]} \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{z}_i$. Then, sample $\mathbf{r} \leftarrow \mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}_0, \mathbf{c}}}$. Output $(\mathbf{r}, (\mathbf{z}_i)_i)$.

Proof (of Lemma A.2). For $(\mathbf{x}', (\mathbf{z}'_i)_i)$, we have

$$\begin{aligned}
\mathcal{P}'_1(\mathbf{x}', (\mathbf{z}'_i)_i) &= \mathbb{P}[\mathbf{x} = \mathbf{x}' | \forall i \in [Q_{\text{hint}}], \mathbf{M}_i \mathbf{x} + \mathbf{y}_i = \mathbf{z}'_i | \mathbf{x} = \mathbf{x}'] \\
&= \mathcal{D}_{\mathbb{Z}^{n(m+d)}, \sqrt{\mathbf{S}}}(\mathbf{x}') \prod_{i \in [Q_{\text{hint}}]} \mathcal{D}_{\mathbb{Z}^{h_i}, \sqrt{\mathbf{S}_i}}(\mathbf{z}'_i - \mathbf{M}_i \mathbf{x}') \\
&= \alpha^{-1} \exp \left(-\pi \left(\mathbf{x}'^T \mathbf{S}^{-1} \mathbf{x}' + \sum_{i \in [Q_{\text{hint}}]} (\mathbf{z}'_i - \mathbf{M}_i \mathbf{x}')^T \mathbf{S}_i^{-1} (\mathbf{z}'_i - \mathbf{M}_i \mathbf{x}') \right) \right)
\end{aligned}$$

where $\alpha = \rho_{\sqrt{\mathbf{S}}}(\mathbb{Z}^{n(m+d)}) \prod_{i \in [Q_{\text{hint}}]} \rho_{\sqrt{\mathbf{S}_i}}(\mathbb{Z}^{h_i})$. Writing $\mathcal{P}'_1(\mathbf{x}', (\mathbf{z}'_i)_i) = \alpha^{-1} e^{-\pi A}$, we can reexpress A as

$$\begin{aligned}
A &= \mathbf{x}'^T \left(\mathbf{S}^{-1} + \sum_{i \in [Q_{\text{hint}}]} \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{M}_i \right) \mathbf{x}' + \sum_{i \in [Q_{\text{hint}}]} (\mathbf{z}'_i{}^T \mathbf{S}_i^{-1} \mathbf{z}'_i - 2\mathbf{x}'^T \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{z}'_i) \\
&= \mathbf{x}'^T \mathbf{S}_0^{-1} \mathbf{x}' - 2\mathbf{x}'^T \mathbf{S}_0^{-1} \mathbf{c}' + \sum_{i \in [Q_{\text{hint}}]} \mathbf{z}'_i{}^T \mathbf{S}_i^{-1} \mathbf{z}'_i,
\end{aligned}$$

where we used the expression of $\mathbf{S}_0$ and defined $\mathbf{c}' = \mathbf{S}_0 \sum_{i \in [Q_{\text{hint}}]} \mathbf{M}_i^T \mathbf{S}_i^{-1} \mathbf{z}'_i$. We then have

$$\mathcal{P}'_1(\mathbf{x}', (\mathbf{z}'_i)_i) = \alpha^{-1} \exp(-\pi((\mathbf{x}' - \mathbf{c}')^T \mathbf{S}_0^{-1} (\mathbf{x}' - \mathbf{c}') - \mathbf{c}'^T \mathbf{S}_0^{-1} \mathbf{c}' + \sum_{i \in [Q_{\text{hint}}]} \mathbf{z}'_i{}^T \mathbf{S}_i^{-1} \mathbf{z}'_i)).$$

As such, the conditional probability of $\mathbf{x} = \mathbf{x}'$ given $(\mathbf{z}_i)_i = (\mathbf{z}'_i)_i$ is

$$\frac{\mathcal{P}'_1(\mathbf{x}', (\mathbf{z}'_i)_i)}{\mathbb{P}[(\mathbf{z}_i)_i = (\mathbf{z}'_i)_i]} = \alpha'^{-1} \rho_{\sqrt{\mathbf{S}_0, \mathbf{c}'}}(\mathbf{x}')$$

where

$$\alpha' = \alpha \mathbb{P}[(\mathbf{z}_i)_i = (\mathbf{z}'_i)_i] \cdot \left(\exp(\pi \mathbf{c}'^T \mathbf{S}_0^{-1} \mathbf{c}') \prod_{i \in [Q_{\text{hint}}]} \rho_{\sqrt{\mathbf{S}_i}}(\mathbf{z}'_i) \right)^{-1}$$

which is a constant with respect to $\mathbf{x}'$, as $\mathbf{c}'$ is fully determined by $(\mathbf{z}'_i)_i$. Summing over all $\mathbf{x}'$ then gives $\alpha' = \rho_{\sqrt{\mathbf{S}_0, \mathbf{c}'}}(\mathbb{Z}^{n(m+d)})$. It thus proves that the distribution of $\mathbf{x}$ given $(\mathbf{z}_i)_i = (\mathbf{z}'_i)_i$ in $\mathcal{P}'_1$ is exactly the distribution of $\mathbf{r}$ given $(\mathbf{z}_i)_i = (\mathbf{z}'_i)_i$ in $\mathcal{P}'_2$. Because the $\mathbf{z}_i$ are identically distributed in both distributions, it holds that the joint distributions are equal and therefore that $\mathcal{P}'_1 = \mathcal{P}'_2$. $\square$

As $\mathcal{P}_1$ and $\mathcal{P}_2$ are obtained from $\mathcal{P}'_1$ and $\mathcal{P}'_2$ by the same (probabilistic) transformation, we have that $\mathcal{P}_1$ and $\mathcal{P}_2$ are identical. It then shows that the input distribution given to the distinguisher $\mathcal{A}$ is well distributed according to MH-M-LWE, up to a statistical distance of at most $\varepsilon/(1-\varepsilon)$. We thus conclude that $\text{Adv}_{\text{M-LWE}}[\mathcal{B}] \geq \text{Adv}_{\text{MH-M-LWE}^1}[\mathcal{A}] - \varepsilon/(1-\varepsilon)$. As a result, we get $\text{Adv}_{\text{MH-M-LWE}^1}[\mathcal{A}] \leq \varepsilon_{\text{M-LWE}} + \varepsilon/(1-\varepsilon)$, which, by taking the supremum, gives $\varepsilon_{\text{MH-M-LWE}^1} \leq \varepsilon_{\text{M-LWE}} + \varepsilon/(1-\varepsilon)$. Further applying Lemma A.1 to $\mathcal{D}_s = \mathcal{D}_{\mathcal{R}^{d+m}, \sqrt{s}}$ and $\mathcal{D}_{y,i} = \mathcal{D}_{\mathbb{Z}^{h_i}, \sqrt{s_i}}$ yields the result.

For the search versions, we use rigorously the same argument. As proven above, the instance fed to $\mathcal{A}$ is within statistical distance $\varepsilon/(1-\varepsilon)$ of the expected one, which proves in the same way that $\varepsilon_{\text{search-MH-M-LWE}^1} \leq \varepsilon_{\text{search-M-LWE}} + \varepsilon/(1-\varepsilon)$. Applying Lemma A.1 for the corresponding search variants then proves that $\varepsilon_{\text{search-MH-M-LWE}^k} \leq \varepsilon_{\text{search-M-LWE}} + \varepsilon/(1-\varepsilon)$ as claimed. $\square$

B Zero-Knowledge Proof

B.1 Optimizations: Tailoring LNP to our Relation

As mentioned in Section 4.3, the relation we have to prove in zero-knowledge can be efficiently proven with the framework of [LNP22] (along with the optimizations of [LN22]). We however tailor the proof framework for our relation by providing some key constraints and optimizations so as to reduce the overall proof size.

Lifting (too) Small Moduli. The sampler of [JS26] leverages the smoothness of the modulus q , which then factors into a product of small (powers of) prime factors. The authors noted that the presence of such small factors called for lifting their zero-knowledge relations (in their anonymous credentials and blind signature schemes) to obtain smaller proofs using [LNP22]. More precisely, relations are proven modulo a proof modulus q_π in [LNP22] which, for better soundness and in turn efficiency, should preferably be prime or a product of few not too small primes. As the knowledge soundness error directly depends on the smallest prime factor of q_π , choosing $q_\pi = qq' \cdot q_1$ would drastically impact the soundness and increase the proof size accordingly. In our construction, this is explained by q' and b_H (which divides q) being typically quite small. It would also place further splitting constraints on q' .

Such situations have been resolved in [LNP22] and subsequent works by lifting the equation modulo q_π and accounting for the lifting term in the approximate range proof component. In more details, we choose q_π to be a sufficiently large prime. Proving an equation of the form $\mathbf{C}\mathbf{x} = \mathbf{u} \bmod p\mathcal{R}$ with $\|\mathbf{x}\|_2 \leq B$ for a smooth p can be done as follows. We first write the equation in $\mathcal{R}$ as

$$\mathbf{C}\mathbf{x} - \mathbf{u} = p\mathbf{j},$$

for some $\mathbf{j}$ over $\mathcal{R}$. The idea is then to prove $\mathbf{C}\mathbf{x} - \mathbf{u} - p\mathbf{j} = \mathbf{0} \bmod q_\pi\mathcal{R}$, and that $\|\mathbf{C}\mathbf{x} - \mathbf{u} - p\mathbf{j}\|_\infty < q_\pi$ to incur no wrap-around modulo q_π so that the equation indeed holds over $\mathcal{R}$. To avoid having to commit to $\mathbf{j}$ in the witness itself, we prove

an approximate bound on $[p^{-1}]_{q_\pi}(\mathbf{C}\mathbf{x} - \mathbf{u})$ using the approximate range proof component of [LNP22]. It thus affect the parameters of this component, and in turn the proof size, but in much lighter way than committing to $\mathbf{j}$ altogether. Because we already use approximate range proofs in ℓ^2 norm to prove the bound on $\mathbf{x}$, we do everything in ℓ^2 and conclude using $\|\cdot\|_\infty \leq \|\cdot\|_2$. Defining these parameters still require knowing an a priori bound on $\mathbf{j}$. Looking at the equation in $\mathcal{R}$, we have

$$\|\mathbf{j}\|_2 \leq \frac{1}{p} (\|\mathbf{C}\|_F \|\mathbf{x}\|_2 + \|\mathbf{u}\|_2) \leq \frac{1}{p} \left(\|\mathbf{C}\|_F B + \frac{p}{2} \sqrt{nm} \right) = B_j$$

Depending on the structure of $\mathbf{C}$, we can derive a bound on its Frobenius norm using p and its dimensions. The bound B_j impacts the approximate range proof bound B_{arp} .

When proving the soundness, the extractor then extracts $\mathbf{x}^*$ such that $\|\mathbf{x}^*\|_2 \leq B$, and $\|[p^{-1}]_{q_\pi}(\mathbf{C}\mathbf{x}^* - \mathbf{u}) \bmod q_\pi\|_2 \leq B_{arp}$. Calling $\mathbf{j}^* = [p^{-1}]_{q_\pi}(\mathbf{C}\mathbf{x}^* - \mathbf{u}) \bmod q_\pi$, we then have $\mathbf{C}\mathbf{x}^* - \mathbf{u} - p\mathbf{j}^* = \mathbf{0} \bmod q_\pi$. Also, based on the above bounds, we have

$$\|\mathbf{C}\mathbf{x}^* - \mathbf{u} - p\mathbf{j}^*\|_\infty \leq \|\mathbf{C}^T\|_\infty \|\mathbf{x}^*\|_2 + \|\mathbf{u}\|_\infty + p\|\mathbf{j}^*\|_2 \leq \|\mathbf{C}^T\|_\infty B + p/2 + pB_{arp}$$

where $\|\mathbf{C}^T\|_\infty$ is the row-norm of $\mathbf{C}$. Hence, if q_π exceeds the quantity on the right-hand side, the equation holds over $\mathcal{R}$. Taking modulo p then shows the desired relation.

Leveraging Chinese Remainders for Smooth Moduli. Our relations sometimes feature smooth moduli of the form $p = p_1 p_2$ for two coprime integers p_1, p_2 (not necessarily prime). Such equations can either be proven modulo p directly as described above, or instead separated into two equations with smaller moduli. One would prove $(\mathbf{C} \bmod p_i \mathcal{R})\mathbf{x} = (\mathbf{u} \bmod p_i \mathcal{R}) \bmod p_i \mathcal{R}$ for $i \in \{1, 2\}$ separately, as well as $\|\mathbf{x}\|_2 \leq B$. In the extraction, one can recover the equation modulo p using the Chinese Remainder Theorem.

This approach has the advantage of relaxing the constraint on q_π as one would need $q_\pi > \Theta(\max(p_1, p_2)B_{arp})$ instead of $q_\pi > \Theta(p_1 p_2 B_{arp})$, opening for a smaller proof modulus. A caveat is that we now have two separate lifting terms and bounds B_{j_1}, B_{j_2} to take into account in the expression of B_{arp} . Hence, B_{arp} can be larger than before.

One therefore has to assess which solution is best for their specific relations and parameters. In our case, we observe this approach is worse for the equations modulo $q_L q'$ (for ct_1, ct_2), but better for Equation (1) modulo $q = q_L q_H$. In the latter, taking the equation modulo q_L gives $\mathbf{A}\mathbf{s} + \mathbf{e} = \mathbf{u}_L \bmod q_L \mathcal{R}$ (for $\mathbf{u}_L = [\mathbf{I}_d | \mathbf{A}] - \mathbf{B}[\mathbf{a}_3] \mathbf{v} \bmod q_L \mathcal{R}$) which simplifies the equation (a feature our scheme actually relies on to avoid having a separate full ciphertext component). The equation modulo q_H becomes $\mathbf{A}\mathbf{s} + \mathbf{e} - t\mathbf{u}'_H = \mathbf{u}_H \bmod q_H \mathcal{R}$, where $\mathbf{u}'_H = q_L \mathbf{G}_H \mathbf{v}_2 \bmod q_H \mathcal{R}$ and $\mathbf{u}_H = [\mathbf{I}_d | \mathbf{A}] - \mathbf{B}[\mathbf{a}_3] \mathbf{v} \bmod q_H \mathcal{R}$.

Elliptic Masking for Inhomogeneous Witnesses. A final optimization we leverage is to use elliptic Gaussian masks with different parameters to mask

different parts of the witness. In the framework of [LNP22], with the bimodal optimization of [LN22], the proof includes a masked opening of the witness of the form $\mathbf{z}_1 = \mathbf{y}_1 + c\mathbf{b}\mathbf{x}$. The mask $\mathbf{y}_1$ is Gaussian, $\mathbf{b}$ is the bimodal sign included in the witness, and c is a challenge. Rejection sampling is also performed so that $\mathbf{z}_1$ follows the same Gaussian as $\mathbf{y}_1$ and does not leak information on $\mathbf{x}$. In prior works, both $\mathbf{y}_1$ and $\mathbf{z}_1$ follow a *spherical* Gaussian of parameter σ_1 , where σ_1 is set as $\alpha_1 B_{cx}$, with B_{cx} a worst-case bound on $\|\mathbf{c}\mathbf{x}\|_2$, and α_1 a slack determining the rejection rate.

This monolithic approach is however suboptimal when the witness $\mathbf{x}$ is heterogeneous, e.g., when part of the vector is much smaller than another. For example, consider $\mathbf{x} = [\mathbf{x}_1^T | \mathbf{x}_2^T]^T$ where $\|\mathbf{x}_i\|_2 \leq B_i$ with $B_1 \ll B_2$. One would thus set $B_{cx} = \eta\sqrt{B_1^2 + B_2^2} \approx \eta B_2$ (we refer to [LNP22, Sec. 2.7] for details on how to bound $\mathbf{c}\mathbf{r}$ from a bound on $\mathbf{r}$). It means that the top part of $\mathbf{z}_1$, i.e., $\mathbf{z}_{1,1} = \mathbf{y}_{1,1} + c\mathbf{b}\mathbf{x}_1$, would employ a mask scaling with ηB_2 to mask a vector of size ηB_1 . As a result, the bit-size of $\mathbf{z}_{1,1}$, which affects the proof size, would be larger than necessary.

Instead, one can use two different parameters $\sigma_{1,1}$ and $\sigma_{1,2}$ for the masks of $\mathbf{x}_1$ and $\mathbf{x}_2$ respectively. In particular, one could choose $\sigma_{1,1} = \eta B_1$ and $\sigma_{1,2} = \eta B_2$, thus leading to a more compact opening $\mathbf{z}_{1,1}$. In our case, due to our reduction from M-LWE to MH-M-LWE, it holds that e_1 and e_2 are much larger than the rest of the witness $(t, \mathbf{s}, \mathbf{e})$, thus calling for this elliptic masking approach.

We however notice two very mild consequences. The first is that doing rejection sampling on elliptic Gaussians is close to doing two separate rejections. Hence, the rejection rates would likely multiply, leading to more frequent rejections overall. This then limits the segmentation of our witness to only very few subvectors. In particular, one cannot deal with the witness entrywise as the rejection rate would become exponential in the dimension of the witness. In our situation, we only consider two subvectors as described above. The second, less damaging, consequence is due to the bimodal rejection sampling. Because we use the same bimodal sign $\mathbf{b}$ for all the subvectors, we have to perform the bimodal rejection sampling on the joint distribution directly, i.e., the bimodal *elliptic* Gaussian. This complexifies the rejection probability that needs to be computed, but it stays acceptable for few subvectors. The latter has already been done in [JS25a, App. A.3.8]. More precisely, for two subvectors, the rejection probability will be

$$\frac{\exp\left(\pi \frac{\|\mathbf{c}\mathbf{x}_1\|_2^2}{\sigma_{1,1}^2} + \pi \frac{\|\mathbf{c}\mathbf{x}_2\|_2^2}{\sigma_{1,2}^2} + \pi \frac{\|\mathbf{R}\tau(\mathbf{x})\|_2^2}{\sigma_3^2}\right)}{M_{1,1}M_{1,2}M_3 \cosh\left(2\pi \frac{\langle \tau(\mathbf{z}_{1,1}), \tau(\mathbf{c}\mathbf{x}_1) \rangle}{\sigma_{1,1}^2} + 2\pi \frac{\langle \tau(\mathbf{z}_{1,2}), \tau(\mathbf{c}\mathbf{x}_2) \rangle}{\sigma_{1,2}^2} + 2\pi \frac{\langle \mathbf{z}_3^2, \mathbf{R}\tau(\mathbf{x}) \rangle}{\sigma_3^2}\right)}$$

where $M_{1,i} = \exp(\pi/\alpha_{1,i}^2)$ is the theoretical rejection rate for subvector i , and the highlighted portions are the approximate range proof terms taken from [JS25a] which we also have in our case (although it would need to be adapted to include the lifting terms as previously described).

Overall, these observations and optimizations allow for tailoring the framework of [LNP22] to the specificities of our relation, that is smooth moduli with

small prime factors, and heterogeneous witnesses, then leading to better proof sizes.

B.2 Security of the Proof System

We now give the formal security result for the proof system ([Prove, Verify](#)) based on the framework (and security results) from [\[LNP22, LN22\]](#) with the optimizations of [Sections B.1](#). We note that the zero-knowledge proof is done over a subring of $\mathcal{R}$, which is itself a power-of-two cyclotomic ring.

Lemma B.1 (Security of ([Prove, Verify](#))). *Let $\tilde{\mathcal{R}}$ be the power-of-two cyclotomic ring of degree $\tilde{n} \leq n$ and let $k = n/\tilde{n}$. We use $\tilde{\tau}$ to denote the coefficient embedding in $\tilde{\mathcal{R}}$. Let $\rho, \tilde{\eta}$ define the challenge space $\mathcal{C}_{\rho, \tilde{\eta}} = \{c \in \tilde{\tau}^{-1}([- \rho, \rho]^{\tilde{n}}) : c^* = c \wedge \sqrt[64]{\|c^{64}\|_1} \leq \tilde{\eta}\}$. Let $M_{1,1}, M_{1,2}, M_2, M_3$ be in $(1, \infty)$, and let $\alpha_x = \sqrt{\pi/\ln M_x}$ for $x \in \{(1,1), (1,2), 2, 3\}$. We call $\mathbf{s}_{1,1} = (t, \mathbf{s}, \mathbf{e}, \mathbf{b}, a_{se})$ the first subwitness of dimension $m_{1,1} = k(2d+1) + 2$ when embedded in $\tilde{\mathcal{R}}$, and $\mathbf{s}_{1,2} = (e_1, e_2, a_{e1}, a_{e2})$ the second subwitness of dimension $m_{1,2} = 2k + 2$ after embedding in $\tilde{\mathcal{R}}$. The elements a_x are ring elements of four-square decompositions to prove ℓ^2 norms exactly. Then, define $B_{s11} = \sqrt{w + B_{se}^2 + 1}$, $B_{s12} = \sqrt{2}B_{e1}$ to be bounds on $\mathbf{s}_{1,1}$ and $\mathbf{s}_{1,2}$ respectively. We also define the following lifting bounds*

$$\begin{aligned} B_{j,L} &= \sqrt{nd/4 + 1/q_L^2} \sqrt{nd}B_{se} + \sqrt{nd}/2 \\ B_{j,H} &= \sqrt{nd/4 + 1/q_H^2} \sqrt{nd}B_{se} + (1+w)\sqrt{nd}/2 \\ B_{j'} &= \sqrt{nd/4 + 1/q'^2} \sqrt{nd}B_{se} + \sqrt{nd}/2 \\ B_{j,ct} &= B_{e1}/q_L q' + n\sqrt{d}B_{se}/2 + \sqrt{w} + \sqrt{n}/2 \end{aligned}$$

We then define $\sigma_{1,1} = \alpha_{1,1}\tilde{\eta}B_{s11}$, $\sigma_{1,2} = \alpha_{1,2}\tilde{\eta}B_{s12}$, $\sigma_2 = \alpha_2\tilde{\eta}\sqrt{\tilde{n}m_2}$, and $\sigma_3 = \alpha_3\sqrt{337}\sqrt{B_{s11}^2 + B_{s12}^2 + B_{j,L}^2 + B_{j,H}^2 + B_{j'}^2 + 2B_{j,ct}^2}$. We also define the parameter $B_{arp} = \frac{2}{\sqrt{26}}c_{256}\sigma_3\sqrt{256}$. Then, let $d_2, m_2, \ell, q_\pi \in \mathbb{N}^\times$ be such that

$$\begin{aligned} q_\pi &\text{ prime s.t. } q_\pi = 5 \bmod 8, \\ q_\pi &> \max(B_{se}^2, B_{e1}^2, w) \\ q_\pi &> B_{arp}^2 - \min(B_{se}^2, B_{e1}^2, w, 1) \\ q_\pi &> \max(41\tilde{n}(m_{1,1} + m_{1,2} + k(3d+2))B_{arp}, w + \sqrt{nw}) \\ q_\pi &> B_{se}\sqrt{1 + ndq_L^2/4} + q_L(1/2 + B_{arp}) \\ q_\pi &> B_{se}\sqrt{1 + ndq_H^2/4} + q_H((1+w)/2 + B_{arp}) \\ q_\pi &> B_{se}\sqrt{1 + ndq'^2/4} + q'(1/2 + B_{arp}) \\ q_\pi &> B_{e1} + q_L q'(B_{se}\sqrt{nd}/2 + 3/2 + B_{arp}) \end{aligned}$$

We also let γ be an even divisor of $q_\pi - 1$, and D be the largest integer such that $2^{D-1}\rho\tilde{n} < \gamma$. Then, (Prove, Verify) is knowledge sound with an extractor running in expected polynomial time and soundness error

$$\varepsilon_{snd} = \frac{2}{|\mathcal{C}_{\rho, \tilde{\eta}}|} + q_\pi^{-\tilde{n}/2} + q_\pi^{-2\ell} + 2^{-128} + \varepsilon_{\text{M-SIS}}^{(snd)}$$

and zero-knowledge with loss $\varepsilon_{zk} = \varepsilon_{\text{M-LWE}}^{(zk)} + \text{negl}(\lambda)$. The term $\varepsilon_{\text{M-SIS}}^{(snd)}$ is the hardness bound of M-SIS $_{\tilde{n}, d_2, m_1+m_2, q_\pi, \beta_\pi}$ where β_π is equal to

$$4\tilde{\eta}\sqrt{4c_{\tilde{n}m_{1,1}}^2\sigma_{1,1}^2\tilde{n}m_{1,1} + 4c_{\tilde{n}m_{1,2}}^2\sigma_{1,2}^2\tilde{n}m_{1,2} + \left(2c_{\tilde{n}m_2}\sigma_2\sqrt{\tilde{n}m_2} + (2^D\tilde{\eta} + \gamma)\sqrt{\tilde{n}d_2}\right)^2},$$

while $\varepsilon_{\text{M-LWE}}^{(zk)}$ is the hardness bound of M-LWE $_{\tilde{n}, m_2-d_2-\lfloor 256/\tilde{n} \rfloor - \ell - 1, m_2, q_\pi, \mathcal{B}_1, \mathcal{B}_1}$ (with $\mathcal{B}_1$ the centered binomial distribution of parameter 1 over $\tilde{\mathcal{R}}$).

The size of the NIZK π is computed from the parameters of the proof system introduced in Lemma B.1 by

$$\begin{aligned} |\pi| = & \tilde{n}d_2 (\lceil \log_2 q_{\pi_2} \rceil - D + 2.25) + \left(\frac{256}{\tilde{n}} + 2\ell + 1 \right) \lceil \log_2 q_\pi \rceil \\ & + \tilde{n} \lceil \log_2 (2\rho + 1) \rceil + \tilde{n}m_{1,1} \left(\frac{1}{2} + \log_2 \sigma_{1,1} \right) + \tilde{n}m_{1,2} \left(\frac{1}{2} + \log_2 \sigma_{1,2} \right) \\ & + \tilde{n}(m_2 - d_2) \left(\frac{1}{2} + \log_2 \sigma_2 \right) + 256 \left(\frac{1}{2} + \log_2 \sigma_3 \right). \end{aligned}$$

We provide in Table C.2 a parameter set that meets all the conditions of Lemma B.1, as well as the soundness and zero-knowledge security and proof size.

C Suggested Parameter Set

Symbol	Description	Value
Parameters of TSampler		
λ	Security parameter	128
n	Signature ring degree	256
d	Module rank	7
q	Modulus	$12949853269 \approx 2^{33.59}$
k_L, k_H	Low and High Gadget lengths	1, 3
b_L, b_H	Low and High Gadget bases	12569, 101
q_L, q_H	Low and High Moduli	12569, 1030301
ε	Smoothing loss for TSampler	2^{-40}
s_L s_H	Gadget sampling widths	42494.647 346.656
s_1 s_2 s_3 s_4 $s_{1,1}$	Gaussian widths in TSampler	69334.299 42494.648 54785.490 600.455 69334.299
w	Hamming weight of tags	5
N	Group size	$8809549056 \approx 2^{33}$
$B_{1,1}$ $B_{1,2}$ B_2	User key bounds in GS.KeyGen	1275374.052 1007755.087 18876.358
β	User key bound	1625578.545
Parameters of KleinSamplePre		
m	Preimage dimension	36
ε_v	Smoothing loss for KleinSamplePre	2^{-131}
s_v	Gaussian width for KleinSamplePre	9199920.595
B_{v1} B_{v2} B_{v3}	Verification bounds on $\mathbf{v}$	233528465.682 282885349.168 72880162.004
Opening Parameters		
w_{op}	Hamming weight of opening keys	208
B_{op}	Euclidean norm of opening keys	14.422
β_{op}	Spectral norm of opening keys	23.366
ε_{op}	Smoothing loss for MH-M-LWE reduction	2^{-163}
α	Gaussian width slack for MH-M-LWE reduction	1.49
s_{se}	Gaussian width for $(\mathbf{s}, \mathbf{e})$	13.124
s_{e1}	Gaussian width for e_1, e_2	924.377
B_{se}	Tail bound on $(\mathbf{s}, \mathbf{e})$	333.129
B_{se}	Tail bound on e_1, e_2	7322.756
q'	Scaling modulus	4
qq' $q_L q'$	Security assumptions moduli	51799413076 50276

Table C.1. Suggested parameter set for the group signature.

Symbol	Description	Value
Proof System Parameters		
λ	Security parameter	128
$\tilde{n}$	Proof system ring degree	64
k	Subring embedding dimension	4
d_2	Module rank	15
q_π	Proof system modulus	15253607492933941
ℓ	Soundness amplification dimension	2
$m_{1,1}$	Subwitness dimensions	62
$m_{1,2}$		10
m_2	Dimension of ABDLOP randomness	67
χ	Distribution of ABDLOP randomness	$\mathcal{B}_1$
ρ	Infinity norm of challenges	8
$\tilde{\eta}$	Manhattan-like norm of challenges	93
$ \mathcal{C}_{\rho, \tilde{\eta}} $	Size of challenge space	$\approx 2^{129}$
$(\alpha_{1,i}, \alpha_2, \alpha_3)$	Rejection sampling slacks	(3.01, 3.01, 3.01)
$(M_{1,i}, M_2, M_3)$	Rejection sampling repetition rates	$(\sqrt{2}, \sqrt{2}, \sqrt{2})$
$\sigma_{1,1}$	First subwitness rejection sampling width	93279.126
$\sigma_{1,2}$	Second subwitness rejection sampling width	2899677.505
σ_2	Second rejection sampling width	18335.280
σ_3	Third rejection sampling width	29915380.299
γ	Mask commitment compression parameter	530478
D	Witness commitment compression parameter	11
B_{s11}	Subwitness bounds	333.138
B_{s12}		10355.941
$B_{j,L}$	Lifting term bounds	298504.843
$B_{j,H}$		298610.673
$B_{j'}$		298525.663
$B_{j,ct}$		112825.489
B_{arp}	Proven approximate range proof bound	123164959.160

Table C.2. Suggested parameter set for proof system ([Prove](#), [Verify](#)).

Assumption	Description	BKZ- β	Hardness Bound
M-LWE ^(op)	Hardness bound $\varepsilon_{\text{M-LWE}}^{(\text{op})}$ (Thm. 5.1 & 5.2)	444	$2^{-146.26}$
M-LWE ^(ct)	Hardness bound $\varepsilon_{\text{M-LWE}}^{(\text{ct})}$ (Thm. 5.1)	565	$2^{-181.65}$
M-LWE ^(MH)	Hardness bound of M-LWE underlying MH-M-LWE (Thm. 3.1)	541	$2^{-174.63}$
MH-M-LWE ¹	Hardness bound $\varepsilon_{\text{MH-M-LWE}^1}$ (Thm. 5.1)	-	$2^{-163.00}$
MH-M-LWE ^{Q_{sign}}	Hardness bound $\varepsilon_{\text{MH-M-LWE}^{Q_{\text{sign}}}}$ (Thm. 5.2)	-	$2^{-131.00}$
M-LWE ^(gpk)	Hardness bound $\varepsilon_{\text{M-LWE}}^{(\text{gpk})}$ (Thm. 5.2)	478	$2^{-156.21}$
M-SIS	Hardness bound $\varepsilon_{\text{M-SIS}}$ (Thm. 5.2)	532	$2^{-172.00}$
M-LWE ^(zk)	Hardness bound $\varepsilon_{\text{M-LWE}}^{(\text{zk})}$ (Lem. B.1)	500	$2^{-162.64}$
M-SIS ^(snd)	Hardness bound $\varepsilon_{\text{M-SIS}}^{(\text{snd})}$ (Lem. B.1)	384	$2^{-128.71}$

Table C.3. Security estimates for the parameters of Tables C.1 and C.2.